FRONTS AT THE SPEED OF LIGHT: RIGIDITY OF PERMUTIVE SECOND-ORDER CELLULAR AUTOMATA

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ABSTRACT. For an elementary rule f, the second-order (Fredkin) automaton F_f on $(\{0,1\}^2)^{\mathbb{Z}}$ is reversible, and the automorphism group of its spacetime subshift X_{F_f} is the centralizer of F_f in the automorphism group of the full 4-shift. It always contains $\langle \sigma, F_f \rangle$, generated by the shift and the automaton. We prove that $\mathrm{Aut}(X_{F_f}) = \langle \sigma, F_f \rangle \cong \mathbb{Z}^2$ for rule 30, for all 16 elementary rules that are permutive in an outer variable and not affine (the classes of 30, 45, 75 and 135 under reflection and complementation), and for the 8 rules in the classes of 63 and 187, which read only two neighbours and are not permutive. More generally, for a second-order automaton of any radius whose local rule is permutive in its leftmost variable, four conditions imply that the automorphism group is $\mathbb{Z}^2$: a stripping condition on the local rule, stronger than non-affineness for radius above 1; the absence of a cellwise complement symmetry; strong connectivity of the light-speed window graphs, for which we give a sufficient finite certificate; and an index condition on spatially periodic orbits. The second condition is necessary. The proof follows the front at which a local difference spreads at the speed of light. A maximum principle shows that every automorphism shifts this front by a constant, and the same holds backwards in time; an automorphism with both offsets zero depends only on a light cone, and it must be the identity. For the rules that read two neighbours, the window graphs are identified with the action of a self-similar group on the 4-ary tree, and the one-sided automorphisms are determined by induction on the anticipation.

1. INTRODUCTION

Let $\mathcal{A}$ be a finite alphabet and σ the left shift on $\mathcal{A}^{\mathbb{Z}}$. By the Curtis–Hedlund–Lyndon theorem [14] the endomorphisms of the full shift are the block maps, and its automorphisms are the invertible block maps. The group $\mathrm{Aut}(\mathcal{A}^{\mathbb{Z}})$ is very large: it contains every finite group and free groups [5], and every countable, locally finite, residually finite group [17], while its center consists of the powers of the shift [23].

A reversible cellular automaton $F \in \mathrm{Aut}(\mathcal{A}^{\mathbb{Z}})$ determines a two-dimensional subshift, its *spacetime* $X_F \subset \mathcal{A}^{\mathbb{Z}^2}$, whose rows are the configurations $F^n(c)$, $n \in \mathbb{Z}$; see Cyr, Franks and Kra [7] for the spacetime of a general shift endomorphism. The automorphism group of X_F is the centralizer of F in $\mathrm{Aut}(\mathcal{A}^{\mathbb{Z}})$ (Lemma 2.2), and it always contains the group $\langle \sigma, F \rangle$. We ask when it contains nothing else. This is a case of the commuting block maps problem of Coven, Hedlund and Rhodes [6]: to find all block maps that commute with a given one.

The automata we study are second-order automata, built by the second-order technique of Fredkin [29, 31]; for radius 1 they are the elementary reversible cellular automata introduced and studied systematically by Takesue [27, 28]. For a Boolean function f of $2r+1$ variables, on the alphabet $\mathcal{A} = \{0,1\}^2$ with cells written (x, y), put

$$F_f(c)_i = \big(y_i,\ x_i \oplus f(y_{i-r}, \ldots, y_{i+r})\big).$$

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Reading y as the current row and x as the previous one, this is the recursion $y_{t+1} = y_{t-1} \oplus f(y_t)$, which is reversible for every f. For $r = 1$ and f an elementary rule, one writes fR for F_f, following Takesue [27]. A rule f is *permutive* in a variable if complementing that variable always changes the value of f [14], and *affine* if it is a sum modulo 2 of some of its variables and a constant. Two rules are in the same *class* if one is obtained from the other by reflecting the neighbourhood, by complementing all cells, or both; the corresponding automata are then conjugate (Section 5.1). For the elementary rules these are the 88 classes of Takesue [27], [28, Table I].

Whether $\mathrm{Aut}(X_{F_f})$ is as small as $\langle \sigma, F_f \rangle$ depends on f. For rule 54 and many other elementary rules, words that F_f fixes in every context act as walls, and permuting the contents of trains of identical segments between walls embeds every finite group in $\mathrm{Aut}(X_{F_f})$ [10]. In this paper we prove the opposite conclusion for two other families of rules: those that are permutive in an outer variable and not affine, and some that read only two neighbours.

Theorem 1.1 (Theorem 3.1)**.** $\mathrm{Aut}(X_{F_{30}}) = \langle \sigma, F_{30} \rangle$*, and* $(a, b) \mapsto \sigma^a F_{30}^b$ *is an isomorphism* $\mathbb{Z}^2 \to \langle \sigma, F_{30} \rangle$*.*

Theorem 1.2 (Theorem 5.1)**.** *Let f be an elementary rule that is permutive in its left or its right variable and is not affine, that is, one of the sixteen rules*

30, 86, 106, 120, 45, 101, 166, 180, 75, 89, 154, 210, 135, 149, 169, 225.

Then $\mathrm{Aut}(X_{F_f}) = \langle \sigma, F_f \rangle \cong \mathbb{Z}^2$*.*

Theorem 1.3 (Theorem 6.1)**.** *Let f be one of the eight rules* 63, 119, 238, 252, 187, 207, 221, 243*, which are not permutive in an outer variable and depend on only two of their three arguments. Then* $\mathrm{Aut}(X_{F_f}) = \langle \sigma \rangle \times \langle F_f \rangle \cong \mathbb{Z}^2$*.*

In all three theorems there is no bound on the radius of the automorphisms, and the proofs are by hand; the finite checks they contain are set out in tables. The rules of Theorems 1.2 and 1.3 make up six of the 88 classes of elementary rules; their place in the classification of all of them is discussed in [11]. In Takesue's study of these automata, 30R and 75R appear as rules with neither an additive conserved quantity on two adjacent cells nor a local conservation law on at most six cells [28, Section 4 and Fig. 4]. Rule 110, which is neither permutive nor of the form of Theorem 1.3, is treated by a different method in [9].

The hypothesis that f is not affine cannot be dropped. For the linear rule 150, which is permutive in both outer variables, the map $\mathrm{id} + \sigma F_{150}$ is an automorphism of $X_{F_{150}}$ outside $\langle \sigma, F_{150} \rangle$ (Remark 5.8), and the automorphism group of the linear rule 90 contains every finite group [10]. The proof shows where nonlinearity enters: only in one derivative identity, in the step that removes the correction term (fact (c) in the proof of Lemma 3.21, and Lemma 4.26 in general).

The method extends to any radius. Fix $r \geq 1$ and a rule of the form $f(y_{i-r}, \ldots, y_{i+r}) = y_{i-r} \oplus g(y_{i-r+1}, \ldots, y_{i+r})$, permutive in its leftmost variable, and let F be the corresponding second-order automaton. Theorem 4.6 uses four hypotheses (Definition 4.5): (N), that a second derivative $\partial_0 \partial_j g$ is nonzero for some $j \geq r$, which for $r = 1$ is non-affineness; (D), that g is not self-dual, so that the cellwise complement κ does not commute with F; (Λ), that the light-speed window graphs Λ_M are strongly connected for every M; and (I), an index condition on the $\mathbb{Z}^2$-orbits of spatially periodic points.

Theorem 1.4 (Theorems 4.6 and 4.7)**.** *Under* (N), (D), (Λ) *and* (I), $\mathrm{Aut}(X_F) = \langle \sigma, F \rangle \cong \mathbb{Z}^2$*. Under* (N), (D) *and* (Λ) *alone,* $\mathrm{Aut}(X_F) \cong \mathbb{Z}^2$*, and* $\langle \sigma, F \rangle$ *is a subgroup of index dividing* $2r$ *with cyclic quotient. If g is self-dual, then* $\kappa \in \mathrm{Aut}(X_F) \setminus \langle \sigma, F \rangle$*, so* (D) *is necessary.*

Hypothesis (Λ) follows from a finite certificate (T_P) on periodic histories that covers all M at once (Proposition 4.38); the certificate is sufficient, and we do not know whether it is necessary. Hypothesis (I) is verified by enumerating spatially periodic points. For $r = 1$ the four hypotheses hold for the eight left-permutive rules of Theorem 1.2, by hand, and this is how that theorem is proved (Section 5). The examples of larger radius are computer-assisted: for $r = 2$, the certificate (T_P) with $P \le 4$ holds for all 65 504 non-affine g, so that (Λ) holds for all of them (Proposition 4.40), and all four hypotheses hold for the thirteen rules of Proposition 4.41; for $r = 3$ they hold for the six rules of Proposition 4.42. We conjecture that (N) can be weakened to non-affineness and that (Λ) holds for every non-affine g (Conjectures 4.43 and 4.44).

The method. The known techniques for proving that an automorphism group is small do not apply to these spacetimes. The complexity arguments of Cyr–Kra and Donoso–Durand–Maass–Petite [8, 12] need subexponential pattern growth, while the rows of X_F are arbitrary configurations. The rigidity of algebraic $\mathbb{Z}^2$-actions, which gives $\mathrm{Aut} = \mathbb{Z}^2$ for Ledrappier's shift [20, 19, 30, 1], needs a group shift, and the rules above are nonlinear. Hochman's theorem that positive entropy forces every finite group into the automorphism group of a $\mathbb{Z}^2$ shift of finite type [15] does not apply, since X_F has zero $\mathbb{Z}^2$-entropy. The finitary Ryan theorems [24, 18] compute centralizers of sets of several automata, not of a single one. For cellular automata that are permutive in both outer variables, the commutant is known to be small, and a bipermutive affine automaton commutes only with affine ones [21, 26]; the automata F_f are not permutive as automata on four letters (Remark 5.8), and these results give no description of the commutant of a nonlinear rule. Conjugacy of reversible cellular automata is undecidable [25], and we know of no general method that determines the centralizer of a given reversible automaton; the arguments below use specific properties of the rule.

What the proofs share with the low-complexity arguments is their object: an automorphism permutes the pairs of points that agree on a half-line (the right-asymptotic pairs), and it must respect the way in which their difference spreads. The lines along which it spreads, of slope ± 1 in spacetime, bound the light cone of [7]; they play the role that nonexpansive directions play in the expansive subdynamics of Boyle and Lind [4].

The property used for permutive rules is the light-speed front. Consider two configurations that agree far to the right. Because the rule is permutive in its left variable and the recursion is permutive in the previous row, the rightmost difference in their spacetime diagrams moves forward in time along a line of slope exactly 1 and backward in time along a line of slope -1, with at most one exceptional row (Proposition 3.3). The main step (Theorem 3.13) is that an automorphism shifts the forward line of every such pair by the same integer. The shift is a function of a finite window of the pair and is invariant under F. Its largest value is stable under arbitrary changes behind the front, by a maximum principle (Lemma 3.11), so the set of words ahead of the front at which the maximum is attained is nonempty and forward-closed in the light-speed window graph Λ_M. When Λ_M is strongly connected, it contains every word, and the shift is constant. The time reversal $\rho(x, y) = (y, x)$ gives the same for the backward line. This gives a homomorphism $\mathrm{Aut}(X_F) \to \mathbb{Z}^2$, injective on $\langle \sigma, F \rangle$. An automorphism with both offsets zero depends only on a light cone (Proposition 3.17), is permutive in both edges of the cone (Proposition 3.19), and so is the identity plus a correction, which the commutation relation forces to vanish (Lemma 3.21); nonlinearity enters only here. A parity argument on spatially periodic points (Lemma 3.23, and the index condition in general) shows that the image of the homomorphism is the image of $\langle \sigma, F \rangle$.

Strong connectivity of Λ_M is proved through a tower of factor graphs, in which each layer adds one bit and the lifting step needs a closed walk with odd holonomy. For rule 30 this walk

is a single loop (Lemma 3.7); in general it comes from a time-periodic solution of the rule in comoving coordinates, supplied by the certificate (T_P).

For the rules that read two neighbours, the rightmost difference is invariant under F, and a maximum principle of the same kind shows that every automorphism moves it by a constant, once the window graphs are known to be strongly connected. Here they are identified with the action of a self-similar group on the 4-ary tree, which is self-replicating and level-transitive (Proposition 6.13). An automorphism with offset zero reads only cells to its right, and an induction on the anticipation shows that the automorphisms with offset zero are exactly the powers of F (Theorem 6.19). In other words, F defines an automorphism of the one-sided full shift on four letters [3] whose centralizer there is $\langle F\rangle$ (Corollary 6.22). The self-similar group is used in the language of [22].

Computations. The proofs of Theorems 1.1–1.3 are by hand. Computations enter Theorem 1.4 only through the certificates for the examples of radius 2 and 3, and every statement that depends on one is marked as computer-assisted. Computations also confirm the theorems independently: an exhaustive SAT enumeration of the block maps of radius at most 4 commuting with F_{30} gives only the maps $\sigma^a F_{30}^b$ and two constant maps [11], in agreement with Theorem 1.1. Appendix A specifies all computations, and the code accompanies the paper.

Organisation. Section 2 fixes notation and proves Lemma 2.2. Section 3 proves Theorem 1.1; it is self-contained and is the model for the rest. Section 4 treats permutive automata of any radius (Theorem 1.4), with examples, and Section 5 derives Theorem 1.2 from it. Section 6 proves Theorem 1.3; it does not depend on Sections 4 and 5. Appendix A describes the computations. Appendix B determines the eventual image of a window graph that follows both configurations of a pair; it is not used in the proofs of the theorems.

2. Preliminaries

Throughout, $\mathcal{A} = \{0,1\}^2$, and a cell $(x,y) \in \mathcal{A}$ is also written as the integer $2x+y \in \{0,1,2,3\}$; thus $3 = (1,1)$ and $0 = (0,0)$. The left shift is $\sigma(c)_i = c_{i+1}$. A *block map of radius* R is a map $h\colon \mathcal{A}^{\mathbb{Z}} \to \mathcal{A}^{\mathbb{Z}}$ of the form $h(c)_i = h_{\mathrm{loc}}(c_{i-R}, \dots, c_{i+R})$ for a local rule $h_{\mathrm{loc}}\colon \mathcal{A}^{2R+1} \to \mathcal{A}$. A block map of radius R is also a block map of every larger radius. By the Curtis–Hedlund–Lyndon theorem, $\mathrm{End}(\mathcal{A}^{\mathbb{Z}})$ is the monoid of block maps and $\mathrm{Aut}(\mathcal{A}^{\mathbb{Z}})$ the group of invertible ones. For a block map F we write

$$\mathcal{C}_R(F) = \{h \text{ a block map of radius} \le R :\ h \circ F = F \circ h\}, \qquad \mathcal{C}(F) = \bigcup_R \mathcal{C}_R(F),$$

for the centralizer of F in the monoid $\mathrm{End}(\mathcal{A}^{\mathbb{Z}})$; the centralizer of F in the group $\mathrm{Aut}(\mathcal{A}^{\mathbb{Z}})$ is its group of invertible elements.

An elementary rule is a map $f\colon \{0,1\}^3 \to \{0,1\}$, identified with its Wolfram number $\sum_{a,b,c} f(a,b,c)\, 2^{4a+2b+c}$. Its second-order automaton F_f is

$$F_f(c)_i = \big(y_i,\ x_i \oplus f(y_{i-1}, y_i, y_{i+1})\big), \tag{1}$$

where $c_i = (x_i, y_i)$. It is reversible, with inverse

$$F_f^{-1}(c)_i = \big(y_i \oplus f(x_{i-1}, x_i, x_{i+1}),\ x_i\big). \tag{2}$$

It has radius 1, and F_f^b has radius at most $|b|$, so $\sigma^a F_f^b \in \mathcal{C}_{|a|+|b|}(F_f)$.

Definition 2.1. The *spacetime subshift* of a reversible cellular automaton F on $\mathcal{A}^{\mathbb{Z}}$ is

$$X_F = \big\{(F^n(c))_{n\in\mathbb{Z}} :\ c \in \mathcal{A}^{\mathbb{Z}}\big\} \subset \mathcal{A}^{\mathbb{Z}^2},$$

with the $\mathbb{Z}^2$-action generated by the horizontal shift σ and the vertical shift τ, $(\tau z)_n = z_{n+1}$. An *automorphism* of X_F is a homeomorphism of X_F commuting with both.

Lemma 2.2. *For reversible F the map $c \mapsto (F^n(c))_{n\in\mathbb{Z}}$ is a homeomorphism $\mathcal{A}^{\mathbb{Z}} \to X_F$ conjugating (σ, F) to (σ, τ), and it induces an isomorphism*

$$\mathrm{Aut}(X_F) \;\cong\; \{h \in \mathrm{Aut}(\mathcal{A}^{\mathbb{Z}}) :\; hF = Fh\}.$$

Under this isomorphism $\langle \sigma, \tau\rangle$ corresponds to $\langle \sigma, F\rangle$.

Proof. The map is continuous, injective (the 0-th row recovers c), and surjective onto X_F by definition, hence a homeomorphism of compact spaces; it carries σ to σ and F to τ since $F^n(F(c)) = F^{n+1}(c)$. A homeomorphism of $\mathcal{A}^{\mathbb{Z}}$ commuting with σ is a block map by Curtis–Hedlund–Lyndon, and commuting with τ means commuting with F. □

So determining $\mathrm{Aut}(X_F)$ means determining the invertible block maps that commute with F, and $\mathcal{C}(F)$ is the monoid of endomorphisms of X_F. It contains non-invertible elements: the constant map onto a fixed point of F is one. For every elementary rule with $f(0,0,0) = f(1,1,1) = 0$, which includes 30, both $(0,0)^{\mathbb{Z}}$ and $(1,1)^{\mathbb{Z}}$ are fixed by F_f, so $\mathcal{C}_R(F_f)$ contains these two constant maps for every R.

3. Rule 30: the automorphism group is $\mathbb{Z}^2$

Throughout this section $f(a,b,c) = a \oplus (b \vee c)$ is rule 30 and $F = F_{30}$, so that

$$F(c)_i = \big(y_i,\; x_i \oplus y_{i-1} \oplus (y_i \vee y_{i+1})\big).$$

By Lemma 2.2, $\mathrm{Aut}(X_F)$ is the group of invertible block maps of $\mathcal{A}^{\mathbb{Z}}$ that commute with F, and we use this description throughout. We prove that for rule 30 this group is as small as possible, namely $\langle \sigma, F\rangle \cong \mathbb{Z}^2$.

Theorem 3.1. $\mathrm{Aut}(X_{F_{30}}) = \langle \sigma, F_{30}\rangle$, *and* $\langle \sigma, F_{30}\rangle \cong \mathbb{Z}^2$. *More precisely, every invertible block map h with $hF_{30} = F_{30}h$ equals $\sigma^a F_{30}^b$ for exactly one $(a,b) \in \mathbb{Z}^2$, namely $a = -\frac{1}{2}(e^+(h) + e^-(h))$ and $b = \frac{1}{2}(e^+(h) - e^-(h))$, where e^+ is the forward offset of Theorem 3.13, which does not depend on the radius bound R used to define it, and $e^-(h) = e^+(\rho h \rho)$ is the backward offset of Corollary 3.16.*

In particular F_{30} has infinite order, and $\sigma^a F_{30}^b = \mathrm{id}$ only for $a = b = 0$; this part is elementary, since by Lemma 3.4 the map $\sigma^a F^b$ shifts the forward intercept of every pair by $b - a$ and the backward intercept by $-a - b$; applied to a single y-flip, whose intercepts are both finite, this shows that $\sigma^a F^b$ moves them unless $a = b = 0$. The theorem concerns automorphisms. The endomorphism monoid $\mathcal{C}(F_{30})$ contains the two constant maps onto the fixed points $(0,0)^{\mathbb{Z}}$ and $(1,1)^{\mathbb{Z}}$ (Section 2), and we do not determine it: the proof below uses the invertibility of h in an essential way (Lemma 3.9).

Strategy. We work with pairs of configurations that agree far to the right, and with the rightmost difference λ_t in row t of their spacetime diagrams. Because rule 30 is permutive in its left argument and the second-order rule is permutive in the previous row, the rightmost difference moves forward in time along a line $i = t + U$ and backward in time along a line $i = V - t$, with at most one exceptional row between them (Proposition 3.3). We call U and V the forward and backward intercepts of the pair. The main step (Theorem 3.13) is that an automorphism h shifts the forward intercept of every *front pair* (a pair whose rightmost difference is a y-bit) by the same integer $e^+(h)$. The shift $E(P)$ is a function of a finite window of the pair, and it is unchanged when the pair is advanced by F (Lemma 3.10). Let $e_{\max}$ be its largest value. A maximum principle (Lemma 3.11) shows that the pairs with $E = e_{\max}$ stay so under arbitrary changes

behind the front, so whether $E(P) = e_{\max}$ depends only on the cells of one configuration at and just ahead of the front (Corollary 3.12). The set of words with this property is nonempty and forward-closed in a finite graph Λ_M that follows one configuration along the light-speed front, and Λ_M is strongly connected (Lemma 3.7), so the set contains every word and the shift is constant. The graph Λ_M is built as a tower of factor graphs in which each layer adds one bit, and the lifting step uses a closed walk along which the new bit changes an odd number of times; for rule 30 this walk is a single loop. Constancy extends from front pairs to all pairs (Lemma 3.15), and through the time reversal $\rho(x, y) = (y, x)$ to the backward intercept. This gives a homomorphism $(e^+, e^-)\colon \mathrm{Aut}(X_F) \to \mathbb{Z}^2$ that is injective on $\langle \sigma, F\rangle$. If $e^+(h) = e^-(h) = 0$, then the value of h at a point of spacetime depends only on the right light cone of that point (Proposition 3.17). It is permutive in both edges of the cone (Proposition 3.19), hence equals the identity plus a correction that depends only on the part of the cone strictly to the right of the apex, and the commutation relation forces the correction to vanish (Lemma 3.21). Finally, a parity argument on the four spatially constant configurations shows that $e^+ \equiv e^-$ (mod 2) (Lemma 3.23), so every automorphism is brought to offsets $(0, 0)$ by an element of $\langle \sigma, F\rangle$. Figure 1 shows the profile of a pair, and Figure 3 the light cone.

Conventions. Bits are combined with $\oplus$, $\vee$, $\wedge$ and $\neg$. For a statement S, $[S] \in \{0, 1\}$ is 1 if S holds and 0 otherwise. Intervals $[m, n]$ are intervals of integers. We use $-\infty$ with the conventions $-\infty + k = -\infty < n$ for all integers k, n, and $\max \emptyset = -\infty$.

3.1. **Diagrams, pairs and profiles.** For $c \in \mathcal{A}^{\mathbb{Z}}$ and $t \in \mathbb{Z}$ let $y^t \in \{0, 1\}^{\mathbb{Z}}$ be the y-row of $F^t(c)$. The x-row of $F(d)$ is the y-row of d, so the x-row of $F^t(c)$ is y^{t-1}, and in particular $c_i = (y_i^{-1}, y_i^0)$. The *diagram* of c is $Y_c\colon \mathbb{Z}^2 \to \{0, 1\}$, $Y_c(t, i) = y_i^t$. By (1), every diagram satisfies

$$Y(t+1, i) \oplus Y(t-1, i) \oplus Y(t, i-1) = Y(t, i) \vee Y(t, i+1) \qquad \text{for all } (t, i) \in \mathbb{Z}^2. \tag{3}$$

The relation is an exclusive or in each of $Y(t+1, i)$, $Y(t-1, i)$ and $Y(t, i-1)$, so each of these three values is determined by the other four. Using the first two, any two consecutive rows determine all rows. Hence $c \mapsto Y_c$ is a bijection from $\mathcal{A}^{\mathbb{Z}}$ onto the set of solutions of (3), and two diagrams that agree on two consecutive rows are equal. Translations act by

$$Y_{\sigma^j F^k c}(t, i) = Y_c(t + k,\, i + j). \tag{4}$$

If h is a block map commuting with F, then $F^t h(c) = h(F^t c)$, so

$$\text{rows } t-1 \text{ and } t \text{ of } Y_{hc} \text{ are the } x\text{-row and the } y\text{-row of } h(F^t c). \tag{5}$$

Time reversal. Let $\rho\colon \mathcal{A}^{\mathbb{Z}} \to \mathcal{A}^{\mathbb{Z}}$ exchange the two bits of every cell, $\rho(c)_i = (y_i, x_i)$. That this exchange reverses time is noted in [27, p. 2500]. From (1), $F(\rho c)_i = \big(x_i,\ y_i \oplus f(x_{i-1}, x_i, x_{i+1})\big)$, so $\rho F \rho(c)_i = \big(y_i \oplus f(x_{i-1}, x_i, x_{i+1}),\ x_i\big) = F^{-1}(c)_i$ by (2). Thus $\rho F \rho = F^{-1}$, and clearly $\rho\sigma = \sigma\rho$. Consequently $\rho h \rho$ is an invertible block map commuting with F whenever h is, and $h \mapsto \rho h \rho$ is an automorphism of the group $\mathrm{Aut}(X_F)$. Since $F^t \rho = \rho F^{-t}$ and the y-row of ρd is the x-row of d,

$$Y_{\rho c}(t, i) = Y_c(-1 - t,\, i). \tag{6}$$

Pairs and profiles. A *pair* is an ordered pair $Q = (c, c')$ of distinct configurations with $c_i = c'_i$ for all sufficiently large i. Its difference diagram is $\Delta_Q = Y_c \oplus Y_{c'}$. If h is an invertible block map commuting with F, then $hQ = (hc, hc')$ is again a pair ($hc \neq hc'$ because h is injective, and if c, c' agree to the right of s and h has radius R, then hc, hc' agree to the right of $s + R$), and so are $F^k Q$, $\sigma^j Q$ and ρQ, defined componentwise.

If rows $t-1$ and t of Δ_Q vanish to the right of a site s, then by (3) rows $t+1$ and $t-2$ vanish to the right of $s+1$. Rows -1 and 0 vanish far to the right by definition, so by induction every

row of Δ_Q does. The *profile* of Q is

$$\lambda_t = \lambda_t(Q) = \max\{i : \Delta_Q(t,i) = 1\} \in \mathbb{Z} \cup \{-\infty\} \qquad (t \in \mathbb{Z}).$$

Lemma 3.2. *For every pair and every $t \in \mathbb{Z}$:*

(1) $\lambda_{t+1} \le \max(\lambda_{t-1}, \lambda_t + 1)$, *with equality if* $\lambda_{t-1} \ne \lambda_t + 1$;
(2) $\lambda_{t-1} \le \max(\lambda_{t+1}, \lambda_t + 1)$, *with equality if* $\lambda_{t+1} \ne \lambda_t + 1$.

Proof. By (3),

$$\Delta(t+1,i) = \Delta(t-1,i) \oplus \Delta(t,i-1) \oplus \big[(Y_c(t,i) \vee Y_c(t,i+1)) \oplus (Y_{c'}(t,i) \vee Y_{c'}(t,i+1))\big].$$

Let $m = \max(\lambda_{t-1}, \lambda_t + 1)$. For $i > m$ all three terms vanish, the last because $i, i+1 > \lambda_t$. Suppose m is finite. At $i = m$ the last term vanishes for the same reason. Since $m \ge \lambda_{t-1}$ and $m - 1 \ge \lambda_t$, we have $\Delta(t-1,m) = [m = \lambda_{t-1}]$ and $\Delta(t,m-1) = [m-1 = \lambda_t]$. At least one of these is 1, because m is one of $\lambda_{t-1}, \lambda_t + 1$, and both are 1 only if $\lambda_{t-1} = \lambda_t + 1$. So $\Delta(t+1,m) = 1$ when $\lambda_{t-1} \ne \lambda_t + 1$. This proves (1). Part (2) is the same argument applied to (3) solved for $Y(t-1,i)$, which has the same form with $t+1$ and $t-1$ exchanged. □

Proposition 3.3. *Let Q be a pair with profile λ_t, and put*

$$I^+ = \{t : \lambda_t \ne -\infty,\ \lambda_t \ge \lambda_{t-1}\}, \qquad I^- = \{t : \lambda_t \ne -\infty,\ \lambda_t \ge \lambda_{t+1}\}.$$

(1) *No two consecutive λ_t equal $-\infty$.*
(2) *If $t \in I^+$, then $t+1 \in I^+$ and $\lambda_{t+1} = \lambda_t + 1$. If $t \in I^-$, then $t-1 \in I^-$ and $\lambda_{t-1} = \lambda_t + 1$. The sets I^+ and I^- are disjoint, and every element of I^- is smaller than every element of I^+.*
(3) *The set $\mathbb{Z} \setminus (I^+ \cup I^-)$ has at most one element. If m is such an element (a* middle row*), then $m-1 \in I^-$, $m+1 \in I^+$ and $\lambda_m < \lambda_{m-1} = \lambda_{m+1}$. If $I^+ = \emptyset$ then $I^- = \mathbb{Z}$, and if $I^- = \emptyset$ then $I^+ = \mathbb{Z}$.*
(4) *Define the* forward intercept $U = U(Q)$ *and the* backward intercept $V = V(Q)$ *by $U = \lambda_t - t$ for $t \in I^+$ and $V = \lambda_t + t$ for $t \in I^-$ (well defined by (2)), with $U = -\infty$ if $I^+ = \emptyset$ and $V = -\infty$ if $I^- = \emptyset$. Then $\lambda_t \le \max(t + U, V - t)$ for all t. Equivalently, $\Delta_Q(t,i) = 1$ implies $i - t \le U$ or $i + t \le V$.*
(5) *Suppose U and V are finite. If $V - U$ is odd, there is no middle row, and $I^+ = [t_+, \infty)$, $I^- = (-\infty, t_+ - 1]$ with $2t_+ - 1 = V - U$. If $V - U$ is even, there is a middle row $m = (V-U)/2$, and $I^+ = [m+1, \infty)$, $I^- = (-\infty, m-1]$.*
(6) *For every t, $U \le \max(\lambda_t - t,\ \lambda_{t-1} - t - 1)$ and $V \le \max(\lambda_t + t,\ \lambda_{t+1} + t - 1)$.*

Proof. (1) If $\lambda_t = \lambda_{t+1} = -\infty$, then Y_c and $Y_{c'}$ agree on two consecutive rows, so $c = c'$, which is excluded.

(2) If $t \in I^+$ then $\lambda_{t-1} \le \lambda_t < \lambda_t + 1$, so Lemma 3.2(1) gives $\lambda_{t+1} = \max(\lambda_{t-1}, \lambda_t + 1) = \lambda_t + 1$; in particular $\lambda_{t+1} \ge \lambda_t$, so $t + 1 \in I^+$. The statement for I^- follows in the same way from Lemma 3.2(2). If $t \in I^+$ then $\lambda_{t+1} > \lambda_t$, so $t \notin I^-$. If $a \in I^+$ and $b \in I^-$ with $a \le b$, then $b \in I^+$ by the first statement, contradicting disjointness.

(3) Let $t \notin I^+ \cup I^-$. If $\lambda_t = -\infty$, then $\lambda_{t\pm1}$ are finite by (1), and $\lambda_{t+1} \ge \lambda_t$, $\lambda_{t-1} \ge \lambda_t$ give $t+1 \in I^+$, $t-1 \in I^-$. If λ_t is finite, then $t \notin I^+$ gives $\lambda_{t-1} > \lambda_t$ (in particular λ_{t-1} is finite), so $t - 1 \in I^-$; and $t \notin I^-$ gives $\lambda_{t+1} > \lambda_t$, so $t + 1 \in I^+$. If $t < t'$ were two such elements, then $t + 1 \in I^+$ and (2) would give $t' \in I^+$. Hence there is at most one. If $I^+ = \emptyset$, no such element can exist (it would have $t + 1 \in I^+$), so $I^- = \mathbb{Z}$; symmetrically for $I^- = \emptyset$.

Now let m be a middle row; we have shown $m \pm 1 \in I^\pm$ and $\lambda_m < \lambda_{m\pm1}$. If $\lambda_m = -\infty$, then $\lambda_{m-1} \ne \lambda_m + 1$ and Lemma 3.2(1) gives $\lambda_{m+1} = \lambda_{m-1}$. If λ_m is finite and $\lambda_{m-1} \ne \lambda_m + 1$, then Lemma 3.2(1) gives $\lambda_{m+1} = \max(\lambda_{m-1}, \lambda_m + 1) = \lambda_{m-1}$, since $\lambda_{m-1} \ge \lambda_m + 1$. If $\lambda_{m-1} = \lambda_m + 1$, then $\lambda_{m+1} \le \lambda_m + 1$ by Lemma 3.2(1), and $\lambda_{m+1} > \lambda_m$, so $\lambda_{m+1} = \lambda_m + 1 = \lambda_{m-1}$.

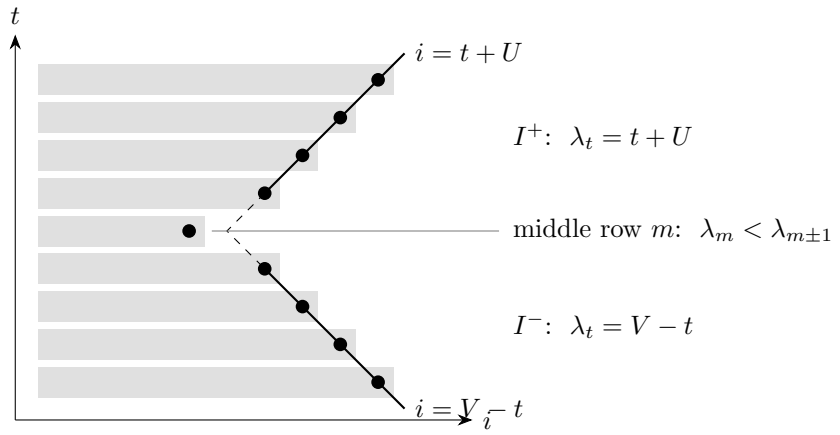


FIGURE 1. The profile of a pair with U, V finite and $V-U$ even (Proposition 3.3). Each row t of the difference diagram vanishes outside the shaded sites $i \le \lambda_t$, and the dot marks the rightmost difference λ_t. It moves along the line $i = t+U$ for $t \in I^+$ and along $i = V - t$ for $t \in I^-$; the two lines meet in the middle row $m = (V-U)/2$. If $V-U$ is odd there is no middle row.

(4) For $t \in I^+$ we have $\lambda_t = t+U$ and for $t \in I^-$ we have $\lambda_t = V - t$. For a middle row m, $\lambda_m < \lambda_{m-1} = V - m + 1$, so $\lambda_m \le V - m$.

(5) Both I^+ and I^- are nonempty. By (2) and (3), $\mathbb{Z}$ is the disjoint union of an interval $I^- = (-\infty, t_-]$, at most one middle row, and an interval $I^+ = [t_+, \infty)$. If there is no middle row, then $t_- = t_+ - 1$; now $t_+ \in I^+$ gives $t_+ + U \ge \lambda_{t_+-1} = V - t_+ + 1$, and $t_+ - 1 \in I^-$ gives $V - t_+ + 1 \ge t_+ + U$, so $V - U = 2t_+ - 1$. If there is a middle row m, then $\lambda_{m+1} = \lambda_{m-1}$ reads $m + 1 + U = V - m + 1$, so $V - U = 2m$. The two cases give different parities of $V-U$, so the parity decides which case occurs.

(6) Put $\mu_t = \max(\lambda_t - t,\ \lambda_{t-1} - t - 1)$. By Lemma 3.2(1), $\lambda_{t+1} - (t+1) \le \max(\lambda_{t-1} - t - 1,\ \lambda_t - t) \le \mu_t$, and $\lambda_t - t - 2 < \lambda_t - t \le \mu_t$, so $\mu_{t+1} \le \mu_t$. If $I^+ \ne \emptyset$, then for $t - 1 \in I^+$ we have $\mu_t = \max(U, U-2) = U$. As μ is nonincreasing, $U \le \mu_t$ for every t; if $I^+ = \emptyset$ there is nothing to prove. Similarly $\nu_t = \max(\lambda_t + t,\ \lambda_{t+1} + t - 1)$ is nondecreasing by Lemma 3.2(2), and $\nu_t = V$ whenever $t + 1 \in I^-$; hence $V \le \nu_t$ for every t. □

Lemma 3.4. *For every pair Q and all $j, k \in \mathbb{Z}$: $U(F^kQ) = U(Q) + k$ and $V(F^kQ) = V(Q) - k$; $U(\sigma^jQ) = U(Q) - j$ and $V(\sigma^jQ) = V(Q) - j$; $U(\rho Q) = V(Q) + 1$ and $V(\rho Q) = U(Q) - 1$.*

Proof. By (4), $\lambda_t(F^kQ) = \lambda_{t+k}(Q)$ and $\lambda_t(\sigma^jQ) = \lambda_t(Q) - j$. So $I^\pm(F^kQ) = I^\pm(Q) - k$, $I^\pm(\sigma^jQ) = I^\pm(Q)$, and the formulas for F^k and σ^j follow. By (6), $\lambda_t(\rho Q) = \lambda_{-1-t}(Q)$. Hence $t \in I^+(\rho Q)$ if and only if $-1 - t \in I^-(Q)$, and then $\lambda_t(\rho Q) = V(Q) - (-1-t) = t + V(Q) + 1$; and $t \in I^-(\rho Q)$ if and only if $-1 - t \in I^+(Q)$, and then $\lambda_t(\rho Q) = (-1-t) + U(Q) = U(Q) - 1 - t$. The values $-\infty$ correspond to empty sets on both sides. □

Definition 3.5. A pair (c, c') is a *front pair at p* if $c_i = c'_i$ for all $i > p$, $y(c_p) \ne y(c'_p)$ and $x(c_p) = x(c'_p)$; equivalently, $\lambda_0 = p > \lambda_{-1}$. For $c \in \mathcal{A}^{\mathbb{Z}}$ and $j \in \mathbb{Z}$, the *y-flip*, *x-flip* and *yx-flip at j* are the pairs (c, c') in which c' is obtained from c by complementing, respectively, the bit $y(c_j)$; the bit $x(c_j)$; both bits $y(c_j)$ and $x(c_{j+1})$.

Lemma 3.6. (1) *If Q is a front pair at p, then $U(Q) = p$, and F^tQ is a front pair at $p + t$ for every $t \ge 0$.*

(2) *A y-flip at j is a front pair at j, with $(U, V) = (j, j-2)$.*

(3) *If Q is an x-flip at j, then FQ is a front pair at j, and $(U,V)=(j-1,j-1)$.*
(4) *A yx-flip at j has $V=j$, and either $U=-\infty$ or $U\le j-1$.*

Proof. (1) Here $0\in I^+$, so $U=\lambda_0=p$, and by Proposition 3.3(2), $\lambda_t=p+t>\lambda_{t-1}$ for all $t\ge 0$. Since $\lambda_0(F^tQ)=\lambda_t$ and $\lambda_{-1}(F^tQ)=\lambda_{t-1}$, this says that F^tQ is a front pair at $p+t$.

(2) Here $\lambda_0=j$ and $\lambda_{-1}=-\infty$, so the pair is a front pair at j and $U=j$ by (1). By Lemma 3.2(2) at $t=-1$, since $\lambda_0\ne\lambda_{-1}+1$, $\lambda_{-2}=\max(\lambda_0,\lambda_{-1}+1)=j$. Thus $-2\in I^-$ and $V=j-2$.

(3) Here $\lambda_{-1}=j$ and $\lambda_0=-\infty$. By Lemma 3.2(1) at $t=0$, $\lambda_1=\max(j,-\infty)=j$. So $\lambda_0(FQ)=j>\lambda_{-1}(FQ)=-\infty$, FQ is a front pair at j, $1\in I^+$ and $U=j-1$. Also $-1\in I^-$ because $\lambda_{-1}\ge\lambda_0$, so $V=j-1$.

(4) Here $\lambda_0=j$ and $\lambda_{-1}=j+1$, so $-1\in I^-$ and $V=j$. By Lemma 3.2(1), $\lambda_1\le j+1$, and by (3), $\Delta(1,j+1)=\Delta(-1,j+1)\oplus\Delta(0,j)\oplus 0=1\oplus 1=0$, the last term vanishing because row 0 of Δ vanishes at $j+1$ and $j+2$. So $\lambda_1\le j$, and Proposition 3.3(6) at $t=1$ gives $U\le\max(\lambda_1-1,\lambda_0-2)\le j-1$. □

3.2. **The light-speed window graph.** Fix $M\ge 0$. Let Λ_M be the directed graph whose vertices are the words $\omega=\omega_0\cdots\omega_M\in\mathcal{A}^{M+1}$, with an edge $\omega\to\omega\cdot\iota$ for every $\iota=(\iota_1,\iota_2)\in\mathcal{A}^2$, where

$$(\omega\cdot\iota)_k=F(\tilde c)_{p+1+k}\qquad(0\le k\le M)$$

and $\tilde c$ is any configuration with $\tilde c_{p+k}=\omega_k$ for $0\le k\le M$, $\tilde c_{p+M+1}=\iota_1$ and $\tilde c_{p+M+2}=\iota_2$. Since F has radius 1, the cells $F(\tilde c)_i$ with $p+1\le i\le p+M+1$ read only $\tilde c$ on $[p,p+M+2]$, so $\omega\cdot\iota$ depends neither on p nor on the other cells of $\tilde c$. A vertex of Λ_M is the word of a single configuration on a cell p (the *front*) and the M cells ahead of it, and an edge advances the front by one site, at the speed of light.

Lemma 3.7. *For every $M\ge 0$, the graph Λ_M is strongly connected.*

The proof occupies the rest of this subsection. It builds Λ_M as a tower of $2M+2$ factor graphs, each of which adds one bit to the previous one (Figure 2).
Layer coordinates. For a vertex ω of Λ_M, placed on the sites $[p,p+M]$, put

$$y[d]=y(\omega_{-d})\ \ (-M\le d\le 0),\qquad x[k]=x(\omega_{-k-1})\ \ (-M-1\le k\le -1).\tag{7}$$

Thus the cell at site $p-d$ is $(x[d-1],y[d])$, and a vertex is the same as an arbitrary assignment of bits to the $2M+2$ *layers* $y[-M],\dots,y[0]$ and $x[-M-1],\dots,x[-1]$. The index of a layer is a comoving coordinate: if c is a configuration and $z_t[d]$ denotes the y-bit of $F^t(c)$ at site $p+t-d$, then the word of $F^t(c)$ on $[p+t,p+t+M]$ has y-layers $z_t[d]$ and x-layers $z_{t-1}[k]$.

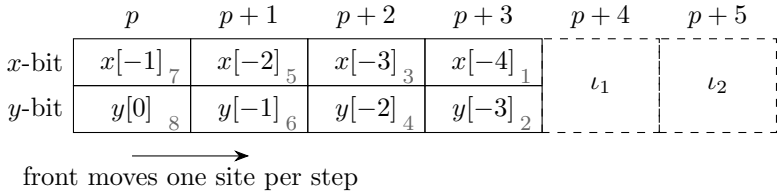


FIGURE 2. A vertex of Λ_M for $M=3$, with its layer coordinates (7) and the two input cells ι_1,ι_2 of an edge. The small numbers give the order $L_1,\dots,L_8$ in which the tower adds the layers: from the cell farthest ahead of the front back to the front, and within a cell first the x-bit, then the y-bit.

Lemma 3.8 (layer equation)**.** *Let ω be a vertex of Λ_M and $\iota = (\iota_1, \iota_2) \in \mathcal{A}^2$, let y', x' be the layers of $\omega' = \omega \cdot \iota$, and put*

$$u = y(\iota_1), \qquad v = x(\iota_1) \oplus y(\omega_M) \oplus \big(y(\iota_1) \vee y(\iota_2)\big).$$

Then:

(1) *$x'[-M-1] = u$, and $x'[k] = y[k]$ for $-M \le k \le -1$;*
(2) *$y'[-M] = v$;*
(3) *for $-M+1 \le d \le 0$,*

$$y'[d] = y[d] \oplus x[d-2] \oplus \big(y[d-1] \vee y[d-2]\big), \tag{8}$$

with the convention $y[-M-1] = u$.

The map $\iota \mapsto (u, v)$ is onto $\{0,1\}^2$. So every vertex ω has exactly four successors, the vertices $\omega \cdot (u, v)$, $(u, v) \in \{0,1\}^2$, given by (1)–(3).

Proof. Let $\tilde{c}$ be as in the definition of $\omega \cdot \iota$. The new cell at a site i is $F(\tilde{c})_i = \big(y(\tilde{c}_i),\, x(\tilde{c}_i) \oplus y(\tilde{c}_{i-1}) \oplus (y(\tilde{c}_i) \vee y(\tilde{c}_{i+1}))\big)$, and the layers of ω' are taken with respect to the new front $p+1$. So $x'[k]$ is the x-bit of the new cell at site $p-k$, which is the old y-bit at that site: this is $y[k]$ if $p-k \le p+M$, and $y(\iota_1) = u$ if $k = -M-1$. Likewise $y'[d]$ is the y-bit of the new cell at site $i = p+1-d$. For $d = -M$ this site is $p+M+1$, and the bit is v. For $d \ge -M+1$ we have $i \le p+M$, and $x(\tilde{c}_i) = x[d-2]$, $y(\tilde{c}_{i-1}) = y[d]$, $y(\tilde{c}_i) = y[d-1]$ and $y(\tilde{c}_{i+1}) = y[d-2]$, where $y[-M-1]$ stands for the y-bit at site $p+M+1$, which is u. Given (u, v), take $y(\iota_1) = u$, any ι_2, and $x(\iota_1) = v \oplus y(\omega_M) \oplus (u \vee y(\iota_2))$. □

Equation (8) is (3) written in comoving coordinates. It is lower triangular: $y'[d]$ reads only layers with index at most d.

The tower. Order the $N = 2M+2$ layers as

$$L_1 = x[-M-1],\ L_2 = y[-M],\ L_3 = x[-M],\ L_4 = y[-M+1],\ \ldots,\ L_{N-1} = x[-1],\ L_N = y[0],$$

that is, $x[d-1]$ followed by $y[d]$ for $d = -M, \ldots, 0$. Let $P_n = \{L_1, \ldots, L_n\}$, and let π_n be the restriction of an assignment of bits to P_n. By Lemma 3.8, the new value of each layer depends only on (u, v) and on layers that precede it or equal it in this order: $x'[k]$ is u or $y[k]$, which precedes $x[k]$; $y'[-M]$ is v; and $y'[d]$ reads $y[d]$ and the earlier layers $x[d-2]$, $y[d-1]$ and $y[d-2]$ (or u). Hence $\pi_n(\omega \cdot (u, v))$ depends only on $\pi_n(\omega)$ and (u, v). Let $\Lambda^{(n)}$ be the graph with vertex set $\{0,1\}^{P_n}$ and an edge $\eta \to \eta \cdot (u, v) := \pi_n(\omega \cdot (u, v))$, labelled (u, v), for every η and (u, v), where ω is any vertex of Λ_M with $\pi_n(\omega) = \eta$. Then:

- $\Lambda^{(N)}$ has the same vertices and the same successor sets as Λ_M, so one is strongly connected if and only if the other is;
- every walk of $\Lambda^{(n)}$ lifts, with the same labels, to a walk of $\Lambda^{(n+1)}$ starting at any vertex that restricts to its initial vertex.

Proof of Lemma 3.7. We show by induction on n that $\Lambda^{(n)}$ is strongly connected for $2 \le n \le N$.

Base. The vertices of $\Lambda^{(2)}$ are the values of $(x[-M-1], y[-M])$, and by Lemma 3.8(1),(2) the edge labelled (u, v) leads from every vertex to (u, v). So $\Lambda^{(2)}$ is a complete digraph with loops.

Predecessors. Every vertex ω_2 of Λ_M, with layers y_2, x_2, has an in-neighbour, and every in-neighbour $\hat{\omega}$ of ω_2, with layers $\hat{y}, \hat{x}$, satisfies $\hat{y}[k] = x_2[k]$ for $-M \le k \le -1$. The second statement is Lemma 3.8(1). For the first, put $u = x_2[-M-1]$, $v = y_2[-M]$, $\hat{y}[k] = x_2[k]$ for $-M \le k \le -1$ and $\hat{y}[0] = 0$, and then, with $\hat{y}[-M-1] = u$,

$$\hat{x}[d-2] = y_2[d] \oplus \hat{y}[d] \oplus \big(\hat{y}[d-1] \vee \hat{y}[d-2]\big) \qquad (-M+1 \le d \le 0),$$

and $\hat{x}[-1] = 0$. The indices $d-2$ are distinct and lie in $[-M-1, -2]$, so this defines a vertex $\hat{\omega}$, and $\hat{\omega} \cdot (u, v) = \omega_2$ by Lemma 3.8.

The step $L_{n+1} = x[k]$, $-M \le k \le -1$. Assume $\Lambda^{(n)}$ is strongly connected. Write the vertices of $\Lambda^{(n+1)}$ as (η, χ) with $\eta \in \{0,1\}^{P_n}$ and χ the bit $x[k]$. By Lemma 3.8, the layer $x[k]$ is read, if at all, only by $y'[k+2]$, which is not in P_{n+1}, and its new value is $x'[k] = y[k]$, with $y[k] \in P_n$. So the edge labelled (u,v) goes from (η,χ) to $(\eta \cdot (u,v),\, y[k](\eta))$. Let (η_1,χ_1) and (η_2,χ_2) be vertices of $\Lambda^{(n+1)}$. Extend (η_2,χ_2) to a vertex ω_2 of Λ_M and take an in-neighbour $\hat\omega$ of ω_2, with $\hat\omega \cdot (u,v) = \omega_2$. Then $\hat\eta = \pi_n(\hat\omega)$ has an edge labelled (u,v) to η_2, and $y[k](\hat\eta) = \hat y[k] = x_2[k] = \chi_2$. Take a walk from η_1 to $\hat\eta$ in $\Lambda^{(n)}$ and lift it to $\Lambda^{(n+1)}$ starting at (η_1,χ_1). It ends at $(\hat\eta,\chi')$ for some χ', and the edge labelled (u,v) then leads to (η_2,χ_2).

The step $L_{n+1} = y[d]$, $-M+1 \le d \le 0$. Assume $\Lambda^{(n)}$ is strongly connected. Here P_n consists of all layers with index at most $d-1$. For an edge of $\Lambda^{(n)}$ from η with label (u,v) put

$$\phi = x[d-2] \oplus \big(y[d-1] \vee y[d-2]\big) \in \mathbb{Z}/2,$$

evaluated on η, with $y[-M-1] = u$. By (8), the edges of $\Lambda^{(n+1)}$ are $(\eta, z) \to (\eta \cdot (u,v),\, z \oplus \phi)$, where z is the bit $y[d]$. The *holonomy* of a walk of $\Lambda^{(n)}$ is the sum in $\mathbb{Z}/2$ of ϕ over its edges; the lift of the walk from (η, z) ends at a vertex whose last bit is z plus the holonomy.

Let $\eta_* \in \{0,1\}^{P_n}$ be the vertex with

$$y[d-1] = x[d-1] = 1, \qquad \text{and all other layers of } P_n \text{ equal to } 0. \tag{9}$$

We claim that the edge from η_* labelled $(u,v) = (0,\, y[-M])$ is a loop, with $\phi = 1$. Indeed, by Lemma 3.8(1), $x'[-M-1] = u = 0$ and $x'[k] = y[k] = x[k]$ for $-M \le k \le d-1$, and by (2), $y'[-M] = v = y[-M]$. For $-M+1 \le i \le d-1$ the layers $y[i-1]$, $y[i-2]$ and $x[i-2]$ have index at most $d-2$, so they vanish (for $i-2 = -M-1$, by $u = 0$), and (8) gives $y'[i] = y[i]$. Finally $\phi = x[d-2] \oplus (y[d-1] \vee y[d-2]) = 0 \oplus (1 \vee y[d-2]) = 1$.

So $\Lambda^{(n)}$ has a closed walk with holonomy 1, namely this loop. Let η_2 be any vertex of $\Lambda^{(n)}$, and choose walks α from η_2 to η_* and β from η_* to η_2. The closed walks at η_2 formed by α followed by β, and by α, the loop and β, have holonomies of different parity. So there are closed walks of both parities at every vertex. Given vertices (η_1, z_1) and (η_2, z_2) of $\Lambda^{(n+1)}$, take a walk from η_1 to η_2 in $\Lambda^{(n)}$ followed by a closed walk at η_2, chosen so that the total holonomy is $z_1 \oplus z_2$. Its lift from (η_1, z_1) ends at (η_2, z_2).

Every layer L_{n+1} with $n \ge 2$ is covered by one of the two steps, so $\Lambda^{(N)}$ is strongly connected, and hence so is Λ_M. □

The form $a \oplus (b \vee c)$ of rule 30 enters this proof only through the loop at η_*: it is a stationary solution of (8) on the layers below d that cannot be continued to the layer $y[d]$. For other rules the loop is replaced by closed walks that come from time-periodic solutions (Section 4.10).

3.3. **The front offset.** In this subsection h is an invertible block map commuting with F, and $R \ge 1$ is an integer bounding the radii of both h and h^{-1}. Lemmas 3.9, 3.10 and 3.11 and Corollary 3.12 use nothing about the rule beyond Lemma 3.2 and its consequences in Section 3.1; the rule enters the proof of Theorem 3.13 only through Lemma 3.7.

Lemma 3.9. *Let P be a front pair at p. Then $U(hP)$ is finite, $|U(hP) - p| \le R$, and $I^+(hP) \supseteq [R+1, \infty)$; that is, $\lambda_t(hP) = t + U(hP)$ for every $t \ge R+1$.*

Proof. Write $\lambda_t = \lambda_t(hP)$, $P = (c, c')$, and let $t \ge 0$. By Lemma 3.6(1), $F^t P$ is a front pair at $p+t$, and by (5), rows $t-1$ and t of Δ_{hP} record where $h(F^t c)$ and $h(F^t c')$ differ in the x-bit and in the y-bit. Since h has radius at most R, these images agree at every site $i > p+t+R$, so $\lambda_t, \lambda_{t-1} \le p+t+R$. Since h^{-1} has radius at most R and $F^t c$, $F^t c'$ differ at site $p+t$, the images differ at some site in $[p+t-R,\, p+t+R]$, so $\max(\lambda_t, \lambda_{t-1}) \ge p+t-R$.

Let $t_+ = \min I^+$, with $t_+ = \infty$ if $I^+ = \emptyset$. Suppose $t_+ \ge 2$. By Proposition 3.3(2),(3), every $t \le t_+ - 2$ lies in I^- (in particular $I^- \ne \emptyset$ and $V = V(hP)$ is finite), and the row $t_+ - 1$ lies in

I^- or is the middle row. In both cases $\lambda_t \le V - t$ for all $t \le t_+ - 1$, by Proposition 3.3(4) (if $t_+ = \infty$, then $I^- = \mathbb{Z}$). As $-1 \in I^-$, the upper bound at $t = 0$ gives $V + 1 = \lambda_{-1} \le p + R$. For $1 \le t \le t_+ - 1$ we get

$$p + t - R \le \max(\lambda_t, \lambda_{t-1}) \le V - t + 1 \le p + R - t,$$

so $t \le R$. If $t_+ = \infty$ this fails at $t = R + 1$, a contradiction; otherwise taking $t = t_+ - 1$ gives $t_+ \le R + 1$. So in all cases $t_+ \le R + 1$, and U is finite. For $t \ge R + 1$ we have $t \in I^+$, so $\lambda_t = t + U \ge \lambda_{t-1}$, and the two bounds give $p + t - R \le t + U \le p + t + R$. □

For a front pair P at p put

$$E(P) = U(hP) - p.$$

This quantity is intrinsic: it does not involve R.

Lemma 3.10. *Let $P = (c, c')$ be a front pair at p.*

1. *$E(P) \in [-R, R]$, and $E(P)$ depends only on the restrictions of c and c' to $[p - 2R, p + 4R + 2]$: if $\tilde{P} = (\tilde{c}, \tilde{c}')$ is a front pair at p with $\tilde{c} = c$ and $\tilde{c}' = c'$ on this interval, then $E(\tilde{P}) = E(P)$.*
2. *$E(\sigma^j P) = E(P)$ for every $j \in \mathbb{Z}$.*
3. *$E(FP) = E(P)$.*
4. *(c', c) is a front pair at p, and $E(c', c) = E(c, c')$.*

Proof. (1) $E(P) \in [-R, R]$ is Lemma 3.9. By that lemma, $R + 1 \in I^+(hP)$, so $U(hP) = \lambda_{R+1}(hP) - R - 1$. By (5), $\lambda_{R+1}(hP)$ is the largest site at which the y-bits of $h(F^{R+1}c)$ and $h(F^{R+1}c')$ differ. Since $F^{R+1}P$ is a front pair at $p + R + 1$, these images agree at all sites beyond $p + 2R + 1$, and $\lambda_{R+1}(hP) = U(hP) + R + 1 \ge p + 1$. So $\lambda_{R+1}(hP)$ is the largest site $i \in [p+1, p+2R+1]$ at which the y-bits of $h(F^{R+1}c)_i$ and $h(F^{R+1}c')_i$ differ. These bits depend only on $F^{R+1}c$ and $F^{R+1}c'$ on $[p+1-R, p+3R+1]$, hence only on c and c' on $[p-2R, p+4R+2]$. The same description applies to $\tilde{P}$, and since $\tilde{P}$ agrees with P on this interval, the same bits are compared. So $\lambda_{R+1}(h\tilde{P}) = \lambda_{R+1}(hP)$ and $E(\tilde{P}) = E(P)$. (2) $\sigma^j P$ is a front pair at $p - j$, and $U(h\sigma^j P) = U(\sigma^j hP) = U(hP) - j$ by Lemma 3.4. (3) FP is a front pair at $p + 1$ by Lemma 3.6(1). Since $hF = Fh$, Lemma 3.4 gives $U(hFP) = U(FhP) = U(hP) + 1$. (4) The conditions of Definition 3.5 are symmetric in c and c'. The difference diagram $\Delta_{h(c',c)} = Y_{hc'} \oplus Y_{hc}$ equals $\Delta_{h(c,c')}$, and U is a function of the difference diagram. □

Front pairs exist, and E takes values in the finite set $[-R, R]$, so

$$e_{\max} = \max\{E(P) : P \text{ a front pair at some } p \in \mathbb{Z}\}$$

exists.

Lemma 3.11 (maximum principle)**.** *Let $P = (c, c')$ be a front pair at p with $E(P) = e_{\max}$.*

1. *If c'' agrees with c' at every site $i \ge p$, then (c, c'') is a front pair at p and $E(c, c'') = e_{\max}$.*
2. *If c'' agrees with c at every site $i \ge p$, then (c'', c') is a front pair at p and $E(c'', c') = e_{\max}$.*

Proof. (1) The pair (c, c'') is a front pair at p, because the conditions of Definition 3.5 involve only sites $i \ge p$, where $c'' = c'$.

Truncation. Let c^* agree with c'' on $[p - 2R, \infty)$ and with c' on $(-\infty, p - 2R)$. Then (c, c^*) is again a front pair at p, and it agrees with (c, c'') on $[p - 2R, p + 4R + 2]$, so $E(c, c^*) = E(c, c'')$ by Lemma 3.10(1). The configurations c' and c^* agree at all sites $i \ge p$ and at all sites $i < p - 2R$, so they differ in finitely many bits, all at sites in $[p - 2R, p - 1]$. If $c^* = c'$ there is nothing to prove, so assume they differ.

Decomposition. Complement the differing bits one at a time, in any order. This gives configurations $c' = c^0, c^1, \ldots, c^N = c^*$ with $N \ge 1$, in which each c^k is obtained from c^{k-1} by

complementing one bit at some site $\ell_k \in [p-2R, p-1]$. So $Q_k = (c^{k-1}, c^k)$ is a y-flip or an x-flip at ℓ_k, in particular a pair.

Bounds for the flips. Let $t \geq R+2$. If Q_k is a y-flip, it is a front pair at ℓ_k (Lemma 3.6(2)), so by Lemma 3.9, applied at time $t \geq R+1$, and the definition of $e_{\max}$,

$$\lambda_t(hQ_k) = \ell_k + t + E(Q_k) \leq p - 1 + t + e_{\max}.$$

If Q_k is an x-flip, then FQ_k is a front pair at ℓ_k (Lemma 3.6(3)). By (4), $\lambda_t(Q) = \lambda_{t-1}(FQ)$ for every pair Q, and $hF = Fh$, so $\lambda_t(hQ_k) = \lambda_{t-1}(FhQ_k) = \lambda_{t-1}(hFQ_k)$. Since $t-1 \geq R+1$, Lemma 3.9 gives

$$\lambda_t(hQ_k) = \ell_k + t - 1 + E(FQ_k) \leq p - 2 + t + e_{\max}.$$

In both cases, row t of Δ_{hQ_k} vanishes at every site $i \geq p + t + e_{\max}$.

Telescoping. Since $\Delta_Q = Y_c \oplus Y_{c'}$ for any pair $Q = (c, c')$,

$$\Delta_{h(c,c^*)} = Y_{hc} \oplus Y_{hc^*} = \big(Y_{hc} \oplus Y_{hc^0}\big) \oplus \bigoplus_{k=1}^{N} \big(Y_{hc^{k-1}} \oplus Y_{hc^k}\big) = \Delta_{hP} \oplus \bigoplus_{k=1}^{N} \Delta_{hQ_k}.$$

This is an identity between sums of diagrams of the configurations $hc, hc^0, \ldots, hc^N$, and it does not use any linearity of h.

No cancellation at the front. Let $t \geq R+2$. By Lemma 3.9, row t of Δ_{hP} has its rightmost 1 exactly at $p + t + e_{\max}$. Every other term vanishes in row t at all sites $\geq p + t + e_{\max}$. Hence row t of $\Delta_{h(c,c^*)}$ also has its rightmost 1 at $p + t + e_{\max}$, that is, $\lambda_t(h(c, c^*)) = p + t + e_{\max}$.

Conclusion. Since (c, c^*) is a front pair at p, Lemma 3.9 gives $\lambda_t(h(c, c^*)) = t + U(h(c, c^*))$ for $t \geq R+1$. Taking $t = R+2$, $U(h(c, c^*)) = p + e_{\max}$, so $E(c, c'') = E(c, c^*) = e_{\max}$.

(2) By Lemma 3.10(4), (c', c) is a front pair at p with $E(c', c) = e_{\max}$. Part (1), applied to (c', c) and c'', shows that (c', c'') is a front pair at p with $E(c', c'') = e_{\max}$. Apply Lemma 3.10(4) again. □

The maximum is used in the bounds for the flips: each flip behind the front has its own front at least one site behind that of hP, because its offset is at most $e_{\max}$. For a pair with $E(P) < e_{\max}$ this bound is not available, and a priori the flips could move the front of $h(c, c^*)$.

Corollary 3.12. *Let $M = 4R+2$. Let $P_1 = (c_1, c_1')$ and $P_2 = (c_2, c_2')$ be front pairs at p_1 and p_2 such that $(c_1)_{p_1+k} = (c_2)_{p_2+k}$ for $0 \leq k \leq M$. Then $E(P_1) = e_{\max}$ if and only if $E(P_2) = e_{\max}$.*

Proof. Put $j = p_2 - p_1$. Then $\sigma^j P_2$ is a front pair at p_1, its first configuration has the same cells as c_2 on $[p_1, p_1 + M]$ read relative to the front, and $E(\sigma^j P_2) = E(P_2)$ by Lemma 3.10(2). So we may assume $p_1 = p_2 = p$. By symmetry it suffices to show that $E(P_1) = e_{\max}$ implies $E(P_2) = e_{\max}$.

Cells right of $p + M$. Let c_3 and c_3' agree with c_1 and c_1', respectively, on $(-\infty, p + M]$, and with c_2 on $(p + M, \infty)$. Since $c_1 = c_1'$ on $(p, p + M]$, the pair $P_3 = (c_3, c_3')$ is a front pair at p, and it agrees with P_1 on $[p - 2R, p + M]$. So $E(P_3) = e_{\max}$ by Lemma 3.10(1).

First application of Lemma 3.11. By hypothesis and construction, c_3 agrees with c_2 at every site $i \geq p$. Lemma 3.11(2) gives $E(c_2, c_3') = e_{\max}$.

Second application. At sites $i > p$ we have $c_3' = c_3 = c_2 = c_2'$. At p, both c_3' and c_2' are obtained from the common cell $(c_3)_p = (c_2)_p$ by complementing its y-bit. So c_3' agrees with c_2' at every site $i \geq p$, and Lemma 3.11(1), applied to (c_2, c_3'), gives $E(c_2, c_2') = e_{\max}$. □

Let $M = 4R+2$, and let $\mathcal{S} \subseteq \mathcal{A}^{M+1}$ be the set of words ω such that $E(P) = e_{\max}$ for some front pair $P = (c, c')$ at some p with $c_{p+k} = \omega_k$ for $0 \leq k \leq M$. By Corollary 3.12, if $\omega \in \mathcal{S}$, then $E(P) = e_{\max}$ for *every* such front pair. Every word occurs as such a word: take any c with $c_{p+k} = \omega_k$ and let c' be c with $y(c_p)$ complemented.

Theorem 3.13 (front offset)**.** *There is an integer $e^+(h) \in [-R, R]$ such that $U(hP) = p + e^+(h)$ for every front pair P at every p. Consequently $\lambda_t(hP) = p + t + e^+(h)$ for $t \geq R + 1$, and $F^t(hP)$ is a front pair at $p + t + e^+(h)$ for every $t \geq R + 2$.*

Proof. Let $M = 4R + 2$ and let $\mathcal{S}$ be as above.

$\mathcal{S}$ is nonempty. The maximum $e_{\max}$ is attained at some front pair $P = (c, c')$ at some p, and the word of c on $[p, p + M]$ lies in $\mathcal{S}$.

$\mathcal{S}$ is forward-closed in Λ_M. Let $\omega \in \mathcal{S}$ and $\iota = (\iota_1, \iota_2) \in \mathcal{A}^2$. Choose c with $c_{p+k} = \omega_k$ for $0 \leq k \leq M$, $c_{p+M+1} = \iota_1$ and $c_{p+M+2} = \iota_2$, and let c' be c with $y(c_p)$ complemented. Then $P = (c, c')$ is a front pair at p whose word is ω, so $E(P) = e_{\max}$ by Corollary 3.12. By Lemma 3.6(1), FP is a front pair at $p + 1$, and $E(FP) = E(P) = e_{\max}$ by Lemma 3.10(3). The first configuration of FP is $F(c)$, whose word on $[p + 1, p + 1 + M]$ is $\omega \cdot \iota$ by the definition of Λ_M. So $\omega \cdot \iota \in \mathcal{S}$.

E is constant. By Lemma 3.7, every word $\omega' \in \mathcal{A}^{M+1}$ is reached in Λ_M by a walk from a word of $\mathcal{S}$, so $\mathcal{S} = \mathcal{A}^{M+1}$. Now let $P = (c, c')$ be any front pair at any p. Its word $c_p \cdots c_{p+M}$ lies in $\mathcal{S}$, so $E(P) = e_{\max}$ by Corollary 3.12.

Let $e^+(h) = e_{\max}$. Then $U(hP) = p + e^+(h)$ for every front pair, and $e^+(h)$ does not depend on R, since $E(P)$ does not. The formula for λ_t is Lemma 3.9. For $t \geq R + 2$, both t and $t - 1$ lie in $I^+(hP)$, so $\lambda_0(F^t hP) = \lambda_t = p + t + e^+(h) > \lambda_{t-1} = \lambda_{-1}(F^t hP)$, which is the front-pair condition. □

Remark 3.14. The theorem is stated for every front pair, with no maturity hypothesis: Lemma 3.9 shows that the image of any front pair is in forward mode from time $R + 1$ on, and the offset is read off at that time from a finite window. The strict front-pair form of the conclusion needs one more step, from $R + 1$ to $R + 2$, because at $t = R + 1$ the x-bits at the front may still differ. Only the intercept U is used below.

3.4. **Offsets of all pairs.**

Lemma 3.15. *Let h be an invertible block map commuting with F, and let Q be any pair. If $U(Q)$ is finite, then $U(hQ) = U(Q) + e^+(h)$. If $U(Q) = -\infty$, then $U(hQ) = -\infty$.*

Proof. Let R be as in Section 3.3 and $e = e^+(h)$. Suppose first that $U(Q)$ is finite. Pick $t_0 \in I^+(Q)$. By Lemma 3.4, $U(F^{t_0}Q) = U(Q) + t_0$ and $U(hF^{t_0}Q) = U(F^{t_0}hQ) = U(hQ) + t_0$, so we may replace Q by $F^{t_0}Q$ and assume $0 \in I^+(Q)$. Write $Q = (c, c')$ and $j = U(Q) = \lambda_0(Q) \geq \lambda_{-1}(Q)$. If $\lambda_{-1}(Q) < j$, then Q is a front pair at j and Theorem 3.13 applies. So assume $\lambda_{-1}(Q) = j$.

Let c'' be c' with $y(c'_j)$ complemented. Then (c'', c') is a y-flip at j, and by Theorem 3.13 and Lemma 3.9, $\lambda_t(h(c'', c')) = t + j + e$ for $t \geq R + 1$. The pair (c, c'') has its y-differences at sites $\leq j - 1$ and its x-differences at sites $\leq j$.

Claim: $\lambda_t(h(c, c'')) \leq t + j - 1 + e$ *for every* $t \geq R + 1$. Let $t \geq R + 1$ and $i > t + j - 1 + e$. The bit $Y_{hc}(t, i)$ is the y-bit of $h(F^t c)_i$, which depends only on c on $[i - R - t, i + R + t]$. Let $\tilde{c}$ agree with c'' on $[i - R - t, \infty)$ and with c on $(-\infty, i - R - t)$. Then $Y_{h\tilde{c}}(t, i) = Y_{hc''}(t, i)$, and c, $\tilde{c}$ differ in finitely many bits: y-bits at sites $\leq j - 1$ and x-bits at sites $\leq j$. Complementing these bits one at a time gives configurations $c = c^0, c^1, \ldots, c^N = \tilde{c}$, in which each $Q_k = (c^{k-1}, c^k)$ is a y-flip at some $\ell \leq j - 1$ or an x-flip at some $\ell \leq j$. In the first case Q_k is a front pair at ℓ, so $U(hQ_k) = \ell + e \leq j - 1 + e$. In the second case FQ_k is a front pair at ℓ (Lemma 3.6(3)), so $U(hQ_k) = U(hFQ_k) - 1 = \ell - 1 + e \leq j - 1 + e$. In both cases hc^{k-1} and hc^k agree at all sites $> j + R$, so rows -1 and 0 of Δ_{hQ_k} vanish beyond $j + R$, and Proposition 3.3(6) at $t = -1$ gives $V(hQ_k) \leq j + R - 1$. By Proposition 3.3(4), $\Delta_{hQ_k}(t, i) = 1$ would require $i - t \leq j - 1 + e$ or $i + t \leq j + R - 1$. The first fails by the choice of i, and the second fails because $i + t > 2t + j - 1 + e \geq 2R + 2 + j - 1 - R = j + R + 1$. Hence $Y_{hc^{k-1}}(t, i) = Y_{hc^k}(t, i)$ for every k, so $Y_{hc}(t, i) = Y_{h\tilde{c}}(t, i) = Y_{hc''}(t, i)$. This proves the claim.

Now $\Delta_{hQ} = \Delta_{h(c,c'')} \oplus \Delta_{h(c'',c')}$. For $t \geq R+1$, the second term has its rightmost 1 in row t at $t+j+e$, and by the claim the first vanishes in row t at all sites $\geq t+j+e$. So $\lambda_t(hQ) = t+j+e$ for all $t \geq R+1$. Hence these t lie in $I^+(hQ)$ and $U(hQ) = j+e$.

If $U(Q) = -\infty$ but $U(hQ)$ were finite, the first part applied to h^{-1} and the pair hQ would give $U(Q) = U(hQ) + e^+(h^{-1})$, which is finite, a contradiction. □

Corollary 3.16. *For $h \in \mathrm{Aut}(X_F)$ put $e^-(h) = e^+(\rho h \rho)$. Then:*

1. *for every pair Q, $V(hQ) = V(Q) + e^-(h)$ if $V(Q)$ is finite, and $V(hQ) = -\infty$ if $V(Q) = -\infty$;*
2. *e^+ and e^- are homomorphisms $\mathrm{Aut}(X_F) \to \mathbb{Z}$;*
3. *$(e^+, e^-)(\sigma^a F^b) = (b-a, -a-b)$ for all $a, b \in \mathbb{Z}$;*
4. *the map $(a,b) \mapsto \sigma^a F^b$ is an isomorphism $\mathbb{Z}^2 \to \langle \sigma, F \rangle$; in particular F has infinite order;*
5. *(e^+, e^-) maps $\langle \sigma, F \rangle$ isomorphically onto $\{(m,n) \in \mathbb{Z}^2 : m \equiv n \pmod 2\}$.*

Proof. (1) $\rho h \rho \in \mathrm{Aut}(X_F)$. By Lemma 3.4, $U(\rho Q) = V(Q) + 1$ and $U(\rho hQ) = V(hQ) + 1$, and $\rho hQ = (\rho h \rho)(\rho Q)$. Apply Lemma 3.15 to $\rho h \rho$ and ρQ.

(2) Let $h_1, h_2 \in \mathrm{Aut}(X_F)$ and let Q be a front pair. Then $U(h_2 Q)$ is finite, so Lemma 3.15 gives $U(h_1 h_2 Q) = U(h_2 Q) + e^+(h_1) = U(Q) + e^+(h_2) + e^+(h_1)$, and therefore $e^+(h_1 h_2) = e^+(h_1) + e^+(h_2)$. For e^-, use $\rho h_1 h_2 \rho = (\rho h_1 \rho)(\rho h_2 \rho)$.

(3) By Lemma 3.4, $U(FQ) = U(Q) + 1$ and $U(\sigma Q) = U(Q) - 1$, so $e^+(F) = 1$ and $e^+(\sigma) = -1$. Also $V(FQ) = V(Q) - 1$ and $V(\sigma Q) = V(Q) - 1$, so by (1) $e^-(F) = e^-(\sigma) = -1$. Now use (2).

(4) σ and F commute, so $(a,b) \mapsto \sigma^a F^b$ is a surjective homomorphism $\mathbb{Z}^2 \to \langle \sigma, F \rangle$. If $\sigma^a F^b = \mathrm{id}$, then by (3) $b - a = 0$ and $-a - b = 0$, so $a = b = 0$.

(5) The linear map $(a,b) \mapsto (b-a, -a-b)$ is injective, and its image is the set of pairs of equal parity: (m,n) is the image of $a = -(m+n)/2$, $b = (m-n)/2$. □

3.5. **Cone causality.** Henceforth $h \in \mathrm{Aut}(X_F)$ satisfies $e^+(h) = e^-(h) = 0$, and $R \geq 1$ bounds the radii of h and h^{-1}. By Lemma 3.15 and Corollary 3.16(1), $U(hQ) = U(Q)$ and $V(hQ) = V(Q)$ for every pair Q, including the value $-\infty$.

Proposition 3.17. *Let $c, c' \in \mathcal{A}^{\mathbb{Z}}$ satisfy $y(c_i) = y(c'_i)$ for all $i \geq 0$ and $x(c_i) = x(c'_i)$ for all $i \geq 1$. Then $y((hc)_0) = y((hc')_0)$.*

Proof. The bit $y((hc)_0) = Y_{hc}(0,0)$ depends only on c on $[-R, R]$. Let c'' agree with c' on $[-R, R]$ and with c elsewhere. Then $Y_{hc''}(0,0) = Y_{hc'}(0,0)$, and c, c'' differ in finitely many bits, namely y-bits at sites in $[-R, -1]$ and x-bits at sites in $[-R, 0]$. Complement them one at a time, $c = c^0, \ldots, c^N = c''$. Each $Q_k = (c^{k-1}, c^k)$ is a y-flip at some $j \leq -1$, with $(U, V) = (j, j-2)$, or an x-flip at some $j \leq 0$, with $(U, V) = (j-1, j-1)$ (Lemma 3.6). Then hQ_k has the same intercepts, and by Proposition 3.3(4), $\Delta_{hQ_k}(0,0) = 1$ would require $0 \leq U$ or $0 \leq V$, which fails in both cases. Hence $Y_{hc^{k-1}}(0,0) = Y_{hc^k}(0,0)$ for all k, and $Y_{hc}(0,0) = Y_{hc''}(0,0) = Y_{hc'}(0,0)$. □

Cone codes. For a diagram Y and a point (t,i), the *cone code* of Y at (t,i) is the sequence $\omega^{(t,i)}(Y) \in \mathcal{A}^{\mathbb{N}}$ with entries

$$\omega_k^{(t,i)} = (\mu_k, \nu_k) = \big(Y(t, i+k),\ Y(t-1, i+k+1)\big) \qquad (k \geq 0),$$

and $\omega(Y) = \omega^{(0,0)}(Y)$. Every $\omega \in \mathcal{A}^{\mathbb{N}}$ is $\omega(Y_c)$ for some c: take $y(c_k) = \mu_k$ and $x(c_{k+1}) = \nu_k$ for $k \geq 0$. By Proposition 3.17 there is a function $\psi \colon \mathcal{A}^{\mathbb{N}} \to \{0,1\}$ with $Y_{hc}(0,0) = \psi(\omega(Y_c))$ for all c, and ψ depends only on $\omega_0, \ldots, \omega_R$. By (4) and $h\sigma^i F^t = \sigma^i F^t h$,

$$Y_{hc}(t,i) = Y_{\sigma^i F^t hc}(0,0) = Y_{h\sigma^i F^t c}(0,0) = \psi\big(\omega(Y_{\sigma^i F^t c})\big) = \psi\big(\omega^{(t,i)}(Y_c)\big). \tag{10}$$

Let σ also denote the shift on $\mathcal{A}^{\mathbb{N}}$, and define $\hat{G}, \hat{H} \colon \mathcal{A}^{\mathbb{N}} \to \mathcal{A}^{\mathbb{N}}$ and the involutions D_1, D_2 by

$$\hat{G}(\omega)_k = \big(\nu_k \oplus f(\mu_k, \mu_{k+1}, \mu_{k+2}),\ \mu_{k+2}\big), \qquad \hat{H}(\omega)_k = \big(\nu_k,\ \mu_{k+2} \oplus f(\nu_k, \nu_{k+1}, \nu_{k+2})\big),$$

D_1 complements μ_0, and D_2 complements both μ_0 and ν_0; neither changes ω_k for $k \geq 1$.

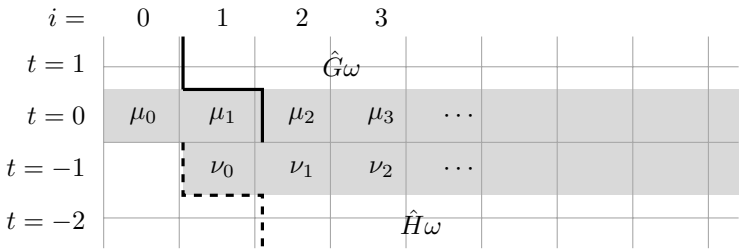


FIGURE 3. The cone code $\omega(Y)$ at the apex $(t, i) = (0, 0)$: the entries $\mu_k = Y(0, k)$ and $\nu_k = Y(-1, k+1)$, $k \geq 0$ (shaded). The cone codes at the apexes $(0, 1)$ and $(0, 2)$ are $\sigma\omega$ and $\sigma^2\omega$. The code $\hat{G}\omega$ at the apex $(1, 1)$ occupies rows 1 and 0 (to the right of the solid line), and the code $\hat{H}\omega$ at the apex $(-1, 1)$ occupies rows -1 and -2 (to the right of the dashed line); both are determined by ω through (3).

Lemma 3.18. (1) *For every diagram Y,*

$$\omega^{(1,1)}(Y) = \hat{G}\omega(Y), \quad \omega^{(-1,1)}(Y) = \hat{H}\omega(Y), \quad \omega^{(0,1)}(Y) = \sigma\omega(Y), \quad \omega^{(0,2)}(Y) = \sigma^2\omega(Y).$$

(2) *For every $\omega \in \mathcal{A}^{\mathbb{N}}$,*

$$\psi(\hat{G}\omega) \oplus \psi(\hat{H}\omega) \oplus \psi(\omega) \oplus \big(\psi(\sigma\omega) \vee \psi(\sigma^2\omega)\big) = 0. \tag{11}$$

(3) *$\hat{G}\sigma = \sigma\hat{G}$, $\hat{H}\sigma = \sigma\hat{H}$, $\hat{G}D_1 = D_1\hat{G}$, $\hat{H}D_1 = \hat{H}$, $\hat{H}D_2 = D_2\hat{H}$ and $\hat{G}D_2 = \hat{G}$.*

Proof. (1) The last two identities are immediate. By (3) at $(0, k+1)$, $Y(1, k+1) = Y(-1, k+1) \oplus Y(0, k) \oplus (Y(0, k+1) \vee Y(0, k+2)) = \nu_k \oplus f(\mu_k, \mu_{k+1}, \mu_{k+2})$, and $Y(0, k+2) = \mu_{k+2}$; this is $\hat{G}$. By (3) at $(-1, k+2)$, $Y(-2, k+2) = Y(0, k+2) \oplus Y(-1, k+1) \oplus (Y(-1, k+2) \vee Y(-1, k+3)) = \mu_{k+2} \oplus f(\nu_k, \nu_{k+1}, \nu_{k+2})$, and $Y(-1, k+1) = \nu_k$; this is $\hat{H}$.

(2) Choose c with $\omega(Y_c) = \omega$. The diagram Y_{hc} satisfies (3) at $(0, 1)$: $Y_{hc}(1, 1) \oplus Y_{hc}(-1, 1) \oplus Y_{hc}(0, 0) = Y_{hc}(0, 1) \vee Y_{hc}(0, 2)$. Now substitute (10) and (1).

(3) $\hat{G}$ and $\hat{H}$ are given by the same rule at every index k, so they commute with σ. The coordinate μ_0 enters $\hat{G}$ only in the first component of $\hat{G}(\omega)_0$, through $f(\mu_0, \mu_1, \mu_2) = \mu_0 \oplus (\mu_1 \vee \mu_2)$, so complementing μ_0 complements the first component of $\hat{G}(\omega)_0$; this is $\hat{G}D_1 = D_1\hat{G}$. In that same component $\nu_0 \oplus \mu_0$ appears, and ν_0 appears nowhere else in $\hat{G}$, so $\hat{G}D_2 = \hat{G}$. The map $\hat{H}$ does not read μ_0 or μ_1, so $\hat{H}D_1 = \hat{H}$. Finally, ν_0 enters $\hat{H}$ only in $\hat{H}(\omega)_0 = (\nu_0,\ \mu_2 \oplus \nu_0 \oplus (\nu_1 \vee \nu_2))$, where complementing it complements both components; this is $\hat{H}D_2 = D_2\hat{H}$. □

3.6. **Both light edges are permutive.**

Proposition 3.19. *For every $\omega \in \mathcal{A}^{\mathbb{N}}$, $\psi(D_1\omega) \neq \psi(\omega)$ and $\psi(D_2\omega) \neq \psi(\omega)$.*

Proof. Choose c with $\omega(Y_c) = \omega$.

Let c' be c with $y(c_0)$ complemented. Then $\omega(Y_{c'}) = D_1\omega$, and $Q = (c, c')$ is a y-flip at 0, with $(U, V) = (0, -2)$ by Lemma 3.6(2). So hQ has $(U, V) = (0, -2)$. Since $V - U$ is even, Proposition 3.3(5) gives a middle row at $m = -1$ and $I^+(hQ) = [0, \infty)$, so $\lambda_0(hQ) = 0 + U = 0$. In particular $\Delta_{hQ}(0, 0) = 1$, that is, $\psi(\omega) \neq \psi(D_1\omega)$.

Let c' be c with $y(c_0)$ and $x(c_1)$ complemented. Then $\omega(Y_{c'}) = D_2\omega$, and $Q = (c, c')$ is a yx-flip at 0, so $V(Q) = 0$ and $U(Q) \in \{-\infty\} \cup (-\infty, -1]$ (Lemma 3.6(4)). Then hQ has the same intercepts. If $U(hQ) = -\infty$, then $I^-(hQ) = \mathbb{Z}$ by Proposition 3.3(3). Otherwise $V - U \geq 1$, and by Proposition 3.3(5) either $I^-(hQ) = (-\infty, t_+ - 1]$ with $t_+ = (V - U + 1)/2 \geq 1$, or $I^-(hQ) = (-\infty, m-1]$ with $m = (V-U)/2 \geq 1$. In every case $0 \in I^-(hQ)$, so $\lambda_0(hQ) = V - 0 = 0$ and $\Delta_{hQ}(0, 0) = 1$, that is, $\psi(\omega) \neq \psi(D_2\omega)$. □

Corollary 3.20. *There is a function $\gamma\colon \mathcal{A}^{\mathbb{N}} \to \{0,1\}$, depending only on the coordinates $0, \dots, R-1$, such that $\psi(\omega) = \mu_0 \oplus \gamma(\sigma\omega)$ for all ω.*

Proof. By Proposition 3.19, $\psi \oplus \mu_0$ is invariant under D_1, so it does not depend on μ_0. It is also invariant under D_2, because D_2 complements both ψ and μ_0. A function independent of μ_0 and invariant under complementing (μ_0, ν_0) is invariant under complementing ν_0. So $\psi \oplus \mu_0$ depends only on $\omega_1, \omega_2, \dots$, and only on $\omega_1, \dots, \omega_R$ because ψ depends only on $\omega_0, \dots, \omega_R$. □

3.7. **The correction term vanishes.**

Lemma 3.21. *Let $\gamma\colon \mathcal{A}^{\mathbb{N}} \to \{0,1\}$ depend on finitely many coordinates, and suppose $\psi(\omega) = \mu_0 \oplus \gamma(\sigma\omega)$ satisfies (11). Then $\gamma \equiv 0$.*

Proof. For $\theta \in \mathcal{A}^{\mathbb{N}}$ write $\theta_k = (\xi_k, \zeta_k)$. Substitute ψ into (11) and put $\theta = \sigma\omega$, so that $\xi_0 = \mu_1(\omega)$ and $\xi_1 = \mu_2(\omega)$. Using $\mu_0(\hat{G}\omega) = \nu_0 \oplus \mu_0 \oplus (\mu_1 \vee \mu_2)$, $\mu_0(\hat{H}\omega) = \nu_0$, $\sigma\hat{G} = \hat{G}\sigma$ and $\sigma\hat{H} = \hat{H}\sigma$, the terms ν_0 and μ_0 cancel, and (11) becomes

$$\gamma(\hat{G}\theta) \oplus \gamma(\hat{H}\theta) \oplus \gamma(\theta) = (\xi_0 \vee \xi_1) \oplus \big((\xi_0 \oplus \gamma(\sigma\theta)) \vee (\xi_1 \oplus \gamma(\sigma^2\theta))\big). \tag{12}$$

This holds for every $\theta \in \mathcal{A}^{\mathbb{N}}$, since ω is arbitrary.

Three facts are used. (a) D_1 and D_2 change only θ_0, both complement ξ_0, and neither changes $\sigma\theta$. (b) The relations of Lemma 3.18(3). (c) Complementing ξ_0 while keeping ξ_1, $\gamma(\sigma\theta)$ and $\gamma(\sigma^2\theta)$ fixed changes the right side of (12) by exactly $\gamma(\sigma^2\theta)$. Indeed, $(z \vee e) \oplus (\neg z \vee e) = \neg e$ for bits z, e, so the change is $\neg\xi_1 \oplus \neg(\xi_1 \oplus \gamma(\sigma^2\theta)) = \gamma(\sigma^2\theta)$.

Put $\delta_i = \gamma \oplus (\gamma \circ D_i)$ for $i = 1, 2$. Evaluate (12) at $D_1\theta$ and add it to (12) at θ. By (a)–(c), $\gamma(\hat{G}D_1\theta) = \gamma(D_1\hat{G}\theta)$ and $\gamma(\hat{H}D_1\theta) = \gamma(\hat{H}\theta)$, so

$$\delta_1(\hat{G}\theta) \oplus \delta_1(\theta) = \gamma(\sigma^2\theta). \tag{13}$$

In the same way, using $\hat{G}D_2 = \hat{G}$ and $\hat{H}D_2 = D_2\hat{H}$,

$$\delta_2(\hat{H}\theta) \oplus \delta_2(\theta) = \gamma(\sigma^2\theta). \tag{14}$$

Say that a function *depends only on* $\theta_0, \dots, \theta_m$ if it takes equal values on sequences that agree in these coordinates; for $m = -1$ this means that it is constant. Let $m \geq -1$ be minimal such that γ depends only on $\theta_0, \dots, \theta_m$, and suppose $m \geq 0$. Since D_1 and D_2 act only on θ_0, the functions δ_1 and δ_2 also depend only on $\theta_0, \dots, \theta_m$.

γ does not depend on ζ_m. For $k \leq m$, $\hat{G}(\theta)_k$ reads only $\xi_k, \xi_{k+1}, \xi_{k+2}$ and ζ_k. So the left side of (13) does not depend on ζ_{m+1} or ζ_{m+2}, and hence neither does $\gamma(\sigma^2\theta)$. Since $(\sigma^2\theta)_m = \theta_{m+2}$ and $\sigma^2\theta$ ranges over all of $\mathcal{A}^{\mathbb{N}}$, γ does not depend on the second component of its coordinate m.

γ does not depend on ξ_m. Now γ, and hence δ_2, depends only on $\theta_0, \dots, \theta_{m-1}$ and ξ_m. So $\delta_2(\hat{H}\theta)$ depends only on $\hat{H}(\theta)_0, \dots, \hat{H}(\theta)_{m-1}$ and on the first component of $\hat{H}(\theta)_m$, which is ζ_m. For $k \leq m-1$, $\hat{H}(\theta)_k$ reads only $\theta_k, \theta_{k+1}, \theta_{k+2}$, all with index at most $m+1$. So the left side of (14) does not depend on θ_{m+2}, and hence neither does $\gamma(\sigma^2\theta)$. Since $\gamma(\sigma^2\theta)$ depends on θ_{m+2} only through ξ_{m+2}, γ does not depend on the first component of its coordinate m.

Thus γ depends only on $\theta_0, \dots, \theta_{m-1}$, contradicting the minimality of m. Hence $m = -1$ and γ is constant. Then $\delta_1 \equiv 0$, and (13) gives $\gamma(\sigma^2\theta) = 0$, so $\gamma \equiv 0$. □

3.8. **Parity and the proof of Theorem 3.1.**

Proposition 3.22. *If $h \in \mathrm{Aut}(X_F)$ satisfies $e^+(h) = e^-(h) = 0$, then $h = \mathrm{id}$.*

Proof. By Corollary 3.20 and Lemmas 3.18(2) and 3.21, $\psi(\omega) = \mu_0$. By (10), $Y_{hc}(t,i) = \mu_0(\omega^{(t,i)}(Y_c)) = Y_c(t,i)$ for all (t,i). So $Y_{hc} = Y_c$, and $hc = c$ for every c. □

Lemma 3.23. *For every $h \in \mathrm{Aut}(X_F)$, $e^+(h) \equiv e^-(h) \pmod 2$.*

Proof. For a cell $s \in \mathcal{A}$ let $\bar{s} = s^{\mathbb{Z}}$ be the constant configuration. Since $f(y,y,y) = y \oplus (y \vee y) = 0$, we have $F(\overline{(x,y)}) = \overline{(y,x)}$. So F permutes the four constant configurations as the transposition exchanging $\overline{(0,1)}$ and $\overline{(1,0)}$, fixing the other two. Every $h \in \mathrm{Aut}(X_F)$ commutes with σ, so it maps the four σ-fixed points bijectively to themselves, and the induced permutation commutes with that of F. The centralizer in $\mathrm{Sym}(4)$ of a transposition $(\alpha\,\beta)$ is $\langle(\alpha\,\beta)\rangle \times \langle(\alpha'\,\beta')\rangle$, where $\{\alpha', \beta'\}$ is the complement of $\{\alpha, \beta\}$. Every element of this group has order at most 2, so h^2 fixes every constant configuration.

Suppose $e^+(h) + e^-(h)$ is odd. Put $a = -(e^+(h) + e^-(h) + 1)/2$ and $b = (e^+(h) - e^-(h) - 1)/2$, which are integers. By Corollary 3.16, $(e^+, e^-)(\sigma^a F^b) = (b - a, -a - b) = (e^+(h), e^-(h) + 1)$, so $h_1 = F^{-b}\sigma^{-a}h$ has offsets $(0, -1)$. Then h_1^2 has offsets $(0, -2) = (e^+, e^-)(\sigma F)$, so $h_1^2(\sigma F)^{-1}$ has offsets $(0,0)$, and $h_1^2 = \sigma F$ by Proposition 3.22. But h_1^2 fixes every constant configuration, while σF exchanges $\overline{(0,1)}$ and $\overline{(1,0)}$. This is a contradiction. □

Proof of Theorem 3.1. Let $h \in \mathrm{Aut}(X_F)$. By Lemma 3.23, $e^+(h) + e^-(h)$ is even, so $a = -\frac{1}{2}(e^+(h) + e^-(h))$ and $b = \frac{1}{2}(e^+(h) - e^-(h))$ are integers, and $(e^+, e^-)(\sigma^a F^b) = (b - a, -a - b) = (e^+(h), e^-(h))$. Hence $h_0 = F^{-b}\sigma^{-a}h$ has $e^+(h_0) = e^-(h_0) = 0$, so $h_0 = \mathrm{id}$ by Proposition 3.22, and $h = \sigma^a F^b$. Uniqueness of (a,b) and $\langle \sigma, F \rangle \cong \mathbb{Z}^2$ are Corollary 3.16(4). By Lemma 2.2, this describes $\mathrm{Aut}(X_{F_{30}})$. □

Remark 3.24. The properties of rule 30 used are the following: the second-order rule is permutive in the previous row and in the left neighbour, so it satisfies Lemma 3.2; the maximum principle (Lemma 3.11), which uses only Lemmas 3.9 and 3.6; the strong connectivity of the light-speed window graph Λ_M (Lemma 3.7), whose proof uses the specific form $a \oplus (b \vee c)$ only through the stationary loop (9); the cone identity (11) forces the correction to vanish (Lemma 3.21); and $f(y,y,y) = 0$, which is used for the parity. Permutivity in the right argument, which rule 30 lacks, is never used.

4. Permutive automata of any radius

In this section we extend Theorem 3.1 to second-order automata of any radius $r \geq 1$ over $\mathbb{F}_2 = \{0,1\}$ whose rule is permutive in its leftmost variable. Most of the proof carries over once the speed 1 of the fronts is replaced by r, and we state each such step at radius r with its proof, or with a precise reference to the proof for rule 30 and the indices that change. Three steps need new arguments. The correction term is removed by a descent on the largest coordinates that it reads, which works when a second derivative of the rule reaches far enough to the right (Section 4.6). The strong connectivity of the light-speed window graph is reduced to a finite certificate that covers all window lengths at once (Section 4.10). And the parity step is replaced by an orbit-counting obstruction on spatially periodic points (Section 4.8), because the offsets determine an automorphism only up to an index $2r$. The elementary rules are treated as the case $r = 1$ in Section 5.

4.1. **Setting, hypotheses and results.** Fix $r \geq 1$ and a Boolean function $g\colon \{0,1\}^{2r} \to \{0,1\}$. Put

$$f(y_{i-r},\dots,y_{i+r}) = y_{i-r} \oplus g(\xi_0,\dots,\xi_{2r-1}), \qquad \xi_k = y_{i-r+1+k} \quad (0 \leq k \leq 2r-1), \tag{15}$$

and define F on $\mathcal{A}^{\mathbb{Z}}$, $\mathcal{A} = \{0,1\}^2$ with cells (x,y), by

$$F(c)_i = \big(y_i,\ x_i \oplus f(y_{i-r},\dots,y_{i+r})\big). \tag{16}$$

For $r = 1$ this is (1) for an elementary rule of the form $f(a,b,c) = a \oplus g(b,c)$; rule 30 is $g(b,c) = b \vee c$. The map F is reversible, with $F^{-1}(c)_i = \big(y_i \oplus f(x_{i-r},\dots,x_{i+r}),\ x_i\big)$, and F and F^{-1} have radius r. Lemma 2.2 holds for every reversible cellular automaton, so $\mathrm{Aut}(X_F)$ is the group of invertible block maps commuting with F, and $\langle \sigma,\tau\rangle$ corresponds to $\langle \sigma, F\rangle$. From now on the letter g denotes this function, and automorphisms are denoted $h, h_1, \dots$, as before.

For $\varphi\colon \{0,1\}^{2r} \to \{0,1\}$ and $0 \leq j \leq 2r-1$, let $\xi^{(j)}$ be ξ with its coordinate ξ_j complemented, and $\partial_j\varphi(\xi) = \varphi(\xi) \oplus \varphi(\xi^{(j)})$. Then $\partial_j\varphi \equiv 0$ if and only if φ does not depend on ξ_j, $\partial_j\partial_k = \partial_k\partial_j$ and $\partial_j\partial_j = 0$. We write $\mathbf{1} = (1,\dots,1)$.

Definition 4.1 (light-speed window graph)**.** Fix $M \geq 0$. The graph Λ_M has vertex set $\mathcal{A}^{M+1}$ and, for every vertex $\omega = \omega_0 \cdots \omega_M$ and every word $\iota = \iota_1 \cdots \iota_{2r} \in \mathcal{A}^{2r}$, an edge $\omega \to \omega \cdot \iota$, where

$$(\omega \cdot \iota)_k = F(\tilde{c})_{p+r+k} \qquad (0 \leq k \leq M)$$

and $\tilde{c}$ is any configuration with $\tilde{c}_{p+k} = \omega_k$ for $0 \leq k \leq M$ and $\tilde{c}_{p+M+j} = \iota_j$ for $1 \leq j \leq 2r$. Since F has radius r, the cells $F(\tilde{c})_i$ with $p+r \leq i \leq p+r+M$ read only $\tilde{c}$ on $[p, p+M+2r]$, so $\omega \cdot \iota$ depends neither on p nor on the other cells of $\tilde{c}$. A vertex is the word of one configuration on a front cell p and the M cells ahead of it, and an edge advances the front by r sites, at the speed of light. For $r = 1$ this is the graph Λ_M of Section 3.2.

The cellwise complement is $\kappa(c)_i = (1 \oplus x_i,\ 1 \oplus y_i)$. We call g *self-dual* if $g(\xi \oplus \mathbf{1}) = g(\xi) \oplus 1$ for every ξ.

Lemma 4.2. *κ is an involution commuting with σ, and $\kappa F = F\kappa$ if and only if g is self-dual.*

Proof. The first statement is clear. By (16), $F(\kappa c)_i = (1 \oplus y_i,\ 1 \oplus x_i \oplus f(y \oplus \mathbf{1}))$ and $\kappa F(c)_i = (1 \oplus y_i,\ 1 \oplus x_i \oplus f(y))$, where y stands for $(y_{i-r},\dots,y_{i+r})$. By (15), $f(y \oplus \mathbf{1}) \oplus f(y) = 1 \oplus g(\xi \oplus \mathbf{1}) \oplus g(\xi)$. This vanishes for all y exactly when g is self-dual. □

Spatially periodic points. For $n \geq 1$ let $\mathrm{Per}_n = \{c \in \mathcal{A}^{\mathbb{Z}} : \sigma^n c = c\}$, a finite set of 4^n points, and $\mathrm{Per} = \bigcup_n \mathrm{Per}_n$. The group $\mathbb{Z}^2$ acts on Per by $(a,b)\cdot c = \sigma^a F^b c$, and $\mathrm{Stab}(c) = \{(a,b) : \sigma^a F^b c = c\}$.

Lemma 4.3. *Let $c \in \mathrm{Per}$ have least spatial period n. Then $\mathrm{Stab}(c) \cap (\mathbb{Z} \times \{0\}) = n\mathbb{Z} \times \{0\}$, and $\mathrm{Stab}(c)$ has finite index in $\mathbb{Z}^2$. All points of a $\mathbb{Z}^2$-orbit have the same stabiliser. If $\mathcal{L}$ is a subgroup of finite index and $\mathcal{L} \cap (\mathbb{Z} \times \{0\}) = n\mathbb{Z} \times \{0\}$, then every orbit with stabiliser $\mathcal{L}$ consists of points of least period n. In particular the number $k(\mathcal{L})$ of $\mathbb{Z}^2$-orbits in Per whose stabiliser is exactly $\mathcal{L}$ is finite, and it is computed exactly by enumerating the points of least period n in Per_n.*

Proof. The first statement is the definition of the least period. F commutes with σ and is injective, so it permutes the finite set Per_n, and $F^\ell c = c$ for some $\ell \geq 1$. So $(n,0), (0,\ell) \in \mathrm{Stab}(c)$. Stabilisers are constant on orbits because $\mathbb{Z}^2$ is abelian. The rest follows. □

Definition 4.4 (index condition)**.** For a subgroup $\mathcal{L} \subseteq \mathbb{Z}^2$ of finite index let $k(\mathcal{L})$ be the number of $\mathbb{Z}^2$-orbits in Per whose stabiliser is exactly $\mathcal{L}$ (Lemma 4.3). For a prime q dividing $2r$ put

$$a_q = 0 \ \text{ if } q \text{ is odd}, \qquad a_2 = r \bmod 2.$$

Condition $(\mathrm{I})_q$ is: *there is a subgroup $\mathcal{L} \subseteq \mathbb{Z}^2$ of finite index with $q \nmid k(\mathcal{L})$ and $(a_q, 1) \notin q\mathbb{Z}^2 + \mathcal{L}$.*

Since $q \mid 0$, a subgroup witnessing $(\mathrm{I})_q$ is the stabiliser of at least one orbit. We call such an $\mathcal{L}$ a *witness*.

Definition 4.5 (hypotheses)**.** (N) *Stripping condition.* $\partial_0 g$ is not constant, and

$$J(g) = \max\{j \in [1, 2r-1] : \ \partial_0 \partial_j g \not\equiv 0\} \quad \text{satisfies} \quad J(g) \geq r.$$

(Since $\partial_0 \partial_0 g = 0$, $\partial_0 g$ is non-constant exactly when this set is nonempty.) For $r \geq 2$ this is strictly stronger than non-affineness of g.

(D) *Not self-dual.* $g(\xi \oplus \mathbf{1}) \neq g(\xi) \oplus 1$ for some ξ; equivalently (Lemma 4.2), κ does not commute with F.

(Λ) *Light-speed connectivity.* Λ_M is strongly connected for every $M \geq 0$.

(I) *Index condition.* $(\mathrm{I})_q$ holds for every prime q dividing $2r$.

For $r = 1$, $\partial_0 \partial_1 g$ is the coefficient of $\xi_0 \xi_1$ in the algebraic normal form of g, so (N) says that f is not affine. For $r = 2$, (N) holds for 64 512 of the 65 504 non-affine g, and (N) and (D) both hold for 64 288 of them. Hypothesis (Λ) is established by the finite computation of Section 4.10, a sufficient certificate that covers all M at once, and (I) by enumerating spatially periodic points (Section 4.8). Both computations are semi-decision procedures: a positive answer proves the hypothesis, and a negative answer at a given size proves nothing.

The offsets $e^{\pm}$ below are those of Proposition 4.20. They are the radius-r versions of the offsets of Corollary 3.16.

Theorem 4.6. *Assume* (N), (D), (Λ) *and* (I). *Then every invertible block map h commuting with F equals $\sigma^a F^b$ for exactly one $(a, b) \in \mathbb{Z}^2$, namely*

$$a = -\tfrac{1}{2}\big(e^+(h) + e^-(h)\big), \qquad b = \tfrac{1}{2r}\big(e^+(h) - e^-(h)\big).$$

Hence $\mathrm{Aut}(X_F) = \langle \sigma, F \rangle \cong \mathbb{Z}^2$.

The next theorem records what each hypothesis contributes. Write $e = (e^+, e^-)$.

Theorem 4.7. (a) *The map $(a, b) \mapsto \sigma^a F^b$ is an isomorphism $\mathbb{Z}^2 \to \langle \sigma, F \rangle$; this needs no hypothesis. Assume* (Λ). *Then every $h \in \mathrm{Aut}(X_F)$ has offsets $e^{\pm}(h) \in \mathbb{Z}$ (Proposition 4.20), and e^+, e^- are homomorphisms $\mathrm{Aut}(X_F) \to \mathbb{Z}$ with*

$$e(\sigma^a F^b) = (rb - a, \ -a - rb).$$

The image of $\langle \sigma, F \rangle$ under e is $L = \{(m, n) \in \mathbb{Z}^2 : m \equiv n \pmod{2r}\}$. It has index $2r$ in $\mathbb{Z}^2$, and $(m, n) \mapsto m - n$ induces $\mathbb{Z}^2/L \cong \mathbb{Z}/2r$.

(b) *Assume* (Λ) *and* (N). *Then* $\ker e = \{\mathrm{id}\}$ *if* (D) *holds, and* $\ker e = \{\mathrm{id}, \kappa\}$ *if g is self-dual. Consequently, under* (Λ), (N) *and* (D), *e is injective,* $\mathrm{Aut}(X_F) \cong \mathbb{Z}^2$, *and $\langle \sigma, F \rangle$ has index dividing $2r$ in* $\mathrm{Aut}(X_F)$, *with cyclic quotient.*

(c) *If g is self-dual, then $\kappa \in \mathrm{Aut}(X_F)$ has order 2, commutes with σ and F, and does not lie in $\langle \sigma, F \rangle$; so* (D) *is necessary for* $\mathrm{Aut}(X_F) = \langle \sigma, F \rangle$. *If moreover* ($\Lambda$) *and* (N) *hold, then κ is central in* $\mathrm{Aut}(X_F)$.

(d) *Assume* (Λ), (N) *and* (D), *and let q be a prime dividing $2r$. Then q divides* $[\mathrm{Aut}(X_F) : \langle \sigma, F \rangle]$ *if and only if $h_1^q = \sigma^a F$ for some $h_1 \in \mathrm{Aut}(X_F)$ and some $a \in \mathbb{Z}$, and in that case necessarily $a \equiv -r \equiv a_q \pmod{q}$. Condition $(\mathrm{I})_q$ rules out $h_1^q = \sigma^a F$ with $a \equiv a_q \pmod{q}$, without any further hypothesis. Hence $(\mathrm{I})_q$ implies that q does not divide the index, and* (I) *removes the index entirely.*

Remark 4.8 (scope). (1) For $r = 1$, (N) is non-affineness, and (D) then holds automatically (Section 5.2). The four hypotheses hold for the eight non-affine left-permutive elementary rules (Section 5).

(2) The reflection $\mathrm{rev}(c)_i = c_{-i}$ conjugates F to the automaton of the reflected rule $f^r(y_{i-r}, \dots, y_{i+r}) = f(y_{i+r}, \dots, y_{i-r})$, since by (16),

$$\mathrm{rev}\, F\, \mathrm{rev}(c)_i = F(\mathrm{rev}\, c)_{-i} = \big(y_i,\ x_i \oplus f(y_{i+r}, \dots, y_{i-r})\big).$$

It satisfies $\mathrm{rev}\,\sigma\,\mathrm{rev} = \sigma^{-1}$, so it maps $\langle \sigma, F\rangle$ onto the corresponding group, and it preserves radii. So the theorem also applies to rules permutive in their rightmost variable, after reflection. Rules permutive in both outer variables are included when the hypotheses hold; an example with $r = 2$ is in Table 1.

(3) If g is affine, then $\partial_0 g$ is constant and (N) fails. This is not an artefact of the method: for rule 150, the map $\mathrm{id} + \sigma F$ is an automorphism outside $\langle \sigma, F\rangle$ (Remark 5.8).

The hypotheses enter as follows. (Λ) gives the offsets (Sections 4.3–4.4). (N) forces the correction term to be constant (Section 4.6). (D) excludes the constant 1, that is, κ (Section 4.7). (I) removes the remaining index (Section 4.8). The proofs of the two theorems are in Section 4.9.

Conventions. Those of Section 3: $[S] \in \{0, 1\}$ is the truth value of S, intervals are intervals of integers, and $-\infty + k = -\infty < n$, $\max \emptyset = -\infty$.

4.2. **Diagrams, profiles and flips.** The definitions of Section 3.1 are used without change: the diagram $Y_c(t, i)$ is the y-bit of $F^t(c)$ at site i, so that the x-row of $F^t(c)$ is row $t - 1$ and $c_i = (Y_c(-1, i), Y_c(0, i))$; pairs $Q = (c, c')$ are ordered pairs of distinct configurations that agree far to the right; $\Delta_Q = Y_c \oplus Y_{c'}$; and $\rho(c)_i = (y_i, x_i)$. By (16), every diagram satisfies

$$Y(t+1, i) \oplus Y(t-1, i) \oplus Y(t, i-r) = g\big(Y(t, i-r+1), \dots, Y(t, i+r)\big) \qquad \text{for all } (t, i). \tag{17}$$

This is an exclusive or in each of $Y(t+1, i)$, $Y(t-1, i)$ and $Y(t, i-r)$, so two consecutive rows determine the diagram, and $c \mapsto Y_c$ is a bijection onto the solutions of (17). Equations (4) and (5) hold as stated, since their proofs do not involve the radius. The computation before (6), with the formula for F^{-1} above, gives $\rho F \rho = F^{-1}$ and (6), so $h \mapsto \rho h \rho$ is an automorphism of $\mathrm{Aut}(X_F)$. If rows $t - 1$ and t of Δ_Q vanish to the right of s, then by (17) rows $t + 1$ and $t - 2$ vanish to the right of $s + r$; hence every row of Δ_Q vanishes far to the right, and the profile $\lambda_t = \lambda_t(Q) = \max\{i : \Delta_Q(t, i) = 1\} \in \mathbb{Z} \cup \{-\infty\}$ is defined.

Lemma 4.9. *For every pair and every t:*

(1) $\lambda_{t+1} \le \max(\lambda_{t-1}, \lambda_t + r)$, *with equality if* $\lambda_{t-1} \ne \lambda_t + r$;
(2) $\lambda_{t-1} \le \max(\lambda_{t+1}, \lambda_t + r)$, *with equality if* $\lambda_{t+1} \ne \lambda_t + r$.

Proof. By (17), $\Delta(t+1, i) = \Delta(t-1, i) \oplus \Delta(t, i-r) \oplus D_i$, where D_i is the difference of the two values of g, which read row t on $[i - r + 1, i + r]$. Let $m = \max(\lambda_{t-1}, \lambda_t + r)$. For $i \ge \lambda_t + r$ we have $i - r + 1 > \lambda_t$, so $D_i = 0$. For $i > m$ all three terms vanish. If m is finite, then $D_m = 0$, $\Delta(t-1, m) = [m = \lambda_{t-1}]$ and $\Delta(t, m-r) = [m = \lambda_t + r]$. At least one of these is 1, and both are 1 only if $\lambda_{t-1} = \lambda_t + r$. This proves (1). Part (2) is the same argument for (17) solved for $Y(t-1, i)$, which has the same form with $t \pm 1$ exchanged. □

Proposition 4.10. *Let Q be a pair, and put*

$$I^+ = \{t : \lambda_t \ne -\infty,\ \lambda_{t-1} < \lambda_t + r\}, \qquad I^- = \{t : \lambda_t \ne -\infty,\ \lambda_{t+1} < \lambda_t + r\}.$$

(1) *No two consecutive λ_t equal $-\infty$.*
(2) *If $t \in I^+$, then $\lambda_{t+1} = \lambda_t + r$ and $t + 1 \in I^+$. If $t \in I^-$, then $\lambda_{t-1} = \lambda_t + r$ and $t - 1 \in I^-$. The sets $I^\pm$ are disjoint, and every element of I^- is smaller than every element of I^+.*
(3) *$\mathbb{Z} \setminus (I^+ \cup I^-)$ has at most one element. If m is such an element (a* middle row*), then $m - 1 \in I^-$, $m + 1 \in I^+$, and $\lambda_{m-1} = \lambda_{m+1} \ge \lambda_m + r$. If $I^+ = \emptyset$ then $I^- = \mathbb{Z}$, and if $I^- = \emptyset$ then $I^+ = \mathbb{Z}$.*

(4) *Put* $U = \lambda_t - rt$ *for* $t \in I^+$ *and* $V = \lambda_t + rt$ *for* $t \in I^-$, *with* $U = -\infty$ *if* $I^+ = \emptyset$ *and* $V = -\infty$ *if* $I^- = \emptyset$ *(the* forward *and* backward intercepts*). Then* $\lambda_t \le \max(rt+U,\, V-rt)$ *for every* t; *that is,* $\Delta_Q(t,i) = 1$ *implies* $i - rt \le U$ *or* $i + rt \le V$.
(5) *Suppose* U, V *are finite. If* $2r \mid V - U$, *there is a middle row* $m = (V-U)/2r$, *and* $I^+ = [m+1, \infty)$, $I^- = (-\infty, m-1]$. *Otherwise there is no middle row, and* $I^+ = [t_+, \infty)$, $I^- = (-\infty, t_+ - 1]$ *with* $t_+ = \lceil (V-U)/2r \rceil$. *In particular* I^+ *and* I^- *are determined by* (U, V), *also when* U *or* V *is* $-\infty$.
(6) *For every* t, $U \le \max\big(\lambda_t - rt,\ \lambda_{t-1} - r(t+1)\big)$ *and* $V \le \max\big(\lambda_t + rt,\ \lambda_{t+1} + r(t-1)\big)$.

Proof. This is the proof of Proposition 3.3 with the step 1 replaced by r, using Lemma 4.9. We give the points where r enters. (1) Two vanishing consecutive rows give $c = c'$. (2) If $t \in I^+$, Lemma 4.9(1) gives $\lambda_{t+1} = \lambda_t + r$, and then $\lambda_t < \lambda_{t+1} + r$, so $t + 1 \in I^+$. Symmetrically for I^-. If $t \in I^+$ then $\lambda_{t+1} = \lambda_t + r$, so $t \notin I^-$; and $a \in I^+$, $b \in I^-$, $a \le b$ would give $b \in I^+$. (3) Let $t \notin I^+ \cup I^-$. If $\lambda_t = -\infty$, then $\lambda_{t\pm1}$ are finite by (1), so $t + 1 \in I^+$ and $t - 1 \in I^-$. If λ_t is finite, then $\lambda_{t-1} \ge \lambda_t + r$ and $\lambda_{t+1} \ge \lambda_t + r$, which give $t - 1 \in I^-$ and $t + 1 \in I^+$. Uniqueness and the last sentence follow from (2) as before. For the values: if $\lambda_{m-1} \ne \lambda_m + r$, Lemma 4.9(1) gives $\lambda_{m+1} = \max(\lambda_{m-1}, \lambda_m + r) = \lambda_{m-1}$; if $\lambda_{m-1} = \lambda_m + r$, then $\lambda_m + r \le \lambda_{m+1} \le \max(\lambda_{m-1}, \lambda_m + r) = \lambda_m + r$. (4) On $I^\pm$ this is the definition. For a middle row, $\lambda_m \le \lambda_{m-1} - r = V - rm$. (5) If there is no middle row, $I^- = (-\infty, t_+ - 1]$ and $I^+ = [t_+, \infty)$. Now $t_+ \in I^+$ reads $V - r(t_+ - 1) < rt_+ + U + r$, that is $V - U < 2rt_+$, and $t_+ - 1 \in I^-$ reads $rt_+ + U < V - r(t_+ - 1) + r$, that is $V - U > 2r(t_+ - 1)$. If there is a middle row m, then $\lambda_{m+1} = \lambda_{m-1}$ reads $r(m+1) + U = V - r(m-1)$, that is $V - U = 2rm$. Divisibility of $V - U$ by $2r$ decides which case occurs. If $U = -\infty$ then $I^- = \mathbb{Z}$, and if $V = -\infty$ then $I^+ = \mathbb{Z}$, by (3). (6) Put $\mu_t = \max(\lambda_t - rt,\ \lambda_{t-1} - r(t+1))$. By Lemma 4.9(1), $\lambda_{t+1} - r(t+1) \le \max(\lambda_{t-1} - r(t+1), \lambda_t - rt) \le \mu_t$, and $\lambda_t - r(t+2) < \mu_t$, so $\mu_{t+1} \le \mu_t$. If $t - 1 \in I^+$, then $\mu_t = \max(U, U - 2r) = U$. Hence $U \le \mu_t$ for all t. The statement for V follows in the same way from Lemma 4.9(2), with $\nu_t = \max(\lambda_t + rt, \lambda_{t+1} + r(t-1))$, which is nondecreasing and equals V when $t + 1 \in I^-$. □

Lemma 4.11. *For every pair* Q *and all* $j, k \in \mathbb{Z}$: $U(F^kQ) = U(Q) + rk$ *and* $V(F^kQ) = V(Q) - rk$; $U(\sigma^jQ) = U(Q) - j$ *and* $V(\sigma^jQ) = V(Q) - j$; $U(\rho Q) = V(Q) + r$ *and* $V(\rho Q) = U(Q) - r$. *The value* $-\infty$ *is preserved in each case.*

Proof. $\lambda_t(F^kQ) = \lambda_{t+k}(Q)$ and $\lambda_t(\sigma^jQ) = \lambda_t(Q) - j$, so $I^\pm(F^kQ) = I^\pm(Q) - k$ and $I^\pm(\sigma^jQ) = I^\pm(Q)$; for $t \in I^+(F^kQ)$, $\lambda_{t+k} - rt = U + rk$. By (6), $\lambda_t(\rho Q) = \lambda_{-1-t}(Q)$, so $t \in I^+(\rho Q)$ iff $-1 - t \in I^-(Q)$, and then $\lambda_t(\rho Q) - rt = V - r(-1-t) - rt = V + r$; similarly $t \in I^-(\rho Q)$ iff $-1 - t \in I^+(Q)$, and then $\lambda_t(\rho Q) + rt = U + r(-1-t) + rt = U - r$. □

A pair (c, c') is a *front pair at* p if $c_i = c'_i$ for $i > p$, $y(c_p) \ne y(c'_p)$ and $x(c_p) = x(c'_p)$; equivalently, $\lambda_0 = p > \lambda_{-1}$. The *y-flip* and the *x-flip at* j complement the bit $y(c_j)$, respectively $x(c_j)$. The *yx-flip at* j complements both $y(c_j)$ and $x(c_{j+r})$.

Lemma 4.12. (1) *If* Q *is a front pair at* p, *then* $U(Q) = p$, *and* F^tQ *is a front pair at* $p + rt$ *for every* $t \ge 0$.
(2) *A y-flip at* j *is a front pair at* j, *and* $(U, V) = (j,\, j - 2r)$.
(3) *If* Q *is an x-flip at* j, *then* FQ *is a front pair at* j, *and* $(U, V) = (j - r,\, j - r)$.
(4) *A yx-flip at* j *has* $V = j$, *and either* $U = -\infty$ *or* $U \le j - 1$.

Proof. (1) $\lambda_{-1} < \lambda_0 < \lambda_0 + r$, so $0 \in I^+$ and $U = p$. By Proposition 4.10(2), $\lambda_t = p + rt$ for $t \ge 0$, so $\lambda_0(F^tQ) = p + rt > p + r(t-1) = \lambda_{-1}(F^tQ)$ for $t \ge 1$. (2) $\lambda_0 = j$ and $\lambda_{-1} = -\infty$, so $U = j$. Lemma 4.9(2) at $t = -1$ gives $\lambda_{-2} = \max(\lambda_0, \lambda_{-1} + r) = j$, and $\lambda_{-1} < \lambda_{-2} + r$, so $-2 \in I^-$ and $V = j - 2r$. (3) $\lambda_{-1} = j$ and $\lambda_0 = -\infty$. Lemma 4.9(1) at $t = 0$ gives $\lambda_1 = j$. So

FQ is a front pair at j, $1 \in I^+$ and $U = j - r$; and $\lambda_0 < \lambda_{-1} + r$ gives $-1 \in I^-$ and $V = j - r$. (4) $\lambda_0 = j$ and $\lambda_{-1} = j + r$, so $\lambda_0 < \lambda_{-1} + r$, $-1 \in I^-$ and $V = \lambda_{-1} - r = j$. For $i > j + r$, all three terms of $\Delta(1, i) = \Delta(-1, i) \oplus \Delta(0, i - r) \oplus D_i$ vanish, since $i - r > j = \lambda_0$. At $i = j + r$, D_{j+r} reads row 0 on $[j + 1, j + 2r]$, where Δ vanishes, so $\Delta(1, j + r) = 1 \oplus 1 \oplus 0 = 0$. Hence $\lambda_1 \le j + r - 1$, and Proposition 4.10(6) at $t = 1$ gives $U \le \max(\lambda_1 - r, \lambda_0 - 2r) \le j - 1$. □

By Proposition 4.10(5), statement (4) determines $I^\pm$ of every pair with the intercepts of a yx-flip at j. This is used in Proposition 4.23.

Proposition 4.13. *$(a, b) \mapsto \sigma^a F^b$ is an isomorphism $\mathbb{Z}^2 \to \langle \sigma, F \rangle$.*

Proof. It is a surjective homomorphism since $\sigma F = F\sigma$. Let Q be a y-flip at 0, so $(U, V) = (0, -2r)$. By Lemma 4.11, $\sigma^a F^b Q$ has $(U, V) = (rb - a, -2r - a - rb)$. If $\sigma^a F^b = \mathrm{id}$, then $rb - a = 0 = a + rb$, so $a = b = 0$. □

4.3. **The front offset.** In this subsection h is an invertible block map commuting with F, $R \ge 1$ bounds the radii of h and h^{-1}, and

$$M_R = 2r(R + 1) + 2R.$$

For $r = 1$, $M_R = 4R + 2$ as in Section 3.3.

Lemma 4.14 (onset)**.** *Let P be a front pair at p, and let $t_+ = \min I^+(hP)$. Then $U(hP)$ is finite, $t_+ \le \lfloor R/r \rfloor + 1$, and $|U(hP) - p| \le R$. In particular $\lambda_t(hP) = rt + U(hP)$ for every $t \ge R + 1$, and for $r \ge 2$ already $t_+ \le R$.*

Proof. Write $\lambda_t = \lambda_t(hP)$ and $P = (c, c')$, and let $t \ge 0$. By Lemma 4.12(1), $F^t P$ is a front pair at $p + rt$. By (5), rows $t - 1$ and t of Δ_{hP} record the x- and y-differences of $h(F^t c)$ and $h(F^t c')$. These agree beyond $p + rt + R$, so $\lambda_t, \lambda_{t-1} \le p + rt + R$; and since h^{-1} has radius at most R, they differ somewhere in $[p + rt - R, p + rt + R]$, so $\max(\lambda_t, \lambda_{t-1}) \ge p + rt - R$.

Suppose $t_+ \ge 2$ (including $t_+ = \infty$ if $I^+ = \emptyset$). By Proposition 4.10(2),(3), every $t \le t_+ - 2$ lies in I^-, and $t_+ - 1$ lies in I^- or is the middle row, so $\lambda_t \le V - rt$ for $t \le t_+ - 1$, with $V = V(hP)$ finite. As $-1 \in I^-$, the upper bound at $t = 0$ gives $V + r = \lambda_{-1} \le p + R$. For $1 \le t \le t_+ - 1$,

$$p + rt - R \le \max(\lambda_t, \lambda_{t-1}) \le V - r(t - 1) \le p + R - rt,$$

so $t \le R/r$. If $t_+ = \infty$ this fails at $t = \lfloor R/r \rfloor + 1$. Otherwise $t = t_+ - 1$ gives $t_+ \le \lfloor R/r \rfloor + 1$. In all cases U is finite and $t_+ \le \lfloor R/r \rfloor + 1 \le R + 1$. For $r \ge 2$, $\lfloor R/r \rfloor + 1 \le R$.

For the bounds on U, take $t = R + 2$. Then $t - 1 \ge t_+$, so t and $t - 1$ lie in I^+, and $\lambda_{t-1} = \lambda_t - r < \lambda_t$. The two bounds give $p + rt - R \le \lambda_t = rt + U \le p + rt + R$. □

For $r = 1$, $t \in I^+$ already gives $\lambda_{t-1} \le \lambda_t$, and the proof of Lemma 3.9 reads the lower bound at $t = R + 1$. For $r \ge 2$, $t \in I^+$ only gives $\lambda_{t-1} < \lambda_t + r$, and the lower bound needs $t - 1 \in I^+$ as well. This is why the proof takes $t = R + 2$, for every r.

For a front pair P at p put $E(P) = U(hP) - p$.

Lemma 4.15. *Let $P = (c, c')$ be a front pair at p.*

1. *$E(P) \in [-R, R]$, and $E(P)$ depends only on c and c' on $[p - 2R, p + M_R]$.*
2. *$E(\sigma^j P) = E(P)$ for every j, and $E(FP) = E(P)$.*
3. *(c', c) is a front pair at p and $E(c', c) = E(c, c')$.*

Proof. (1) The bound is Lemma 4.14. Put $t_0 = R + 1$. By that lemma $t_0 \in I^+(hP)$, so $U(hP) = \lambda_{t_0}(hP) - rt_0$, and $\lambda_{t_0}(hP) \in [p + rt_0 - R, p + rt_0 + R]$. By (5), $\lambda_{t_0}(hP)$ is the largest site in this interval at which the y-bits of $h(F^{t_0} c)$ and $h(F^{t_0} c')$ differ. These bits depend only on $F^{t_0} c$ and $F^{t_0} c'$ on $[p + rt_0 - 2R, p + rt_0 + 2R]$, and, since F^{t_0} has radius rt_0, only on c and c' on $[p - 2R, p + 2rt_0 + 2R] = [p - 2R, p + M_R]$. The same description applies to any front pair at p

that agrees with P on this interval. (2) $\sigma^j P$ is a front pair at $p-j$ and $U(h\sigma^j P) = U(hP) - j$ by Lemma 4.11. FP is a front pair at $p+r$ and $U(hFP) = U(FhP) = U(hP) + r$. (3) The definition is symmetric, and $\Delta_{h(c',c)} = \Delta_{h(c,c')}$. □

Front pairs exist and E takes values in $[-R, R]$, so $e_{\max} = \max\{E(P)\}$ over all front pairs exists.

Lemma 4.16 (maximum principle)**.** *Let $P = (c, c')$ be a front pair at p with $E(P) = e_{\max}$. If c'' agrees with c' at every site $i \geq p$, then (c, c'') is a front pair at p and $E(c, c'') = e_{\max}$. The same holds with the roles of c and c' exchanged.*

Proof. We follow the proof of Lemma 3.11. The pair (c, c'') is a front pair at p. Let c^* agree with c'' on $[p-2R, \infty)$ and with c' elsewhere. By Lemma 4.15(1), $E(c, c^*) = E(c, c'')$. The configurations c' and c^* differ in finitely many bits, at sites in $[p-2R, p-1]$; if they are equal there is nothing to prove. Complementing these bits one at a time gives $c' = c^0, \dots, c^N = c^*$, and each $Q_k = (c^{k-1}, c^k)$ is a y-flip or an x-flip at some $\ell_k \leq p-1$.

Let $t \geq R+2$. If Q_k is a y-flip, it is a front pair at ℓ_k, and Lemma 4.14 gives $\lambda_t(hQ_k) = \ell_k + rt + E(Q_k) \leq p - 1 + rt + e_{\max}$. If Q_k is an x-flip, FQ_k is a front pair at ℓ_k (Lemma 4.12(3)), and $\lambda_t(hQ_k) = \lambda_{t-1}(hFQ_k) = \ell_k + r(t-1) + E(FQ_k) \leq p - 1 - r + rt + e_{\max}$, since $t - 1 \geq R+1$. In both cases row t of Δ_{hQ_k} vanishes at all sites $\geq p + rt + e_{\max}$, while row t of Δ_{hP} has its rightmost 1 exactly at $p + rt + e_{\max}$ (Lemma 4.14). The telescoping identity $\Delta_{h(c,c^*)} = \Delta_{hP} \oplus \bigoplus_k \Delta_{hQ_k}$ of the proof of Lemma 3.11, which uses no linearity of h, gives $\lambda_t(h(c, c^*)) = p + rt + e_{\max}$ for $t \geq R+2$. By Lemma 4.14 these t lie in $I^+(h(c, c^*))$, so $U(h(c, c^*)) = p + e_{\max}$. The last statement follows from Lemma 4.15(3), as in Lemma 3.11(2). □

Corollary 4.17. *Let $P_1 = (c_1, c_1')$ and $P_2 = (c_2, c_2')$ be front pairs at p_1 and p_2 with $(c_1)_{p_1+k} = (c_2)_{p_2+k}$ for $0 \leq k \leq M_R$. Then $E(P_1) = e_{\max}$ if and only if $E(P_2) = e_{\max}$.*

Proof. The proof of Corollary 3.12 applies verbatim with $M = M_R$, using Lemma 4.15(1),(2) and Lemma 4.16 in place of Lemmas 3.10 and 3.11. □

Theorem 4.18 (front offset)**.** *If Λ_{M_R} is strongly connected, there is an integer $e^+(h) \in [-R, R]$ such that $U(hP) = p + e^+(h)$ for every front pair P at every p. Then $\lambda_t(hP) = p + rt + e^+(h)$ for $t \geq R+1$, and $e^+(h)$ does not depend on R.*

Proof. Let $\mathcal{S} \subseteq \mathcal{A}^{M_R+1}$ be the set of words ω such that $E(P) = e_{\max}$ for some front pair $P = (c, c')$ at some p with $c_{p+k} = \omega_k$ for $0 \leq k \leq M_R$. By Corollary 4.17, $E(P) = e_{\max}$ then holds for every such front pair. Every word occurs, with c' the y-flip of c at p. The set $\mathcal{S}$ is nonempty. It is forward-closed in Λ_{M_R}: for $\omega \in \mathcal{S}$ and $\iota \in \mathcal{A}^{2r}$, choose c with word ω on $[p, p + M_R]$ and ι on $[p + M_R + 1, p + M_R + 2r]$, and let c' be its y-flip at p. Then FP is a front pair at $p + r$ with $E(FP) = E(P) = e_{\max}$, and the word of $F(c)$ on $[p + r, p + r + M_R]$ is $\omega \cdot \iota$. By strong connectivity, $\mathcal{S} = \mathcal{A}^{M_R+1}$, so $E \equiv e_{\max}$. Put $e^+(h) = e_{\max}$. It is independent of R because E is. The formula for λ_t is Lemma 4.14. □

4.4. **Offsets of all pairs.**

Lemma 4.19. *Let h and R be as in Section 4.3, and assume that Λ_{M_R} is strongly connected. Let Q be any pair. If $U(Q)$ is finite, then $U(hQ) = U(Q) + e^+(h)$. If $U(Q) = -\infty$, then $U(hQ) = -\infty$.*

Proof. Write e^+ for $e^+(h)$. Theorem 4.18 applies to h and to h^{-1}. Let $U(Q)$ be finite and $t_0 \in I^+(Q)$. Since $U(F^{t_0}Q) = U(Q) + rt_0$ and $U(hF^{t_0}Q) = U(hQ) + rt_0$, we may assume $0 \in I^+(Q)$. Write $Q = (c, c')$ and $j = \lambda_0(Q) = U(Q)$, so that $\lambda_{-1}(Q) < j + r$. If $\lambda_{-1}(Q) < j$, then Q is a front pair at j and Theorem 4.18 applies. So assume $j \leq \lambda_{-1}(Q) \leq j + r - 1$.

Let c'' be c' with $y(c'_j)$ complemented. Then (c'', c') is a y-flip at j, and $\lambda_t(h(c'', c')) = rt+j+e^+$ for $t \geq R+1$. The pair (c, c'') has y-differences only at sites $\leq j-1$ and x-differences only at sites $\leq j+r-1$, and $c \neq c''$.

Claim: $\lambda_t(h(c, c'')) \leq rt+j-1+e^+$ *for every* $t \geq R+1$. Let $t \geq R+1$ and $i > rt+j-1+e^+$. The bit $Y_{hc}(t,i)$ depends only on c on $[i-R-rt,\ i+R+rt]$. Let $\tilde{c}$ agree with c'' on $[i-R-rt, \infty)$ and with c elsewhere, so $Y_{h\tilde{c}}(t,i) = Y_{hc''}(t,i)$. Complementing the finitely many differing bits of c and $\tilde{c}$ one at a time gives $c = c^0, \dots, c^N = \tilde{c}$, where each $Q_k = (c^{k-1}, c^k)$ is a y-flip at some $\ell \leq j-1$ or an x-flip at some $\ell \leq j+r-1$. For a y-flip, $U(hQ_k) = \ell + e^+ \leq j-1+e^+$. For an x-flip, FQ_k is a front pair at ℓ, so $U(hQ_k) = U(hFQ_k) - r = \ell - r + e^+ \leq j-1+e^+$. In both cases hc^{k-1} and hc^k agree beyond $\ell+R \leq j+r-1+R$, so rows -1 and 0 of Δ_{hQ_k} vanish beyond $j+r-1+R$. Proposition 4.10(6) at $t=-1$ gives $V(hQ_k) \leq \max(\lambda_{-1}-r, \lambda_0-2r) \leq j+R-1$. By Proposition 4.10(4), $\Delta_{hQ_k}(t,i) = 1$ would need $i-rt \leq j-1+e^+$ or $i+rt \leq j+R-1$. The first fails by the choice of i. The second fails because $i+rt > 2rt+j-1+e^+ \geq 2r(R+1)+j-1-R$, and $2r(R+1)-R \geq R$, that is, $2r(R+1) \geq 2R$. So $Y_{hc^{k-1}}(t,i) = Y_{hc^k}(t,i)$ for all k, and $Y_{hc}(t,i) = Y_{hc''}(t,i)$. This proves the claim.

Now $\Delta_{hQ} = \Delta_{h(c,c'')} \oplus \Delta_{h(c'',c')}$, and for $t \geq R+1$ the second term has its rightmost 1 in row t at $rt+j+e^+$ while the first vanishes there and beyond. So $\lambda_t(hQ) = rt+j+e^+$ for $t \geq R+1$. For $t \geq R+2$ this gives $\lambda_{t-1}(hQ) < \lambda_t(hQ) + r$, so $t \in I^+(hQ)$ and $U(hQ) = j+e^+$.

If $U(Q) = -\infty$ but $U(hQ)$ were finite, the first part for h^{-1} and the pair hQ would make $U(Q)$ finite. □

Proposition 4.20 (offsets)**.** *Assume* (Λ). *For* $h \in \mathrm{Aut}(X_F)$ *let* $e^+(h)$ *be the integer of Theorem 4.18, and put* $e^-(h) = e^+(\rho h \rho)$. *Then:*

1. *for every pair* Q, $U(hQ) = U(Q) + e^+(h)$ *and* $V(hQ) = V(Q) + e^-(h)$, *where* $-\infty + k = -\infty$;
2. e^+ *and* e^- *are homomorphisms* $\mathrm{Aut}(X_F) \to \mathbb{Z}$, *and* $e(\sigma^a F^b) = (rb-a,\ -a-rb)$;
3. $e(\langle \sigma, F \rangle) = L = \{(m,n) : m \equiv n \pmod{2r}\}$, *a subgroup of index* $2r$ *with* $\mathbb{Z}^2/L \cong \mathbb{Z}/2r$ *via* $(m,n) \mapsto m-n$.

Proof. Under (Λ), Lemma 4.19 applies to every h and every R. (1) The statement for U is that lemma. For V, apply it to $\rho h \rho$ and ρQ, and use $U(\rho Q) = V(Q) + r$ and $\rho h \rho\, \rho Q = \rho h Q$ (Lemma 4.11). (2) Additivity follows from (1) applied to a front pair, as in Corollary 3.16(2), and from $\rho h_1 h_2 \rho = (\rho h_1 \rho)(\rho h_2 \rho)$. By Lemma 4.11, $e(F) = (r, -r)$ and $e(\sigma) = (-1, -1)$. (3) $m-n = 2rb$ on the image. Conversely, if $m \equiv n \pmod{2r}$, then $m+n$ is even, and $a = -(m+n)/2$, $b = (m-n)/2r$ give $e(\sigma^a F^b) = (m,n)$. The homomorphism $(m,n) \mapsto m-n \bmod 2r$ is onto $\mathbb{Z}/2r$ with kernel L. □

4.5. **Cone causality and the two edges.** In this subsection $h \in \mathrm{Aut}(X_F)$ *preserves both intercepts*: $U(hQ) = U(Q)$ and $V(hQ) = V(Q)$ for every pair Q, including the value $-\infty$. Under (Λ) this means $e^+(h) = e^-(h) = 0$, by Proposition 4.20(1). Let $R \geq 1$ bound the radii of h and h^{-1}.

Proposition 4.21. *If* $y(c_i) = y(c'_i)$ *for all* $i \geq 0$ *and* $x(c_i) = x(c'_i)$ *for all* $i \geq r$, *then* $y((hc)_0) = y((hc')_0)$.

Proof. As in Proposition 3.17, let c'' agree with c' on $[-R, R]$ and with c elsewhere, and complement the differing bits of c and c'' one at a time. They are y-bits at sites $j \leq -1$, giving y-flips with $(U,V) = (j, j-2r)$, and x-bits at sites $j \leq r-1$, giving x-flips with $(U,V) = (j-r, j-r)$ (Lemma 4.12). In all cases $U < 0$ and $V < 0$. The images under h have the same intercepts, so by Proposition 4.10(4) their difference diagrams vanish at $(0,0)$. Hence $Y_{hc}(0,0) = Y_{hc''}(0,0) = Y_{hc'}(0,0)$. □

Cone codes. The cone code of a diagram Y at (t,i) is $\omega^{(t,i)}(Y) \in \mathcal{A}^{\mathbb{N}}$ with entries

$$\omega_k^{(t,i)} = (\mu_k, \nu_k) = \big(Y(t, i+k),\ Y(t-1, i+k+r)\big) \qquad (k \geq 0),$$

and $\omega(Y) = \omega^{(0,0)}(Y)$. Every $\omega \in \mathcal{A}^{\mathbb{N}}$ is $\omega(Y_c)$ for some c: put $y(c_k) = \mu_k$ and $x(c_{k+r}) = \nu_k$. By Proposition 4.21 there is $\psi\colon \mathcal{A}^{\mathbb{N}} \to \{0,1\}$, depending only on $\omega_0, \dots, \omega_R$, with $Y_{hc}(0,0) = \psi(\omega(Y_c))$, and the computation (10) gives

$$Y_{hc}(t,i) = \psi\big(\omega^{(t,i)}(Y_c)\big) \qquad \text{for all } (t,i). \tag{18}$$

Let σ also denote the shift of $\mathcal{A}^{\mathbb{N}}$, and define

$$\hat{G}(\omega)_k = \big(\nu_k \oplus \mu_k \oplus g(\mu_{k+1}, \dots, \mu_{k+2r}),\ \mu_{k+2r}\big),$$
$$\hat{H}(\omega)_k = \big(\nu_k,\ \mu_{k+2r} \oplus \nu_k \oplus g(\nu_{k+1}, \dots, \nu_{k+2r})\big).$$

D_1 complements μ_0, and D_2 complements μ_0 and ν_0.

Lemma 4.22. (1) *$\omega^{(1,r)}(Y) = \hat{G}\omega(Y)$, $\omega^{(-1,r)}(Y) = \hat{H}\omega(Y)$ and $\omega^{(0,k)}(Y) = \sigma^k\omega(Y)$ for every diagram Y and $k \geq 0$.*

(2) *For every $\omega \in \mathcal{A}^{\mathbb{N}}$,*

$$\psi(\hat{G}\omega) \oplus \psi(\hat{H}\omega) \oplus \psi(\omega) \oplus g\big(\psi(\sigma\omega), \dots, \psi(\sigma^{2r}\omega)\big) = 0. \tag{19}$$

(3) *$\hat{G}\sigma = \sigma\hat{G}$, $\hat{H}\sigma = \sigma\hat{H}$, $\hat{G}D_1 = D_1\hat{G}$, $\hat{H}D_1 = \hat{H}$, $\hat{H}D_2 = D_2\hat{H}$ and $\hat{G}D_2 = \hat{G}$.*

Proof. (1) By (17) at $(0, r+k)$, $Y(1, r+k) = Y(-1, r+k) \oplus Y(0,k) \oplus g(Y(0,k+1), \dots, Y(0,k+2r)) = \nu_k \oplus \mu_k \oplus g(\mu_{k+1}, \dots, \mu_{k+2r})$, and $Y(0, 2r+k) = \mu_{k+2r}$. By (17) at $(-1, 2r+k)$, $Y(-2, 2r+k) = Y(0, 2r+k) \oplus Y(-1, r+k) \oplus g(Y(-1, r+k+1), \dots, Y(-1, 3r+k)) = \mu_{k+2r} \oplus \nu_k \oplus g(\nu_{k+1}, \dots, \nu_{k+2r})$, and $Y(-1, r+k) = \nu_k$. The last identity is immediate. (2) Y_{hc} satisfies (17) at $(0,r)$: $Y_{hc}(1,r) \oplus Y_{hc}(-1,r) \oplus Y_{hc}(0,0) = g(Y_{hc}(0,1), \dots, Y_{hc}(0,2r))$. Substitute (18) and (1). (3) Both maps are given by the same rule at every index. In $\hat{G}$, μ_0 occurs only in the first component of $\hat{G}(\omega)_0$, through $\nu_0 \oplus \mu_0$, since g reads $\mu_{k+1}, \dots, \mu_{k+2r}$ and the second component reads μ_{k+2r}. And ν_0 occurs nowhere else in $\hat{G}$. In $\hat{H}$, μ occurs only with indices $\geq 2r \geq 2$, and ν_0 occurs only in $\hat{H}(\omega)_0 = (\nu_0,\ \mu_{2r} \oplus \nu_0 \oplus g(\nu_1, \dots, \nu_{2r}))$, where complementing it complements both components. □

Proposition 4.23. *For every ω, $\psi(D_1\omega) \neq \psi(\omega)$ and $\psi(D_2\omega) \neq \psi(\omega)$.*

Proof. Choose c with $\omega(Y_c) = \omega$. If c' is the y-flip of c at 0, then $\omega(Y_{c'}) = D_1\omega$, and hQ, $Q = (c, c')$, has $(U, V) = (0, -2r)$. Since $2r \mid V - U$, Proposition 4.10(5) gives a middle row $m = -1$ and $I^+(hQ) = [0, \infty)$, so $\lambda_0(hQ) = U = 0$ and $\Delta_{hQ}(0,0) = 1$. If c' is the yx-flip of c at 0, which complements $y(c_0)$ and $x(c_r)$, then $\omega(Y_{c'}) = D_2\omega$, and hQ has $V = 0$ and $U \in \{-\infty\} \cup (-\infty, -1]$ (Lemma 4.12(4)). If $U = -\infty$, then $I^-(hQ) = \mathbb{Z}$. Otherwise $V - U \geq 1$, and Proposition 4.10(5) gives $I^-(hQ) = (-\infty, m-1]$ with $m = (V-U)/2r \geq 1$, or $I^-(hQ) = (-\infty, t_+ - 1]$ with $t_+ = \lceil (V-U)/2r \rceil \geq 1$. In every case $0 \in I^-(hQ)$, so $\lambda_0(hQ) = V = 0$ and $\Delta_{hQ}(0,0) = 1$. □

Corollary 4.24. *There is $\gamma\colon \mathcal{A}^{\mathbb{N}} \to \{0,1\}$, depending only on the coordinates $0, \dots, R-1$, with $\psi(\omega) = \mu_0 \oplus \gamma(\sigma\omega)$.*

Proof. The proof of Corollary 3.20 applies verbatim. □

4.6. **The stripping descent.**

Proposition 4.25. *Let $\gamma\colon \mathcal{A}^{\mathbb{N}} \to \{0,1\}$ depend on finitely many coordinates, and suppose that $\psi(\omega) = \mu_0 \oplus \gamma(\sigma\omega)$ satisfies (19). If (N) holds, then γ is constant. The constant $\gamma \equiv 0$ always satisfies (19), and $\gamma \equiv 1$ does so if and only if g is self-dual.*

The proof occupies the rest of this subsection. For $\theta \in \mathcal{A}^{\mathbb{N}}$ write $\theta_k = (\xi_k, \zeta_k)$, and let D_1, D_2, $\hat{G}$, $\hat{H}$ act on θ by the formulas above, with (μ, ν) renamed (ξ, ζ). Thus

$$\hat{G}(\theta)_k = \big(\zeta_k \oplus \xi_k \oplus g(\xi_{k+1}, \dots, \xi_{k+2r}),\ \xi_{k+2r}\big), \qquad \hat{H}(\theta)_k = \big(\zeta_k,\ \xi_{k+2r} \oplus \zeta_k \oplus g(\zeta_{k+1}, \dots, \zeta_{k+2r})\big).$$

Coordinates read. A function φ on $\mathcal{A}^{\mathbb{N}}$ that depends on finitely many coordinates *reads* the coordinate ξ_k (or ζ_k) if complementing that coordinate changes the value of φ at some point. If φ depends only on the coordinates in a finite set S and does not read $s \in S$, then it depends only on $S \setminus \{s\}$. Hence φ depends only on the coordinates it reads, and a function that reads no coordinate is constant.

Step 1: the stripped identity. Put $\theta = \sigma\omega$, so $\xi_k = \mu_{k+1}(\omega)$ and $\zeta_k = \nu_{k+1}(\omega)$, and

$$\Gamma_k = \gamma(\sigma^{k+1}\theta) \qquad (0 \le k \le 2r-1).$$

Then $\psi(\sigma^{k+1}\omega) = \xi_k \oplus \Gamma_k$, $\psi(\omega) = \mu_0 \oplus \gamma(\theta)$, $\psi(\hat{H}\omega) = \nu_0 \oplus \gamma(\hat{H}\theta)$ and $\psi(\hat{G}\omega) = \nu_0 \oplus \mu_0 \oplus g(\xi) \oplus \gamma(\hat{G}\theta)$, where $g(\xi) = g(\xi_0, \dots, \xi_{2r-1})$ and we used $\sigma\hat{G} = \hat{G}\sigma$, $\sigma\hat{H} = \hat{H}\sigma$. The bits μ_0 and ν_0 cancel in (19), which becomes

$$\gamma(\hat{G}\theta) \oplus \gamma(\hat{H}\theta) \oplus \gamma(\theta) = g(\xi) \oplus g(\xi \oplus \Gamma), \qquad \xi \oplus \Gamma = (\xi_0 \oplus \Gamma_0, \dots, \xi_{2r-1} \oplus \Gamma_{2r-1}), \tag{20}$$

for every $\theta \in \mathcal{A}^{\mathbb{N}}$, since ω is arbitrary. The Γ_k do not read θ_0.

Step 2: two difference equations. Put $\varpi = \partial_0 g$ and $\delta_i = \gamma \oplus \gamma \circ D_i$ for $i = 1, 2$. Since $\partial_0\varpi = 0$ and $\partial_j\varpi = \partial_j\partial_0 g \equiv 0$ for $j > J = J(g)$, ϖ is a function of $\xi_1, \dots, \xi_J$ only, which we write $\varpi(\xi_1, \dots, \xi_J)$. Put

$$\Omega(\theta) = \varpi(\xi_1, \dots, \xi_J) \oplus \varpi(\xi_1 \oplus \Gamma_1, \dots, \xi_J \oplus \Gamma_J).$$

Add (20) at $D_1\theta$ to (20) at θ. The map D_1 complements ξ_0 and fixes every Γ_k, so the right sides add up to $\varpi(\xi) \oplus \varpi(\xi \oplus \Gamma) = \Omega(\theta)$. On the left, $\gamma(\hat{G}D_1\theta) = \gamma(D_1\hat{G}\theta)$ and $\gamma(\hat{H}D_1\theta) = \gamma(\hat{H}\theta)$ by Lemma 4.22(3). Doing the same with D_2, using $\hat{G}D_2 = \hat{G}$ and $\hat{H}D_2 = D_2\hat{H}$, gives the second equation:

$$\delta_1(\hat{G}\theta) \oplus \delta_1(\theta) = \Omega(\theta), \tag{21}$$

$$\delta_2(\hat{H}\theta) \oplus \delta_2(\theta) = \Omega(\theta). \tag{22}$$

Step 3: read-ranges. Suppose γ is not constant. Let a be the largest k such that γ reads ξ_k, and b the largest k such that γ reads ζ_k, with $a = -1$ (respectively $b = -1$) if there is no such k. Then $(a, b) \ne (-1, -1)$. Since D_i only complements coordinates of θ_0, δ_1 and δ_2 read ξ_k only for $k \le a$ and ζ_k only for $k \le b$. The first component of $\hat{G}(\theta)_k$ reads ζ_k and $\xi_k, \dots, \xi_{k+2r}$, and its second component reads ξ_{k+2r}. The first component of $\hat{H}(\theta)_k$ reads ζ_k, and its second component reads ξ_{k+2r} and $\zeta_k, \dots, \zeta_{k+2r}$. Hence:

- the left side of (21) reads ζ_k only for $k \le \max(a, b)$, and ξ_k only for $k \le \max(a, b) + 2r$;
- the left side of (22) reads ξ_k only for $k \le \max(a, b + 2r)$ if $b \ge 0$, and only for $k \le a$ if $b = -1$; it reads ζ_k only for $k \le \max(a, b + 2r)$.

Only two of these four ranges are used: the ζ-range of (21), in move (A) below, and the ξ-range of (22), in move (B). The function Γ_k reads ξ_{k+1+i} for $i \le a$ and ζ_{k+1+i} for $i \le b$. So the coordinates ζ_{J+1+b} (if $b \ge 0$) and ξ_{J+1+a} (if $a \ge 0$) enter Ω only through Γ_J. They are read by no Γ_k with $k < J$, since those read indices $\le J + b$, respectively $\le J + a$. And ξ_{J+1+a} is not one of $\xi_1, \dots, \xi_J$, because $a \ge 0$.

Step 4: steering.

Lemma 4.26. *Assume $\partial_0 g$ is not constant, and let $J = J(g)$. If $b \ge 0$, then Ω reads ζ_{J+1+b}. If $a \ge 0$, then Ω reads ξ_{J+1+a}.*

Proof. Let z be ζ_{J+1+b} or ξ_{J+1+a}, respectively. This is the coordinate ζ_b, respectively ξ_a, of $\sigma^{J+1}\theta$, which γ reads. So the coordinates of $\theta_{\geq J+1}$ other than z can be fixed so that complementing z complements $\Gamma_J = \gamma(\sigma^{J+1}\theta)$. Fix them in this way, and put $\theta_J = (0,0)$.

The function $\partial_J\varpi = \partial_J\partial_0 g$ is not identically 0 by the definition of J. It does not depend on ξ_J, nor on ξ_k for $k > J$ or $k = 0$, so it is a function of $\xi_1, \dots, \xi_{J-1}$. Choose $\mathbf{w} = (w_1, \dots, w_{J-1}) \in \{0,1\}^{J-1}$ with $\partial_J\varpi(\mathbf{w}) = 1$; for $J = 1$, $\partial_1\varpi$ is the constant 1. Now choose θ_k for $k = J-1, J-2, \dots, 1$, in this order. At step k the function Γ_k reads only $\theta_{k+1}, \theta_{k+2}, \dots$, which are already fixed, and it does not read z. Put $\zeta_k = 0$ and $\xi_k = w_k \oplus \Gamma_k$. Then $\eta_k := \xi_k \oplus \Gamma_k = w_k$ for $1 \leq k \leq J-1$. Choose θ_0 arbitrarily.

Complementing z changes neither $\xi_1, \dots, \xi_J$ nor $\Gamma_1, \dots, \Gamma_{J-1}$ (Step 3), and complements Γ_J. So it changes Ω by

$$\varpi(\mathbf{w}, \eta_J) \oplus \varpi(\mathbf{w}, \eta_J \oplus 1) = \partial_J\varpi(\mathbf{w}) = 1. \qquad \square$$

Step 5: the moves, and where they get stuck. Equations (21) and (22) hold identically, so a coordinate read by Ω must be read by the left side of each. Lemma 4.26 and the ranges of Step 3 give two contradictions:

(A) If $b \geq 0$ and $J + 1 + b > \max(a, b)$, then Ω reads ζ_{J+1+b} but the left side of (21) does not. So γ cannot read ζ_b, contradicting the choice of b.

(B) If $a \geq 0$, and either $b = -1$ or $J + 1 + a > \max(a, b + 2r)$, then Ω reads ξ_{J+1+a} but the left side of (22) does not. So γ cannot read ξ_a, contradicting the choice of a.

Since $J+1+b > b$ and $J+1+a > a$, (A) applies unless $b = -1$ or $a \geq J+1+b$, and (B) applies unless $a = -1$, or $b \geq 0$ and $a \leq b + 2r - J - 1$. If $b = -1$, then $a \geq 0$, and (B) applies. If $b \geq 0$ and (A) does not apply, then $a \geq J+1+b \geq 0$, and (B) does not apply only if $a \leq b+2r-J-1$. Hence both moves fail exactly on the *stuck region*

$$\big\{(a,b) :\ b \geq 0,\quad J+1+b \leq a \leq b+2r-J-1\big\}.$$

It is nonempty if and only if $J + 1 \leq 2r - J - 1$, that is, $J \leq r - 1$. Under (N), $J \geq r$, so the stuck region is empty and a non-constant γ is impossible. Thus γ is constant.

The same argument in the language of a descent: if γ depends only on $\theta_0, \dots, \theta_m$, then the moves strip the top ζ-coordinate while $a - b \leq J$, and the top ξ-coordinate once $a - b \geq 2r - J$; for $J \geq r$ one of the two always applies. (For instance, if $J = r \geq 2$, $a = m$ and $b = m - 1$, then (B) does not apply, and (A) removes $\zeta_{m-1}, \zeta_{m-2}, \dots$ until $a - b \geq 2r - J$; only then does (B) remove ξ_m.) For $r = 1$, where $J = 1$, this is the descent in the proof of Lemma 3.21. If $J \leq r - 1$, the descent may stop at a pair (a, b) in the stuck region, and the argument gives nothing. If $\partial_0 g$ is constant, then $\Omega \equiv 0$, and the argument gives nothing either.

Step 6: the constants. If $\gamma \equiv \epsilon \in \{0, 1\}$, then $\Gamma = \epsilon\mathbf{1}$ and (20) reads $\epsilon = g(\xi) \oplus g(\xi \oplus \epsilon\mathbf{1})$ for all ξ. For $\epsilon = 0$ this always holds, and for $\epsilon = 1$ it says that g is self-dual. Since (20) is equivalent to (19) for $\psi = \mu_0 \oplus \gamma \circ \sigma$, this proves Proposition 4.25.

4.7. **The kernel of the offsets.**

Proposition 4.27. *Assume* (N). *If $h \in \mathrm{Aut}(X_F)$ preserves both intercepts of every pair, then $h = \mathrm{id}$, or g is self-dual and $h = \kappa$. Conversely, if g is self-dual, then $\kappa \in \mathrm{Aut}(X_F)$ preserves both intercepts of every pair.*

Proof. By Corollary 4.24, Lemma 4.22(2) and Proposition 4.25, $\psi(\omega) = \mu_0 \oplus \epsilon$ for a constant ϵ, with $\epsilon = 0$ unless g is self-dual. By (18), $Y_{hc} = Y_c \oplus \epsilon$. Rows -1 and 0 of the diagram are the two bits of the cells, so $h = \mathrm{id}$ if $\epsilon = 0$ and $h = \kappa$ if $\epsilon = 1$. Conversely, if g is self-dual, then κ is an involutive block map commuting with F (Lemma 4.2), so $F^t\kappa c = \kappa F^t c$ and $Y_{\kappa c} = Y_c \oplus 1$. Hence $\Delta_{\kappa Q} = \Delta_Q$ for every pair. $\square$

4.8. **The index step.**

Lemma 4.28 (orbit counting)**.** *Let q be a prime, $a \in \mathbb{Z}$, and let $\mathcal{L}$ be a subgroup of finite index with $q \nmid k(\mathcal{L})$. If some $h_1 \in \mathrm{Aut}(X_F)$ satisfies $h_1^q = \sigma^a F$, then $(a,1) \in q\mathbb{Z}^2 + \mathcal{L}$. Consequently $(\mathrm{I})_q$ excludes $h_1^q = \sigma^a F$ with $a \equiv a_q \pmod q$.*

Proof. Since h_1 commutes with σ and F, it maps Per to itself and each $\mathbb{Z}^2$-orbit onto a $\mathbb{Z}^2$-orbit, $h_1(v \cdot c) = v \cdot h_1(c)$. Since h_1 is injective, $\mathrm{Stab}(h_1 c) = \mathrm{Stab}(c)$, and distinct orbits have distinct images. So h_1 induces a permutation π of the finite set $\mathcal{O}_\mathcal{L}$ of orbits with stabiliser $\mathcal{L}$, which has $k(\mathcal{L})$ elements. The power π^q is induced by $\sigma^a F$, which maps every orbit to itself, so $\pi^q = \mathrm{id}$ and every cycle of π has length 1 or q. Since $q \nmid k(\mathcal{L})$, some orbit O is fixed by π. Fix $c_0 \in O$ and write $h_1(c_0) = v_0 \cdot c_0$ with $v_0 \in \mathbb{Z}^2$. Then $h_1(v \cdot c_0) = (v + v_0) \cdot c_0$ for every v, so $h_1^q(c_0) = (qv_0) \cdot c_0$, while $\sigma^a F(c_0) = (a,1) \cdot c_0$. Hence $(a,1) - qv_0 \in \mathrm{Stab}(c_0) = \mathcal{L}$. If $a \equiv a_q \pmod q$, then $(a_q, 1) = (a,1) - (a - a_q, 0) \in q\mathbb{Z}^2 + \mathcal{L}$, contrary to $(\mathrm{I})_q$. □

Lemma 4.29 (the root)**.** *Assume (Λ), and that $e = (e^+, e^-)$ is injective on $\mathrm{Aut}(X_F)$. Then $\mathrm{Aut}(X_F) \cong e(\mathrm{Aut}(X_F)) \cong \mathbb{Z}^2$, and $\mathrm{Aut}(X_F)/\langle \sigma, F\rangle \cong e(\mathrm{Aut}(X_F))/L$ is cyclic of order dividing $2r$. For a prime q, q divides the index if and only if $h_1^q = \sigma^a F$ for some $h_1 \in \mathrm{Aut}(X_F)$ and $a \in \mathbb{Z}$. In that case necessarily $a \equiv -r \equiv a_q \pmod q$.*

Proof. By Proposition 4.20, $L \subseteq e(\mathrm{Aut}(X_F)) \subseteq \mathbb{Z}^2$, and L has index $2r$. So $e(\mathrm{Aut}(X_F))$ is free abelian of rank 2, and the quotient embeds in $\mathbb{Z}^2/L \cong \mathbb{Z}/2r$, which is cyclic. If q divides the order of the quotient, then the quotient contains the subgroup of order q of $\mathbb{Z}/2r$. So there is $v = (m,n) \in e(\mathrm{Aut}(X_F))$ with $m - n \equiv 2r/q \pmod{2r}$. Then $q(m-n) \equiv 0 \pmod{2r}$, so $qv \in L$, and $qv = e(\sigma^a F^b)$ with $2rb = q(m-n) \equiv 2r \pmod{2rq}$, that is, $b \equiv 1 \pmod q$. Replacing v by $v - \frac{b-1}{q}\, e(F)$ we may assume $b = 1$. Then $qm = r - a$ and $qn = -a - r$, so $a = -qn - r \equiv -r \pmod q$. If q is odd, then $q \mid r$ and $-r \equiv 0 = a_q$. If $q = 2$, then $-r \equiv r \bmod 2 = a_2$. Let $h_1 = e^{-1}(v)$. Then $e(h_1^q) = e(\sigma^a F)$, so $h_1^q = \sigma^a F$ by injectivity. Conversely, if $h_1^q = \sigma^a F$, then $e(h_1) = (m,n)$ with $qm = r - a$ and $qn = -a - r$, so $a \equiv -r \equiv a_q \pmod q$ by the computation above, and $q(m-n) = 2r$, so the class of $e(h_1)$ in $\mathbb{Z}^2/L \cong \mathbb{Z}/2r$ is $2r/q$, which has order q. Hence q divides the order of the quotient. □

For $r = 1$ the only prime is $q = 2$, with $a_2 = 1$, and for rule 30 the orbit count of Lemma 4.28, applied to the 2-cycle of F on the constant configurations, is the parity argument of Lemma 3.23. For $r \geq 2$ the index $2r$ can have odd prime factors, which the parity argument of Lemma 3.23 does not see.

4.9. **Proofs of Theorems 4.6 and 4.7.**

Proof of Theorem 4.7. (a) The isomorphism is Proposition 4.13, and the rest is Proposition 4.20. (b) Under (Λ), $h \in \ker e$ if and only if h preserves both intercepts of every pair (Proposition 4.20(1)). Proposition 4.27 then gives $\ker e = \{\mathrm{id}\}$ under (D) and $\ker e = \{\mathrm{id}, \kappa\}$ for self-dual g. The consequences are Lemma 4.29. (c) By Lemma 4.2, $\kappa \in \mathrm{Aut}(X_F)$, it has order 2, and it commutes with σ and F. By Proposition 4.13, $\langle \sigma, F\rangle$ is torsion-free, so $\kappa \notin \langle \sigma, F\rangle$. If (Λ) and (N) hold and $h \in \mathrm{Aut}(X_F)$, then $e(h\kappa h^{-1}) = e(\kappa) = 0$ by Proposition 4.27, so $h\kappa h^{-1} \in \ker e = \{\mathrm{id}, \kappa\}$ by (b). Since $h\kappa h^{-1} \neq \mathrm{id}$, it equals κ. Without (Λ) and (N) we only know that κ commutes with σ and F. (d) By (b), e is injective, and the first two statements are Lemma 4.29. The third is Lemma 4.28, which uses no hypothesis. Combining them, $(\mathrm{I})_q$ implies $q \nmid [\mathrm{Aut}(X_F) : \langle \sigma, F\rangle]$. Under (I) this holds for every prime $q \mid 2r$, and the index divides $2r$, so it is 1. □

Proof of Theorem 4.6. By Theorem 4.7(b),(d), e is injective and $e(\mathrm{Aut}(X_F)) = e(\langle \sigma, F\rangle) = L$. Let $h \in \mathrm{Aut}(X_F)$. Then $e(h) = e(\sigma^a F^b)$ for some (a,b), so $h = \sigma^a F^b$. Since $e(\sigma^a F^b) =$

$(rb-a,-a-rb)$, we get $a=-\frac{1}{2}(e^+(h)+e^-(h))$ and $b=\frac{1}{2r}(e^+(h)-e^-(h))$. Uniqueness of (a,b) is Proposition 4.13, and Lemma 2.2 translates the result to $\mathrm{Aut}(X_F)$. □

4.10. **The light-speed graph: a tower and a finite certificate.** We prove that a finite computation, the certificate (T_P), implies (Λ) for every M. The method is that of the proof of Lemma 3.7: a tower of factor graphs of Λ_M, in which the stationary loop (9) is replaced by closed walks that come from time-periodic histories. Fix $M \geq 0$.
Layer coordinates. For a vertex ω of Λ_M, on the sites $[p, p+M]$, put

$$y[d] = y(\omega \text{ at site } p-d) \ \ (-M \leq d \leq 0), \qquad x[j] = x(\omega \text{ at site } p-j-r) \ \ (-M-r \leq j \leq -r).$$

So the cell at site $p-d$ is $(x[d-r], y[d])$, and ω is the same as an arbitrary assignment of bits to these $2(M+1)$ layers. Put

$$I_u = [-M-r,\, -M-1], \qquad B_y = [-M,\, \min(0, -M+r-1)],$$
$$B_x = [-M-r,\, \min(-r, -M-1)].$$

The *base layers* are $y[d]$ for $d \in B_y$ and $x[j]$ for $j \in B_x$. If $M \geq r-1$ there are r of each.

Lemma 4.30 (layer equation)**.** *Let ω be a vertex, $\iota \in \mathcal{A}^{2r}$, and $\omega' = \omega \cdot \iota$ with layers y', x'. Let u_k $(k \in I_u)$ be the y-bit of the input cell at site $p-k \in [p+M+1, p+M+r]$, and v_d $(d \in B_y)$ the y-bit of the new cell at site $p+r-d$. With the convention $y[k] = u_k$ for $k \in I_u$:*

(1) $x'[j] = y[j]$ *for* $-M-r \leq j \leq -r$;
(2) $y'[d] = v_d$ *for* $d \in B_y$;
(3) $y'[d] = y[d] \oplus x[d-2r] \oplus g\big(y[d-1], y[d-2], \ldots, y[d-2r]\big)$ *for* $-M+r \leq d \leq 0$.

For every vertex ω and every $(u,v) \in \{0,1\}^{I_u} \times \{0,1\}^{B_y}$ there is ι with these values. So the successors of ω in Λ_M are the vertices $\omega \cdot (u,v)$ given by (1)–(3).

Proof. The new front is $p+r$. (1) $x'[j]$ is the new x-bit at site $p-j$, which is the old y-bit there. This is $y[j]$ if $j \geq -M$, and u_j if $j \in I_u$. (2) is the definition of v_d. (3) $y'[d]$ is the new y-bit at $i = p+r-d$, which is $x_i \oplus y_{i-r} \oplus g(y_{i-r+1}, \ldots, y_{i+r})$. For $d \geq -M+r$ the site i is in $[p, p+M]$, and $x_i = x[d-2r]$ with $d-2r \geq -M-r$. Next, $y_{i-r} = y[d]$, and $y_{i-r+1+k}$ is the y-bit at site $p-(d-1-k)$, which is $y[d-1-k]$. Its index $d-1-k \geq -M-r$ is a layer or lies in I_u. For the last statement, set the y-bits of the input cells at $p+M+1, \ldots, p+M+r$ to u. For $d \in B_y$ the site $p+r-d$ lies in $[p+M+1, p+M+r]$. The x-bit of the input cell there is read by F only in the new y-bit at that site, through $\oplus$, so it can be chosen to make that bit v_d. □

The tower. Order the $N = 2(M+1)$ layers as $L_1, \ldots, L_N$: first the base layers, in any order, then, for $k = -M, -M+1, \ldots, 0$ in turn, the layer $y[k]$ if $k \geq -M+r$, followed by $x[k]$ if $-M \leq k \leq -r$. Each layer occurs exactly once. Let $P_n = \{L_1, \ldots, L_n\}$ and π_n the restriction to P_n.

Lemma 4.31. *For each n, $\pi_n(\omega \cdot (u,v))$ depends only on $\pi_n(\omega)$ and (u,v). Let $\Lambda^{(n)}$ be the graph on $\{0,1\}^{P_n}$ with edges $\eta \to \eta \cdot (u,v) := \pi_n(\omega \cdot (u,v))$ for any ω with $\pi_n(\omega) = \eta$. Then $\Lambda^{(N)}$ has the same successor sets as Λ_M. Every walk of $\Lambda^{(n)}$ lifts, with the same labels, to a walk of $\Lambda^{(n+1)}$ from any vertex restricting to its initial vertex.*

Proof. By Lemma 4.30, the new value of each layer depends only on (u,v) and on layers that precede it or equal it. The new values of the base layers $x[j]$ $(j \in B_x \subseteq I_u)$ and $y[d]$ $(d \in B_y)$ are u_j and v_d. The layer $x'[k]$ $(k \geq -M)$ is $y[k]$, which precedes $x[k]$. The layer $y'[d]$ $(d \geq -M+r)$ reads $y[d]$, the layer $x[d-2r]$, which is a base layer or comes at index $d-2r < d$, and the layers $y[d-1], \ldots, y[d-2r]$ or u. The rest follows from Lemma 4.30, since strong connectivity depends only on successor sets. □

At the step $L_{n+1} = y[d]$ with $d \geq -M + r$, the set P_n consists of all layers with index $\leq d-1$, since every base layer has index $\leq -M + r - 1 \leq d - 1$.

Lemma 4.32. *Let n_0 be the number of base layers. Then $\Lambda^{(n_0)}$ is a complete digraph with loops.*

Proof. From every vertex, the edge labelled (u, v) leads to the assignment $x[j] = u_j$ $(j \in B_x)$, $y[d] = v_d$ $(d \in B_y)$. □

If $M \leq r - 1$, every layer is a base layer, so Λ_M itself is complete.

Lemma 4.33 (predecessors)**.** *Every vertex ω_2 of Λ_M has an in-neighbour. Every in-neighbour $\hat{\omega}$ of ω_2 satisfies $\hat{y}[j] = x_2[j]$ for $-M \leq j \leq -r$.*

Proof. The second statement is Lemma 4.30(1). For the first, we construct $\hat{\omega}$ and (u, v) with $\hat{\omega} \cdot (u, v) = \omega_2$. Put $u_j = x_2[j]$ for $j \in I_u$ with $j \leq -r$ (other u_j are arbitrary), $v_d = y_2[d]$ for $d \in B_y$, $\hat{y}[j] = x_2[j]$ for $-M \leq j \leq -r$, and $\hat{y}[d] = 0$ for $\max(-M, -r + 1) \leq d \leq 0$. With the convention $\hat{y}[k] = u_k$ for $k \in I_u$, all values $\hat{y}[k]$, $-M - r \leq k \leq 0$, are now fixed. For $-M + r \leq d \leq 0$ put

$$\hat{x}[d - 2r] = y_2[d] \oplus \hat{y}[d] \oplus g\big(\hat{y}[d-1], \ldots, \hat{y}[d-2r]\big).$$

The indices $d - 2r$ are distinct and lie in $[-M - r, -2r]$, so this is consistent, and the remaining layers $\hat{x}[j]$ are arbitrary. By Lemma 4.30, conditions (1), (2) and (3) give $x' = x_2$, $y' = y_2$ on B_y, and $y' = y_2$ on $[-M + r, 0]$. □

The property used below is that a predecessor with a prescribed bit $y[j]$ exists: it is any predecessor of a successor whose $x[j]$ is that bit. The number of predecessors plays no role.

Lemma 4.34 (memory steps)**.** *Let $L_{n+1} = x[j]$ with $-M \leq j \leq -r$. If $\Lambda^{(n)}$ is strongly connected, so is $\Lambda^{(n+1)}$.*

Proof. Write vertices of $\Lambda^{(n+1)}$ as (η, χ) with χ the bit $x[j]$. By Lemma 4.30, $x[j]$ is read only by $y'[j + 2r]$, which is not in P_{n+1}, and $x'[j] = y[j]$ with $y[j] \in P_n$. So the edge labelled (u, v) goes from (η, χ) to $(\eta \cdot (u, v), y[j](\eta))$. Let (η_1, χ_1) and (η_2, χ_2) be vertices. Extend (η_2, χ_2) to a vertex ω_2 of Λ_M, and take an in-neighbour $\hat{\omega}$ of ω_2, with label (u, v), by Lemma 4.33. Then $\hat{y}[j] = x_2[j] = \chi_2$, and $\hat{\eta} = \pi_n(\hat{\omega})$ has an edge labelled (u, v) to η_2. Take a walk from η_1 to $\hat{\eta}$ in $\Lambda^{(n)}$, lift it from (η_1, χ_1) to a walk ending at some $(\hat{\eta}, \chi')$, and append the edge labelled (u, v), which leads to $(\eta_2, \hat{y}[j]) = (\eta_2, \chi_2)$. □

Lemma 4.35 ($\mathbb{Z}/2$ steps)**.** *Let $L_{n+1} = y[d]$ with $-M + r \leq d \leq 0$. For an edge of $\Lambda^{(n)}$ from η with label (u, v) put*

$$\phi = x[d - 2r] \oplus g\big(y[d-1], \ldots, y[d-2r]\big),$$

evaluated on η with $y[k] = u_k$ for $k \in I_u$. Then the edges of $\Lambda^{(n+1)}$ are $(\eta, z) \to (\eta \cdot (u, v),\ z \oplus \phi)$. If $\Lambda^{(n)}$ is strongly connected and some closed walk of $\Lambda^{(n)}$ has odd total $\sum \phi$ (its holonomy*), then $\Lambda^{(n+1)}$ is strongly connected.*

Proof. The first statement is Lemma 4.30(3), all of whose inputs other than $y[d]$ are in P_n or in (u, v). For the second, let ς be a closed walk at some η with odd holonomy, and let η_2 be any vertex. Choose walks α from η_2 to η and β from η to η_2. The closed walks $\alpha\beta$ and $\alpha\varsigma\beta$ at η_2 have holonomies of different parity. So there are closed walks of both parities at every vertex. Given (η_1, z_1) and (η_2, z_2), take a walk from η_1 to η_2 followed by a closed walk at η_2 of the right parity, and lift it from (η_1, z_1). □

Periodic histories. The closed walks come from time-periodic solutions of (17) in comoving coordinates.

Definition 4.36. Let $D \in \mathbb{Z}$ and $T \geq 1$. A *periodic history below D* of period T is a function $Y\colon \mathbb{Z} \times (-\infty, D-1] \to \{0,1\}$, $(t,i) \mapsto Y_t(i)$, with $Y_{t+T} = Y_t$, such that

$$Y_{t+1}(i) = Y_t(i) \oplus Y_{t-1}(i-2r) \oplus g\big(Y_t(i-1), \dots, Y_t(i-2r)\big) \qquad \text{for all } t \in \mathbb{Z} \text{ and } i \leq D-1. \tag{23}$$

Its *increment* at level D is $\phi_t = Y_{t-1}(D-2r) \oplus g(Y_t(D-1), \dots, Y_t(D-2r))$, and its *holonomy* is $\ell(Y) = \sum_{t=0}^{T-1} \phi_t \in \mathbb{Z}/2$.

Equation (23) is (17) for $Y_t(i) = Y_c(t, p + rt - i)$, and it is invariant under shifts of i. If Y extends to a periodic history below $D+1$ of the same period, then $\phi_t = Y_{t+1}(D) \oplus Y_t(D)$ and $\ell(Y) = 0$. So odd holonomy means that the history cannot be continued one layer up with the same period.

Lemma 4.37. *Let $-M + r \leq d \leq 0$, let $L_{n+1} = y[d]$, and let Y be a periodic history below d of period T. For $t \in \mathbb{Z}$ let $\eta_t \in \{0,1\}^{P_n}$ be given by $y[i] = Y_t(i)$ for $-M \leq i \leq d-1$ and $x[j] = Y_{t-1}(j)$ for $-M - r \leq j \leq \min(d-1, -r)$. Then $\Lambda^{(n)}$ has an edge $\eta_t \to \eta_{t+1}$ labelled $u_k = Y_t(k)$ ($k \in I_u$), $v_k = Y_{t+1}(k)$ ($k \in B_y$), whose value of ϕ is the increment ϕ_t of Y at level d. So $\eta_0 \to \eta_1 \to \cdots \to \eta_T = \eta_0$ is a closed walk with holonomy $\ell(Y)$.*

Proof. We check Lemma 4.30 on the layers of P_n, with $y[k] = u_k = Y_t(k)$ for $k \in I_u$. By (1), $x'[j] = Y_t(j)$, which is $x[j]$ of η_{t+1}. By (2), $y'[k] = Y_{t+1}(k)$ for $k \in B_y$. By (3), for $-M + r \leq k \leq d-1$, $y'[k] = Y_t(k) \oplus Y_{t-1}(k-2r) \oplus g(Y_t(k-1), \dots, Y_t(k-2r)) = Y_{t+1}(k)$ by (23). The value of ϕ on this edge is $Y_{t-1}(d-2r) \oplus g(Y_t(d-1), \dots, Y_t(d-2r)) = \phi_t$. Finally, $\eta_T = \eta_0$ by periodicity. □

The certificate. Fix $P \geq 1$. A *column state* is a $2r$-tuple $s = (s_1, \dots, s_{2r})$ of functions $s_j\colon \mathbb{Z}/P \to \{0,1\}$; there are 2^{2rP} of them. Think of s_j as the time series of layer $D-2r-1+j$ of a P-periodic history, so that s_1 is the oldest layer $D-2r$ and s_{2r} is layer $D-1$. Put

$$\phi(s)_t = s_1(t-1) \oplus g\big(s_{2r}(t), s_{2r-1}(t), \dots, s_1(t)\big), \qquad \pi(s) = \sum_{t \in \mathbb{Z}/P} \phi(s)_t \in \mathbb{Z}/2.$$

If $\pi(s) = 0$, the *successors* of s are the two states $(s_2, \dots, s_{2r}, z)$ with $z(t+1) = z(t) \oplus \phi(s)_t$ for all $t \in \mathbb{Z}/P$; these are consistent because $\pi(s) = 0$. If $\pi(s) = 1$, then s has no successor. Let Im_0 be the set of all column states, Im_{m+1} the set of successors of elements of Im_m, and $S_\infty = \bigcap_m \mathrm{Im}_m$.

(T_P) *Some $s \in S_\infty$ has $\pi(s) = 1$.*

Proposition 4.38.
(1) *$\mathrm{Im}_{m+1} \subseteq \mathrm{Im}_m$ for all m, and the sequence is constant, equal to S_∞, from some $m_0 \leq 2^{2rP}$ on. Every element of S_∞ is a successor of an element of S_∞. In particular S_∞, and hence the truth of (T_P), is computed in finitely many steps.*
(2) *For every $s \in S_\infty$ and every $D \in \mathbb{Z}$ there is a periodic history Y below D of period P whose layers $D-2r, \dots, D-1$ are $s_1, \dots, s_{2r}$, and $\ell(Y) = \pi(s)$.*
(3) *If (T_P) holds for some $P \geq 1$, then Λ_M is strongly connected for every $M \geq 0$. That is, (Λ) holds.*

Proof. (1) $\mathrm{Im}_1 \subseteq \mathrm{Im}_0$, and taking successors is monotone, so the sequence decreases by induction. A decreasing sequence of subsets of a set with 2^{2rP} elements is constant from some $m_0 \leq 2^{2rP}$ on. Then $S_\infty = \mathrm{Im}_{m_0+1}$ is the set of successors of elements of $\mathrm{Im}_{m_0} = S_\infty$.

(2) By (1), choose $s^{(0)} = s$ and $s^{(-1)}, s^{(-2)}, \dots \in S_\infty$ with $s^{(-m)}$ a successor of $s^{(-m-1)}$. Define $Y(D-1-m)$, as a function of t, to be the last component of $s^{(-m)}$. Consecutive states overlap

in $2r-1$ components, so by induction $s^{(-m)} = (Y(D-2r-m), \dots, Y(D-1-m))$. For the layer $i = D-1-m$, the successor relation from $s^{(-m-1)} = (Y(i-2r), \dots, Y(i-1))$ reads $Y_{t+1}(i) = Y_t(i) \oplus Y_{t-1}(i-2r) \oplus g(Y_t(i-1), \dots, Y_t(i-2r))$, which is (23). The increment of Y at level D is $\phi(s)$, so $\ell(Y) = \pi(s)$.

(3) Let $M \geq 0$. We show by induction on n that $\Lambda^{(n)}$ is strongly connected for $n_0 \leq n \leq N$. For $n = n_0$ this is Lemma 4.32. A layer $L_{n+1} = x[j]$ with $-M \leq j \leq -r$ is covered by Lemma 4.34. For a layer $L_{n+1} = y[d]$ with $-M + r \leq d \leq 0$, take $s \in S_\infty$ with $\pi(s) = 1$, and apply (2) with $D = d$. Lemma 4.37 then gives a closed walk of $\Lambda^{(n)}$ with odd holonomy, and Lemma 4.35 applies. By Lemma 4.31, Λ_M is strongly connected. The same state s serves every d and every M, because (23) is invariant under shifts of the layer index. □

Remark 4.39. (1) The certificate is a finite computation on the 2^{2rP} states, described in Appendix A.2. Increasing P gives a semi-decision procedure for a sufficient condition. We do not claim that (Λ) implies (T_P) for some P.

(2) For rule 30, the stationary loop (9) is the periodic history of period 1 given by the column state $(s_1, s_2) = (0, 1)$, which lies in S_∞ and has $\pi = 1$. Certificates for the other elementary classes, small enough to check by hand, are given in Proposition 5.4.

4.11. **Examples.** The statements of this subsection are computer-assisted, except those for $r = 1$; the computations are specified in Appendix A.2. A function g of $2r$ variables is given either by a formula or by its *truth table*, the integer whose binary digit of weight $2^{\xi_0 + 2\xi_1 + \cdots + 2^{2r-1}\xi_{2r-1}}$ is $g(\xi_0, \dots, \xi_{2r-1})$, written in hexadecimal; for $r = 2$, ξ_3 is `ff00` and $\xi_0\xi_2$ is `a0a0`. A witness $\mathcal{L} = \langle (n, 0), (s, b) \rangle$ for $(\mathrm{I})_q$ is written (n, b, s); its points have least spatial period n, and $k(\mathcal{L})$ is computed exactly by Lemma 4.3.

Radius 1. The four hypotheses hold for the eight non-affine left-permutive elementary rules, by hand (Section 5). Two affine rules serve as controls. Rule 90 ($g = \xi_1$) fails (Λ): Λ_M is not strongly connected for $1 \leq M \leq 8$, and (T_P) fails for $P \leq 5$. Rule 150 ($g = \xi_0 \oplus \xi_1$) satisfies (T_1) but fails (N), and $(\mathrm{I})_2$ has no witness in Per_n for $n \leq 4$. This is consistent with the automorphism $\mathrm{id} + \sigma F$ of Remark 5.8.

Radius 2. Here $\xi_k = y_{i-1+k}$ and $f = y_{i-2} \oplus g(y_{i-1}, y_i, y_{i+1}, y_{i+2})$, the index of $\langle \sigma, F \rangle$ divides 4, and (I) is $(\mathrm{I})_2$ with $a_2 = 0$, that is, $(0, 1) \notin 2\mathbb{Z}^2 + \mathcal{L}$. There are 65 536 functions g, of which 32 are affine.

Proposition 4.40 (computer-assisted, radius 2)**.** *Every non-affine* $g\colon \{0,1\}^4 \to \{0,1\}$ *satisfies* (T_P) *for some* $P \leq 4$*. Hence* (Λ) *holds for all* 65 504 *non-affine rules of radius* 2*. The least such* P *is* 1 *for* 57 482 *of them,* 2 *for* 6 586*,* 3 *for* 1 392*, and* 4 *for* 44*; the smallest truth table with least* $P = 4$ *is* `0339`*. Among the* 32 *affine* g*,* (T_1) *holds for* 30*, and* (T_P) *fails for* $P \leq 4$ *for* $g = \xi_3$ *and* $g = 1 \oplus \xi_3$*.*

So Conjecture 4.44 below holds for $r = 2$. Combined with Theorem 4.7(b) and the count after Definition 4.5: for each of the 64 288 functions g of four variables that satisfy (N) and (D), $\mathrm{Aut}(X_F) \cong \mathbb{Z}^2$, and $\langle \sigma, F \rangle$ has index 1, 2 or 4 in it. For $g = \xi_3$ the graph Λ_M is not strongly connected for $2 \leq M \leq 5$.

Proposition 4.41 (computer-assisted, radius 2)**.** *For the rules* $\xi_0\xi_2$*,* $\xi_0\xi_3$ *and* $\xi_3 \oplus \xi_0\xi_1\xi_2$*, and for the ten functions* g *of Table 2, the hypotheses* (N)*,* (D)*,* (Λ) *and* (I) *hold. Hence* $\mathrm{Aut}(X_F) = \langle \sigma, F \rangle \cong \mathbb{Z}^2$ *for these thirteen rules. Here* (N) *and* (D) *are read off the algebraic normal form,* (Λ) *is the certificate* (T_P) *with* $P \leq 2$ *(Proposition 4.38), and* (I) *is a witness in* Per_n *with* $n \leq 4$ *(Lemma 4.28).*

In the column "extra" of Table 1, the extra automorphisms for $\mathrm{maj}(\xi_0, \xi_2, \xi_3)$ are the maps $\kappa\sigma^a F^b$ of radius at most R, as Theorem 4.7(c) predicts. The two rules of that table with (N)

TABLE 1. Radius-2 rules given by formulas. $J = J(g)$ ("–" if $\partial_0 g$ is constant); s.d. = self-dual; P is the least period with (T_P); the column $(\mathrm{I})_2$ gives a witness (n, b, s), which has $k(\mathcal{L}) = 1$ in every row; "strip" is the number of functions $\gamma \not\equiv 0$ of $\theta_0, \ldots, \theta_m$ that satisfy (20), for $m = 1$ and $m = 2$ (the count stops at 20); "extra" is the number of invertible block maps of radius at most R that commute with F and do not lie in $\langle \sigma, F \rangle$, for $R = 1$ and $R = 2$.

g	J	s.d.	P	$(\mathrm{I})_2$	strip	extra	status
$\xi_0\xi_2$	2	no	1	$(1,2,0)$	$0, 0$	0/0	Thm. 4.6 applies
$\xi_0\xi_3$	3	no	1	$(1,2,0)$	$0, 0$	0/0	applies
$\xi_3 \oplus \xi_0\xi_1\xi_2$ (bipermutive)	2	no	1	$(2,1,1)$	$0, 0$	0/0	applies
$\xi_0\xi_1$	1	no	1	$(1,2,0)$	$0, 0$	0/0	(N) fails
$\xi_1\xi_2$	–	no	1	$(1,2,0)$	$0, 0$	0/0	(N) fails
$\mathrm{maj}(\xi_0, \xi_2, \xi_3)$	3	yes	1	$(1,2,0)$	$1, 1$	3/7	(D) fails
$\xi_0 \oplus \xi_3$ (affine)	–	no	1	none, $n \le 5$	$15, \ge 20$	0/4	(N) fails; no witness, $n \le 5$
ξ_1 (affine)	–	yes	1	$(1,2,0)$	$\ge 20, \ge 20$	5/–	(N), (D) fail

TABLE 2. Ten further radius-2 rules to which Theorem 4.6 applies, given by their truth tables. None is self-dual. Columns as in Table 1; $k = k(\mathcal{L})$ for the witness shown. The last rule also has the witness $(3, 4, 0)$ with $k = 1$.

g	J	P	$(\mathrm{I})_2$	k	g	J	P	$(\mathrm{I})_2$	k
`ad44`	3	1	$(1,2,0)$	1	`5ff8`	3	1	$(4,2,2)$	1
`34ff`	3	2	$(1,4,0)$	1	`5980`	3	1	$(4,4,2)$	1
`3f7b`	3	2	$(1,4,0)$	1	`4f98`	3	1	$(2,1,1)$	1
`baec`	3	1	$(1,2,0)$	1	`e98e`	3	1	$(1,2,0)$	1
`fa9b`	3	1	$(4,6,2)$	1	`6152`	2	1	$(3,2,0)$	7

failing and g not affine have no extra automorphism of radius at most 2 and no non-constant solution of (20) on $\theta_0, \theta_1, \theta_2$.

Radius 3. Here $2r = 6$, the primes are $q = 2$ ($a_2 = 1$) and $q = 3$ ($a_3 = 0$), and the index divides 6.

Proposition 4.42 (computer-assisted, radius 3)**.** *For the six functions g of six variables in Table 3, the hypotheses* (N), (D), (Λ) *and* (I) *hold, so* $\mathrm{Aut}(X_F) = \langle \sigma, F \rangle \cong \mathbb{Z}^2$. *All six have $J(g) = 5$ and none is self-dual; every witness shown has $k(\mathcal{L}) = 1$.*

TABLE 3. Six radius-3 rules to which Theorem 4.6 applies, given by their truth tables. P is the least period with (T_P); the witnesses (n, b, s) for $(\mathrm{I})_2$ and $(\mathrm{I})_3$ have $k(\mathcal{L}) = 1$. The witness $(5, 50, 0)$ lies in $\mathbb{Z} \times 2\mathbb{Z}$, so it avoids $(1, 1)$ modulo $2\mathbb{Z}^2$.

g	P	$(\mathrm{I})_2$	$(\mathrm{I})_3$
`02f8aca0c96b3825`	1	$(1,4,0)$	$(2,3,0)$
`9207dbf3d68f59b4`	1	$(1,2,0)$	$(2,3,0)$
`6241805bcece3963`	2	$(1,4,0)$	$(2,3,0)$
`dfc519edc1a14412`	1	$(1,2,0)$	$(3,3,0)$
`822cd8c8f65243a3`	1	$(5,50,0)$	$(1,3,0)$
`8557d3586049b4bb`	1	$(4,4,2)$	$(1,3,0)$

4.12. **Two conjectures.** Hypothesis (N) is used only in Proposition 4.25, through the emptiness of the stuck region.

Conjecture 4.43. *Theorem 4.6 remains true with* (N) *replaced by the condition that g is not affine. This would follow from the conclusion of Proposition 4.25 for every non-affine g.*

The implication holds because (N) is used only through Proposition 4.25. The converse is not clear: a non-constant solution γ of (20) need not come from an automorphism, since the triangular map it defines need not be injective. For affine g the conclusion of Theorem 4.6 fails in general: rule 150 at $r = 1$ and $\xi_0 \oplus \xi_3$ at $r = 2$ (Table 1) are not self-dual and have automorphisms outside $\langle \sigma, F \rangle$.

Conjecture 4.44. (Λ) *holds for every non-affine g, of every radius.*

The conjecture holds for $r = 1$, with (T_P) for some $P \leq 2$ (Proposition 5.4, and its images under κ), and for $r = 2$, with $P \leq 4$ (Proposition 4.40). For affine g, (Λ) can fail (rule 90, and $g = \xi_3$ at $r = 2$) or hold (rule 150, and $g = \xi_1$, $\xi_0 \oplus \xi_3$ at $r = 2$). A uniform proof of (Λ), in the spirit of the stationary loop (9), would need a periodic history with odd holonomy for every non-affine g, and we do not have one. Together, Conjectures 4.43 and 4.44 would reduce Theorem 4.6 for non-affine, non-self-dual g to the index condition (I).

Remark 4.45. We do not treat the alphabets $(\mathbb{Z}/p)^2$ with p odd. The profile and maximum-principle steps use only the group structure of the differences, but the remaining steps use $p = 2$.

5. The elementary permutive rules

We now apply Theorem 4.6 with $r = 1$ to the elementary rules that are permutive in an outer variable and not affine. There are 16 such rules, in four conjugacy classes (Lemma 5.2).

Theorem 5.1. *Let f be one of the sixteen rules*

30, 86, 106, 120, 45, 101, 166, 180, 75, 89, 154, 210, 135, 149, 169, 225,

that is, an elementary rule that is permutive in its left or its right variable and is not affine. Then $\mathrm{Aut}(X_{F_f}) = \langle \sigma, F_f \rangle \cong \mathbb{Z}^2$: every invertible block map commuting with F_f equals $\sigma^a F_f^b$ for exactly one $(a, b) \in \mathbb{Z}^2$.

For rule 30 this is Theorem 3.1. The proof is by hand. The hypotheses of Theorem 4.6 are checked in Section 5.2: (N) and (D) are read off the rule, (Λ) is the certificate (T_1) or (T_2) on 4 or 16 column states, and (I) comes from the F-orbits of the 4 or 16 configurations of spatial period 1 or 2.

5.1. **The sixteen rules.** An elementary rule f is *left-permutive* if $f(a, b, c) = a \oplus g(b, c)$ for a Boolean function g, and *right-permutive* if $f(c, b, a)$ is left-permutive. For a left-permutive rule, F_f is the automaton (16) with $r = 1$ and $(\xi_0, \xi_1) = (b, c)$. Every g has a unique algebraic normal form

$$g(b, c) = g_{11}bc \oplus g_{10}b \oplus g_{01}c \oplus g_{00}, \qquad g_{11}, g_{10}, g_{01}, g_{00} \in \{0, 1\},$$

and f is affine if and only if $g_{11} = 0$. The Wolfram number of $f = a \oplus g(b, c)$ is $240 - 15N_g$, where $N_g = \sum_{b,c} g(b, c) 2^{2b+c}$. Indeed, the bits with $a = 0$ form N_g, and the bits with $a = 1$ form $16(15 - N_g)$.

Two conjugacies act on second-order automata, the Boolean conjugation and the reflection of [28, Section 2]. The cellwise complement κ of Section 4.1 conjugates F_f to $F_{\kappa f}$ with $\kappa f(a, b, c) = f(1 \oplus a, 1 \oplus b, 1 \oplus c)$, and it commutes with σ: by (1), $\kappa F_f \kappa(c)_i = \big(y_i,\ x_i \oplus f(1 \oplus y_{i-1}, 1 \oplus y_i, 1 \oplus y_{i+1})\big)$.

The reflection $\mathrm{rev}(c)_i = c_{-i}$ conjugates F_f to F_{f^r} with $f^r(a,b,c) = f(c,b,a)$ (Remark 4.8(2)), and $\mathrm{rev}\,\sigma\,\mathrm{rev} = \sigma^{-1}$. Both conjugacies preserve the radius of block maps. Hence $h \mapsto \theta h \theta^{-1}$, for $\theta \in \{\kappa, \mathrm{rev}\}$, maps $\mathrm{Aut}(X_{F_f})$ isomorphically onto the corresponding group for the conjugate rule, and it maps $\langle \sigma, F_f \rangle$ onto $\langle \sigma, F_{f'} \rangle$.

Lemma 5.2. *The elementary rules that are permutive in their left or right variable and are not affine are the* 16 *rules in Table 4. They form four classes under κ and reflection. Each class contains two left-permutive rules, which κ exchanges, and their two reflections, which are right-permutive and not left-permutive. On the coefficients of a left-permutive rule, κ acts by*

$$(g_{11}, g_{10}, g_{01}, g_{00}) \longmapsto (g_{11},\ g_{10}+g_{11},\ g_{01}+g_{11},\ g_{10}+g_{01}+g_{00}+g_{11}+1).$$

Proof. $f = a \oplus g$ is affine if and only if $g_{11} = 0$. It is also permutive in c if and only if $g(b,c) \oplus g(b, 1 \oplus c) = g_{11} b \oplus g_{01}$ is constant equal to 1, which forces $g_{11} = 0$. So the non-affine left-permutive rules are the eight with $g_{11} = 1$, and none of them is right-permutive. Their reflections are the eight non-affine right-permutive rules, and none of those is left-permutive. Next,

$$\kappa f(a,b,c) = (1 \oplus a) \oplus g(1 \oplus b, 1 \oplus c) = a \oplus \big(1 \oplus g_{11}(1+b)(1+c) \oplus g_{10}(1+b) \oplus g_{01}(1+c) \oplus g_{00}\big),$$

and expanding gives the displayed action. For $g_{11} = 1$ it is $(g_{10}, g_{01}, g_{00}) \mapsto (1+g_{10},\ 1+g_{01},\ g_{10}+g_{01}+g_{00})$. This map has no fixed point, and it commutes with reflection. So the classes are $\{f, \kappa f, f^r, (\kappa f)^r\}$, four of size 4. The numbers in the table come from $240 - 15N_g$ and from f^r. □

TABLE 4. The non-affine rules that are permutive in an outer variable. Each left-permutive rule $f = a \oplus g(b,c)$ is listed with g and (g_{10}, g_{01}, g_{00}) (here $g_{11} = 1$), next to its reflection. Rows in the same block form one class under κ and reflection.

left-permutive f	$g(b,c)$	(g_{10}, g_{01}, g_{00})	reflection f^r
30	$b \vee c$	$(1,1,0)$	86
120	$b \wedge c$	$(0,0,0)$	106
45	$b \vee \neg c$	$(0,1,1)$	101
180	$b \wedge \neg c$	$(1,0,0)$	166
75	$\neg b \vee c$	$(1,0,1)$	89
210	$\neg b \wedge c$	$(0,1,0)$	154
135	$\neg(b \wedge c)$	$(0,0,1)$	149
225	$\neg(b \vee c)$	$(1,1,1)$	169

Proposition 5.3. *For every elementary rule f that is permutive in its left or its right variable, affine or not, $(a,b) \mapsto \sigma^a F_f^b$ is an isomorphism $\mathbb{Z}^2 \to \langle \sigma, F_f \rangle$.*

Proof. Reflection conjugates $\sigma^a F_f^b$ to $\sigma^{-a} F_{f^r}^b$, so we may assume that f is left-permutive, and then this is Proposition 4.13 with $r = 1$. □

5.2. **The hypotheses for** 30**,** 45**,** 75 **and** 135**.** By the conjugacies, it suffices to prove Theorem 5.1 for $f \in \{30, 45, 75, 135\}$. In this subsection $f = a \oplus g(b,c)$ is one of these four rules, $F = F_f$, and $r = 1$.

(N) and (D).. Here $\partial_0\partial_1 g = g_{11} = 1$, so $J(g) = 1 = r$ and (N) holds. For $g_{11} = 1$,

$$g(1 \oplus b, 1 \oplus c) \oplus g(b, c) \oplus 1 = b \oplus c \oplus g_{10} \oplus g_{01},$$

which is not identically 0. So g is not self-dual, and (D) holds. Both statements hold for every non-affine left-permutive elementary rule.

(Λ). For $r = 1$ a column state of period P is a pair $s = (s_1, s_2)$ of functions $\mathbb{Z}/P \to \{0, 1\}$, and

$$\phi(s)_t = s_1(t-1) \oplus g\big(s_2(t), s_1(t)\big).$$

If $\pi(s) = \sum_t \phi(s)_t = 0$, the successors of s are the two states (s_2, z) with $z(t+1) = z(t) \oplus \phi(s)_t$.

Proposition 5.4. *The certificate* (T_1) *holds for* $f = 30$, 45 *and* 75, *and* (T_2) *holds for* $f = 135$. *Hence, for* $f \in \{30, 45, 75, 135\}$ *and every* $M \geq 0$, *the light-speed window graph* Λ_M *is strongly connected.*

Proof. The last statement follows from the first by Proposition 4.38(3). By the definition of S_∞, a state lies in S_∞ as soon as it lies on a cycle of the successor relation, or is a successor of a state of S_∞.

For $P = 1$ a state is a pair of bits (s_1, s_2), $\phi(s) = \pi(s) = s_1 \oplus g(s_2, s_1)$, and a state with $\phi(s) = 0$ has the successors $(s_2, 0)$ and $(s_2, 1)$. The values of ϕ are:

(s_1, s_2)	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
30: $s_1 \oplus (s_2 \vee s_1)$	0	1	0	0
45: $s_1 \oplus (s_2 \vee \neg s_1)$	1	1	1	0
75: $s_1 \oplus (\neg s_2 \vee s_1)$	1	0	0	0
135: $s_1 \oplus \neg(s_2 \wedge s_1)$	1	1	0	1

For 30, the state $(0,0)$ is its own successor, and its successor $(0,1)$ has $\pi = 1$. For 45, the state $(1,1)$ is its own successor, and its successor $(1,0)$ has $\pi = 1$. For 75, the states $(0,1)$ and $(1,0)$ are successors of each other, and the successor $(0,0)$ of $(1,0)$ has $\pi = 1$. So (T_1) holds in these three cases. (For 135 only $(1,0)$ has a successor, its successors $(0,0)$ and $(0,1)$ have none, and $S_\infty = \emptyset$: (T_1) fails.)

For $f = 135$ and $P = 2$, write a function $z\colon \mathbb{Z}/2 \to \{0,1\}$ as the word $z(0)z(1)$, and a state as (s_1, s_2). Here $g(b,c) = \neg(b \wedge c)$ and $\phi(s) = \big(s_1(1) \oplus g(s_2(0), s_1(0)),\ s_1(0) \oplus g(s_2(1), s_1(1))\big)$. Then:

s	$\phi(s)$	$\pi(s)$	successors
$(01,01)$	$(1\oplus 1,\ 0\oplus 0) = (0,0)$	0	$(01,00)$, $(01,11)$
$(01,11)$	$(1\oplus 1,\ 0\oplus 0) = (0,0)$	0	$(11,00)$, $(11,11)$
$(11,00)$	$(1\oplus 1,\ 1\oplus 1) = (0,0)$	0	$(00,00)$, $(00,11)$
$(00,00)$	$(0\oplus 1,\ 0\oplus 1) = (1,1)$	0	$(00,01)$, $(00,10)$
$(00,01)$	$(0\oplus 1,\ 0\oplus 1) = (1,1)$	0	$(01,01)$, $(01,10)$
$(01,00)$	$(1\oplus 1,\ 0\oplus 1) = (0,1)$	1	none

The first five states form a cycle $(01,01) \to (01,11) \to (11,00) \to (00,00) \to (00,01) \to (01,01)$, so they lie in S_∞, and so does the successor $(01,00)$ of $(01,01)$, which has $\pi = 1$. So (T_2) holds. □

By Proposition 4.38(2), each of these certificates is a periodic history with odd holonomy. For 30 it is the stationary loop (9).

(I).. For $r = 1$ the only prime dividing $2r$ is 2, and $a_2 = 1$. For $n \geq 1$, a configuration in Per_n is written by its x-row and y-row over one period, $(x_0 \cdots x_{n-1};\, y_0 \cdots y_{n-1})$.

Lemma 5.5. *For $f = 30, 135, 45, 75$, the set Per_n contains the following F-orbit O of even length ℓ, with $\sigma(O) = O$ and $\sigma = F^j$ on O, and O is the only F-orbit of length ℓ in Per_n:*

f	n	*orbit* O	ℓ	j	*other orbit lengths in* Per_n
30	1	$(0;1) \to (1;0)$	2	0	$1, 1$
135	1	$(0;0) \to (0;1) \to (1;1) \to (1;0)$	4	0	*none*
45	2	$(01;01) \to (01;10) \to (10;10) \to (10;01)$	4	2	$3, 8, 1$
75	2	$(00;01) \to (01;00) \to (00;10) \to (10;00)$	4	2	$3, 2, 2, 2, 1, 1, 1$

Proof. On Per_1, $F\big((x;y)\big) = (y;\, x \oplus f(y,y,y))$. For 30, $f(y,y,y) = 0$, which gives the transposition shown and two fixed points. For 135, $f(y,y,y) = 1$, which gives the 4-cycle shown. (For 45 and 75, $f(0,0,0) = 1$ and $f(1,1,1) = 0$, which gives the 3-cycle $(0;0) \to (0;1) \to (1;0)$ and the fixed point $(1;1)$; so Per_1 contains no orbit of even length.)

On Per_2, $F(x_0x_1; y_0y_1) = (y_0y_1;\, x_0 \oplus y_1 \oplus g(y_0,y_1),\ x_1 \oplus y_0 \oplus g(y_1,y_0))$. Applying this to the 16 points gives the following orbits. For 45 ($g = b \vee \neg c$): the 3-cycle $(00;00) \to (00;11) \to (11;00)$; the 8-cycle $(00;01) \to (01;11) \to (11;01) \to (01;00) \to (00;10) \to (10;11) \to (11;10) \to (10;00)$; the 4-cycle O; and the fixed point $(11;11)$. For 75 ($g = \neg b \vee c$): the 3-cycle $(00;00) \to (00;11) \to (11;00)$; the 4-cycle O; the 2-cycles $\{(01;10),(10;01)\}$, $\{(01;11),(11;01)\}$ and $\{(10;11),(11;10)\}$; and the fixed points $(01;01)$, $(10;10)$ and $(11;11)$. In both cases $\sigma(x_0x_1; y_0y_1) = (x_1x_0; y_1y_0)$ maps the first element of O to the third, so $\sigma = F^2$ on O. □

Lemma 5.6. *Condition* $(\mathrm{I})_2$ *holds for* $f \in \{30, 45, 75, 135\}$.

Proof. Let O, n, ℓ and j be as in Lemma 5.5, and let $\mathcal{L}$ be the stabiliser of the points of O. The orbit O is a single $\mathbb{Z}^2$-orbit, since $\sigma(O) = O$. Its points have least spatial period n, and for $c \in O$ we have $\sigma^a F^b c = F^{aj+b} c$, so

$$\mathcal{L} = \{(a,b) \in \mathbb{Z}^2 :\ \ell \mid aj + b\}.$$

For 30 and 135 ($n = 1$, $j = 0$) this is $\mathbb{Z} \times \ell\mathbb{Z}$ with $\ell = 2, 4$, and for 45 and 75 ($n = 2$, $j = 2$, $\ell = 4$) it is $\{(a,b) : 4 \mid 2a + b\} = \langle (2,0), (-1,2) \rangle$. In each case $\mathcal{L} \subseteq \mathbb{Z} \times 2\mathbb{Z}$, so $2\mathbb{Z}^2 + \mathcal{L} \subseteq \mathbb{Z} \times 2\mathbb{Z}$ does not contain $(a_2, 1) = (1, 1)$.

It remains to see that $k(\mathcal{L}) = 1$. Let O' be a $\mathbb{Z}^2$-orbit with stabiliser $\mathcal{L}$. Since $\mathcal{L} \cap (\mathbb{Z} \times \{0\}) = n\mathbb{Z} \times \{0\}$, its points have least period n (Lemma 4.3), and since $\mathcal{L} \cap (\{0\} \times \mathbb{Z}) = \{0\} \times \ell\mathbb{Z}$, each of them has F-period exactly ℓ. So O' contains an F-orbit of length ℓ in Per_n, which is O by Lemma 5.5. Hence $O' = O$, $k(\mathcal{L}) = 1$ is odd, and $\mathcal{L}$ is a witness for $(\mathrm{I})_2$. In the notation (n, b, s) of Section 4.11 these witnesses are $(1, 2, 0)$ for 30, $(1, 4, 0)$ for 135, and $(2, 2, 1)$ for 45 and 75. □

5.3. **Proof of Theorem 5.1.**

Proof of Theorem 5.1. By Lemma 5.2 and the conjugacies κ and rev, which map $\mathrm{Aut}(X_{F_f})$ onto $\mathrm{Aut}(X_{F_{f'}})$ and $\langle \sigma, F_f \rangle$ onto $\langle \sigma, F_{f'} \rangle$, it suffices to take $f \in \{30, 45, 75, 135\}$. For these rules (N) and (D) hold by the computation above, (Λ) by Proposition 5.4, and (I) by Lemma 5.6. So Theorem 4.6 with $r = 1$ gives that every invertible block map commuting with F_f equals $\sigma^a F_f^b$ for exactly one (a, b), and $\langle \sigma, F_f \rangle \cong \mathbb{Z}^2$. □

Remark 5.7 (what depends on the class). The proof of Theorem 4.6 is the proof of Theorem 3.1 with two steps replaced, and only these two depend on the class. The stationary loop (9) in the proof of Lemma 3.7 is replaced by a periodic history with odd holonomy, and the parity argument of Lemma 3.23, which uses $f(y,y,y) = 0$, is replaced by the orbit count of Lemma 4.28. For the class of 135, for instance, the inputs are the six column states of period 2 in the proof

of Proposition 5.4 and the 4-cycle of F on Per_1. Nonlinearity enters through $g_{11} = 1$ only, in Lemma 4.26: for $r = 1$ it is the identity $\partial_0 g(b, c) \oplus \partial_0 g(b \oplus \epsilon_1, c \oplus \epsilon_2) = g_{11}\epsilon_2$, which for rule 30 is fact (c) in the proof of Lemma 3.21.

5.4. **Affine permutive rules.**

Remark 5.8. For an affine left-permutive rule ($g_{11} = 0$), hypothesis (N) fails: $\partial_0 g$ is constant, the right sides of (21) and (22) vanish, and the descent of Section 4.6 no longer forces the correction term to be constant. This failure is genuine. For rule 150, let X act on $\mathbb{F}_2^{\mathbb{Z}}$ as σ, so that $F = F_{150}$ acts on column vectors $(x, y)^{\mathsf{T}}$ as the matrix

$$A = \begin{pmatrix} 0 & 1 \\ 1 & X + 1 + X^{-1} \end{pmatrix}$$

over the ring $\mathbb{F}_2[X, X^{-1}]$. The map $h = \mathrm{id} + \sigma F$, with matrix $I + XA$, commutes with F, because F is linear and commutes with σ. It is invertible, because $\det(I + XA) = 1 + X(X + 1 + X^{-1}) + X^2 = X$ is a unit of $\mathbb{F}_2[X, X^{-1}]$. And it does not lie in $\langle \sigma, F \rangle$, because $\det(\sigma^a F^b) = X^{2a}(\det A)^b = X^{2a}$ has even degree, while $\det h = X$ has odd degree. So $\mathrm{Aut}(X_{F_{150}}) \neq \langle \sigma, F_{150} \rangle$.

The front-offset part of the argument does not detect linearity: Theorem 4.18 holds for every left-permutive rule for which the graphs Λ_M are strongly connected. For most affine left-permutive elementary rules they are: a direct construction of Λ_M from the local rule (computer-assisted; Appendix A.3) shows that Λ_M is strongly connected for $M \leq 4$ for the rules 15, 60, 105, 150, 195 and 240, and not strongly connected for 90 (for $1 \leq M \leq 8$) and 165 (for $2 \leq M \leq 4$). So the failure of this step for rule 90 is a feature of that rule, not a reflection of its linearity. For the two-copy window graph of Appendix B, a direct construction of $\Gamma_{w,0}$ shows that the eventual image, of size 2^{3w+2}, is not strongly connected for rule 90 at $w = 1$ and $w = 2$, and for rule 150 at $w = 3$; for rule 150 it is strongly connected at $w = 1, 2$.

For cellular automata that are permutive in both outer variables, Moore and Boykett [21] and Salo and Törmä [26] show that the commutant is small, and that a bipermutive affine automaton commutes only with affine ones. These results do not apply to F_f directly: as a cellular automaton on the alphabet $\mathcal{A}$, F_f is not permutive in either neighbour, because $F_f(c)_i$ does not depend on the x-bits of $c_{i\pm1}$.

6. Rules that read two neighbours: the classes of 63 and 187

The rules of the classes $\{63, 119, 238, 252\}$ and $\{187, 207, 221, 243\}$ are not permutive in an outer variable, and the front method of Section 3 does not apply to them. They have another feature: the local rule ignores one of the two outer neighbours. We show that this alone makes the automorphism group as small as possible.

Theorem 6.1. *Let f be one of the eight rules*

$$63,\ 119,\ 238,\ 252, \qquad 187,\ 207,\ 221,\ 243,$$

and $F = F_f$. Then $\mathrm{Aut}(X_F) = \langle \sigma \rangle \times \langle F \rangle \cong \mathbb{Z}^2$: every invertible block map commuting with F equals $\sigma^a F^b$ for exactly one $(a, b) \in \mathbb{Z}^2$. There is no bound on the radius.

The proof has two parts. In the first (Sections 6.1–6.2) the rightmost difference between two configurations is shown to be invariant under F, every automorphism h is shown to move it by a constant $e(h)$, and h is reduced to $\sigma^{e(h)}h$, an automorphism that, like its inverse, reads only cells to its right. The constancy of $e(h)$ comes from a maximum principle and the strong connectivity of a window graph W_M, which we prove for every M by identifying the window graphs with the action of a self-similar group on the 4-ary tree (Section 6.3). In the second part (Section 6.4) the automorphisms h with $e(h) = 0$ are shown to be exactly the powers of F, by an induction

on the anticipation. A consequence is that F defines an automorphism of the one-sided full shift $\mathcal{A}^{\mathbb{N}}$ whose centralizer in $\mathrm{Aut}(\mathcal{A}^{\mathbb{N}})$ is generated by it (Corollary 6.22). The proof is by hand. It contains three finite checks, each set out in a table: the cell permutations commuting with F (Lemma 6.2), two section identities in the tree group (Lemma 6.11), and the level-1 groups generated by two permutations of four letters (Lemma 6.12). They are marked *(finite check)* where they occur.

6.1. **The half-radius form.** Throughout this section $\mathrm{Aut}(X_F)$ is identified with the group of invertible block maps commuting with F (Lemma 2.2). Let $\kappa(c)_i = (1 \oplus x_i, 1 \oplus y_i)$ be the cellwise complement and $\mathrm{rev}(c)_i = c_{-i}$ the reflection; both are involutions. By (1), $\kappa F_f \kappa = F_{f'}$ with $f'(a,b,c) = f(\neg a, \neg b, \neg c)$, and $\mathrm{rev}\, F_f\, \mathrm{rev} = F_{f'}$ with $f'(a,b,c) = f(c,b,a)$; moreover $\kappa\sigma\kappa = \sigma$ and $\mathrm{rev}\,\sigma\,\mathrm{rev} = \sigma^{-1}$. So conjugation by κ, and by rev, maps $\mathrm{Aut}(X_{F_f})$ onto $\mathrm{Aut}(X_{F_{f'}})$ and $\langle \sigma, F_f \rangle$ onto $\langle \sigma, F_{f'} \rangle$. Rule 63 is $\neg(a \wedge b)$, its reflection 119 is $\neg(b \wedge c)$, and κ maps 119 to $238 = b \vee c$, whose reflection is $252 = a \vee b$. Rule 187 is $\neg b \vee c$, its reflection 243 is $a \vee \neg b$, and κ maps 187 to $221 = b \vee \neg c$, whose reflection is $207 = \neg a \vee b$. So it suffices to prove Theorem 6.1 for $f = 119$ and $f = 187$. Both have the form

$$f(a,b,c) = g(b,c), \qquad g(y,\nu) = \begin{cases} \neg(y \wedge \nu) & (f = 119), \\ \neg y \vee \nu & (f = 187). \end{cases}$$

For a cell $z = (x, y)$ write $x(z) = x$, $y(z) = y$, and, for $\nu \in \{0,1\}$,

$$\phi_\nu(x,y) = \big(y,\ x \oplus g(y,\nu)\big), \qquad \psi_\nu(x,y) = \big(y \oplus g(x,\nu),\ x\big).$$

By (1) and (2),

$$F(c)_i = \phi_{y(c_{i+1})}(c_i), \qquad F^{-1}(d)_i = \psi_{x(d_{i+1})}(d_i). \tag{24}$$

Each ϕ_ν and each ψ_ν is a permutation of the four cells. Table 5 lists the values of ϕ_0 and ϕ_1, with the cells written as $z = 2x + y$; all the finite checks of this section are read off from it.

f			$z=0$	1	2	3	cycles
187	ϕ_0	$(g(y,0) = \neg y)$	1	2	0	3	(0 1 2)
	ϕ_1	$(g(y,1) = 1)$	1	3	0	2	(0 1 3 2)
119	ϕ_0	$(g(y,0) = 1)$	1	3	0	2	(0 1 3 2)
	ϕ_1	$(g(y,1) = \neg y)$	1	2	0	3	(0 1 2)

TABLE 5. The permutations $\phi_\nu(x,y) = (y,\ x \oplus g(y,\nu))$ of the cells $z = 2x + y$: the entry in column z is $\phi_\nu(z)$.

We use the following facts, all immediate from (24) and the definition of g.

(a) $F(c)_i$ depends only on c_i and $y(c_{i+1})$, and $F^{-1}(d)_i$ only on d_i and $x(d_{i+1})$.
(b) $x(F(c)_i) = y(c_i)$ and $y(F^{-1}(d)_i) = x(d_i)$.
(c) For a cell z, the map $\nu \mapsto \phi_\nu(z)$ is injective if and only if $y(z) = 1$, and $\nu \mapsto \psi_\nu(z)$ is injective if and only if $x(z) = 1$. (In both rules $g(0,\cdot) \equiv 1$ and $g(1,\cdot)$ is a bijection of $\{0,1\}$.)

Lemma 6.2 (finite check)**.** *For $f \in \{119, 187\}$ the only permutation π of $\mathcal{A}$ such that the cellwise map $c \mapsto (\pi(c_i))_i$ commutes with F is the identity.*

Proof. On a constant configuration $z^{\mathbb{Z}}$, F acts as the permutation $C(z) = \phi_{y(z)}(z)$ of $\mathcal{A}$. By Table 5, $C = (0\,1\,3\,2)$ for 187 and $C = (0\,1\,2)$ for 119. A cellwise π commuting with F maps constant configurations to constant configurations, so $\pi C = C\pi$. In $\mathrm{Sym}(\mathcal{A})$ the 4-cycles form a conjugacy class of 6 elements and the 3-cycles one of 8 elements, so the centralizer of C has order $24/6 = 4$ for 187 and $24/8 = 3$ for 119; in both cases it is $\langle C\rangle$. It remains to see that $\pi = C^j \neq \mathrm{id}$ does not commute with F. By (24), if π commutes with F, then

$$\pi\big(\phi_{y(c_1)}(c_0)\big) = \phi_{y(\pi c_1)}(\pi c_0) \qquad \text{for all cells } c_0, c_1.$$

Table 6 gives, for each $\pi = C^j \neq \mathrm{id}$, cells c_0, c_1 for which the two sides differ; each evaluation is read off from Table 5. □

f	π	(c_0, c_1)	$\pi(\phi_{y(c_1)}(c_0))$	$\phi_{y(\pi c_1)}(\pi c_0)$
187	$C = (0\,1\,3\,2)$	$(1,0)$	$\pi(\phi_0(1)) = \pi(2) = 0$	$\phi_{y(1)}(3) = \phi_1(3) = 2$
	$C^2 = (0\,3)(1\,2)$	$(0,1)$	$\pi(\phi_1(0)) = \pi(1) = 2$	$\phi_{y(2)}(3) = \phi_0(3) = 3$
	$C^3 = (0\,2\,3\,1)$	$(1,0)$	$\pi(\phi_0(1)) = \pi(2) = 3$	$\phi_{y(2)}(0) = \phi_0(0) = 1$
119	$C = (0\,1\,2)$	$(1,0)$	$\pi(\phi_0(1)) = \pi(3) = 3$	$\phi_{y(1)}(2) = \phi_1(2) = 0$
	$C^2 = (0\,2\,1)$	$(1,0)$	$\pi(\phi_0(1)) = \pi(3) = 3$	$\phi_{y(2)}(0) = \phi_0(0) = 1$

TABLE 6. Proof of Lemma 6.2: for each nontrivial power π of C, a pair of cells (c_0, c_1) at which $\pi \circ F \neq F \circ \pi$.

The exact right edge. A *pair* is an ordered pair $Q = (c, c')$ of distinct configurations that agree at all sufficiently large sites, and its *right edge* is $\lambda(Q) = \max\{i : c_i \neq c'_i\}$. As in Section 3, $hQ = (hc, hc')$, FQ and $\sigma^j Q$ are pairs whenever h is an invertible block map.

Lemma 6.3. *$\lambda(FQ) = \lambda(F^{-1}Q) = \lambda(Q)$ for every pair Q, and $\lambda(\sigma^j Q) = \lambda(Q) - j$.*

Proof. Let $p = \lambda(Q)$. For $i > p$, $F(c)_i$ and $F(c')_i$ depend only on cells at sites $\geq i > p$, where $c = c'$. At p, $F(c)_p = \phi_\nu(c_p)$ and $F(c')_p = \phi_\nu(c'_p)$ with the same $\nu = y(c_{p+1}) = y(c'_{p+1})$, and ϕ_ν is injective, so they differ. The same argument with ψ applies to F^{-1}. The last statement is clear. □

So the right edge of a difference is stationary, in both time directions and in every context. This replaces the light-speed front of the permutive rules.

6.2. **The offset.** In this subsection $h \in \mathrm{Aut}(X_F)$ and $R \geq 1$ bounds the radii of h and h^{-1}. For a pair Q with $\lambda(Q) = p$ put $E(Q) = \lambda(hQ) - p$.

Lemma 6.4. *Let $Q = (c, c')$ be a pair with $\lambda(Q) = p$.*

1. *$E(Q) \in [-R, R]$, and $E(Q)$ depends only on the restrictions of c and c' to $[p-2R, p+2R]$.*
2. *$E(\sigma^j Q) = E(Q)$, $E(FQ) = E(Q)$ and $E(c', c) = E(c, c')$.*

Proof. (1) hc and hc' agree at every site $i > p + R$. If they agreed at every site $i \geq p - R$, then $c_p = h^{-1}(hc)_p = h^{-1}(hc')_p = c'_p$, since h^{-1} has radius at most R. So $\lambda(hQ) \in [p - R, p + R]$, and it is the largest site in this interval at which hc and hc' differ. These cells depend only on c and c' on $[p - 2R, p + 2R]$. (2) By Lemma 6.3 and $hF = Fh$, $\lambda(hFQ) = \lambda(FhQ) = \lambda(hQ)$ and $\lambda(FQ) = \lambda(Q)$; the shift is similar, and λ is symmetric in the two configurations. □

Let $e_{\max}$ be the largest value of E over all pairs.

Lemma 6.5 (maximum principle)**.** *Let $Q = (c, c')$ be a pair with $\lambda(Q) = p$ and $E(Q) = e_{\max}$. If c'' agrees with c' at every site $i \geq p$, then (c, c'') is a pair with $\lambda(c, c'') = p$ and $E(c, c'') = e_{\max}$. The same holds with the roles of c and c' exchanged.*

Proof. Clearly $\lambda(c, c'') = p$. Let c^* agree with c'' on $[p - 2R, \infty)$ and with c' elsewhere. By Lemma 6.4(1), $E(c, c^*) = E(c, c'')$. The configurations c' and c^* differ at finitely many sites, all in $[p-2R, p-1]$. Change them one site at a time, $c' = c^0, c^1, \dots, c^n = c^*$, so that $Q_k = (c^{k-1}, c^k)$ is a pair with $\lambda(Q_k) = \ell_k \leq p - 1$. Then $\lambda(hQ_k) = \ell_k + E(Q_k) \leq p - 1 + e_{\max}$, so hc^{k-1} and hc^k agree at all sites $\geq p + e_{\max}$, for every k. Hence hc^* agrees with hc' at all sites $\geq p + e_{\max}$. Since hc and hc' differ at $p + e_{\max}$ and agree beyond it, so do hc and hc^*, that is, $E(c, c^*) = e_{\max}$. The last statement follows from $E(c', c) = E(c, c')$. □

Unlike Lemma 3.11, this is instantaneous: no time evolution is needed, because the right edge of a difference does not move.

Corollary 6.6. *Let $M = 2R$. Whether $E(Q) = e_{\max}$ for a pair $Q = (c, c')$ with $\lambda(Q) = p$ depends only on the word $c_p \cdots c_{p+M}$ and the cell c'_p.*

Proof. By Lemma 6.4(2) we may compare pairs $Q_1 = (c_1, c'_1)$ and $Q_2 = (c_2, c'_2)$ with the same right edge p, with $c_1 = c_2$ on $[p, p + M]$ and $(c'_1)_p = (c'_2)_p$, and suppose $E(Q_1) = e_{\max}$. Let c_3, c'_3 agree with c_1, c'_1 on $(-\infty, p + M]$ and with c_2 beyond. Then $Q_3 = (c_3, c'_3)$ is a pair with right edge p that agrees with Q_1 on $[p - 2R, p + 2R]$, so $E(Q_3) = e_{\max}$. Since $c_3 = c_2$ on $[p, \infty)$, Lemma 6.5 with the roles exchanged gives $E(c_2, c'_3) = e_{\max}$. Since $c'_3 = c'_2$ on $[p, \infty)$, Lemma 6.5 gives $E(c_2, c'_2) = e_{\max}$. □

Definition 6.7. Fix $M \geq 0$. For $\nu \in \{0, 1\}$ and a word $\omega = \omega_0 \cdots \omega_M \in \mathcal{A}^{M+1}$ let $F_\nu(\omega)$ be the word of $F(\tilde{c})$ on $[p, p + M]$, where $\tilde{c}$ has the word ω on $[p, p + M]$ and $y(\tilde{c}_{p+M+1}) = \nu$; by (24), $F_\nu(\omega)_k = \phi_{y(\omega_{k+1})}(\omega_k)$ for $k < M$ and $F_\nu(\omega)_M = \phi_\nu(\omega_M)$. The *window graph* Λ^0_M (the front does not move; this is not the light-speed graph Λ_M of Section 3.2) has vertex set $\mathcal{A}^{M+1}$ and the edges $\omega \to F_\nu(\omega)$, $\nu \in \{0, 1\}$. The *two-copy window graph* W_M has as vertices the pairs (ω, z) with $\omega \in \mathcal{A}^{M+1}$, $z \in \mathcal{A}$, $z \neq \omega_0$, and the edges

$$(\omega, z) \longrightarrow \big(F_\nu(\omega),\ \phi_{y(\omega_1)}(z)\big), \qquad \nu \in \{0, 1\},$$

with $\phi_{y(\omega_1)}$ replaced by ϕ_ν if $M = 0$.

A vertex (ω, z) of W_M records a pair (c, c') with right edge p: the word of c on $[p, p + M]$, which is also that of c' on $[p + 1, p + M]$, and the cell $c'_p = z$. An edge applies F to both, with the input ν entering from the right; the new second cell is $F(c')_p = \phi_{y(c_{p+1})}(z)$, which differs from $F(c)_p$.

Proposition 6.8. *Suppose that W_{2R} is strongly connected. Then E is constant: there is $e(h) \in [-R, R]$ with $\lambda(hQ) = \lambda(Q) + e(h)$ for every pair Q.*

Proof. Let $M = 2R$ and let $\mathcal{S}$ be the set of vertices (ω, z) of W_M such that $E(Q) = e_{\max}$ for some, equivalently (Corollary 6.6) every, pair $Q = (c, c')$ with right edge p, word ω of c on $[p, p + M]$ and $c'_p = z$. Every vertex is realised by some pair, so $\mathcal{S} \neq \emptyset$. If $(\omega, z) \in \mathcal{S}$ and $\nu \in \{0, 1\}$, choose such a pair with $y(c_{p+M+1}) = \nu$. Then FQ has right edge p (Lemma 6.3), $E(FQ) = E(Q) = e_{\max}$, and FQ realises the successor of (ω, z) along the edge labelled ν. So $\mathcal{S}$ is forward-closed, and by strong connectivity it is everything. Put $e(h) = e_{\max}$. □

Corollary 6.9. *Suppose that W_M is strongly connected for every M. Then:*

(1) *$e\colon \mathrm{Aut}(X_F) \to \mathbb{Z}$ is a homomorphism, with $e(\sigma) = -1$ and $e(F) = 0$, and $e(h)$ does not depend on R;*

(2) *if* $e(h) = 0$*, then there are* $A \geq 0$ *and* $K\colon \mathcal{A}^{A+1} \to \mathcal{A}$ *with* $h(c)_i = K(c_i, \dots, c_{i+A})$ *for all* c *and* i*, and* K *is a bijection in its first argument when the others are fixed; the same holds for* h^{-1}*;*
(3) $\mathrm{Aut}(X_F) = \langle\sigma\rangle \times \ker e$*, the isomorphism being* $h \mapsto (\sigma^{-e(h)}, \sigma^{e(h)}h)$*.*

Proof. (1) $\lambda(h_1h_2Q) = \lambda(h_2Q) + e(h_1) = \lambda(Q) + e(h_2) + e(h_1)$; the values on σ and F are Lemma 6.3. (2) Let $e(h) = 0$ and let R bound the radii of h and h^{-1}. If c and c' differ only at sites $< i$, then $\lambda(hc, hc') = \lambda(c, c') < i$, so $h(c)_i = h(c')_i$: the cell $h(c)_i$ does not depend on the cells left of i, and so $h(c)_i = K(c_{[i,i+A]})$ with $A \leq R$. If c and c' differ only at i, then $\lambda(hc, hc') = i$, so $h(c)_i \neq h(c')_i$; hence K is injective, so bijective, in its first argument. Since $e(h^{-1}) = -e(h) = 0$, the same holds for h^{-1}. (3) σ commutes with every block map, $e(\sigma^a h) = e(h) - a$, and $\langle\sigma\rangle \cap \ker e = \{\mathrm{id}\}$. □

For $h \in \ker e$ the least A in (2) is the *anticipation* $A(h)$. If $A(h) = 0$, then h is a cellwise permutation commuting with F, so $h = \mathrm{id}$ by Lemma 6.2.

6.3. **The window graphs as a self-similar group.** We write words of Λ^0_M backwards: the word $\omega_0 \cdots \omega_M$ on $[p, p+M]$ is the vertex $\bar\omega = \omega_M\omega_{M-1}\cdots\omega_0$ of level $M+1$ of the rooted 4-ary tree $\mathcal{A}^*$, so that the first letter is the cell next to the incoming bit. Let s_0 and s_1 be the maps of $\mathcal{A}^*$ to itself given by

$$s_\nu(xw) = \phi_\nu(x)\, s_{y(x)}(w) \qquad (x \in \mathcal{A},\ w \in \mathcal{A}^*),$$

so that $s_\nu(\bar\omega) = \overline{F_\nu(\omega)}$ by Definition 6.7. These are the maps computed by the two states of the Mealy automaton of Figure 4: in the state s_ν the automaton reads a letter x, writes $\phi_\nu(x)$ and passes to the state $s_{y(x)}$. Each s_ν is a bijection on every level, and we put $G = \langle s_0, s_1\rangle$. We use the standard language of self-similar groups [22]. For $t \in G$ and a letter x there are $t(x) \in \mathcal{A}$ and $t|_x \in G$, the *section* of t at x, with $t(xw) = t(x)\, t|_x(w)$ for all $w \in \mathcal{A}^*$; for a word $u = x_1 \cdots x_m$ the section at u is $t|_u = (\cdots(t|_{x_1})|_{x_2}\cdots)|_{x_m}$, so that $t(uw) = t(u)\, t|_u(w)$. The sections of the generators do not depend on the generator: $s_\nu|_x = s_{y(x)}$. Sections satisfy the product rule $(tt')|_x = t|_{t'(x)}\, t'|_x$ and $t^{-1}|_x = (t|_{t^{-1}(x)})^{-1}$. The *level-*1 *image* of a subgroup of G is its image in $\mathrm{Sym}(\mathcal{A})$ under $t \mapsto (x \mapsto t(x))$; that of s_ν is ϕ_ν.

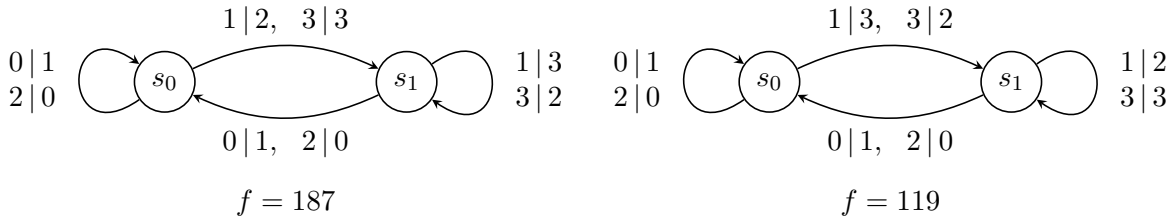


FIGURE 4. The Mealy automata defining s_0 and s_1. An arrow from s_ν labelled $x \mid z$ means that in the state s_ν the letter x is replaced by $z = \phi_\nu(x)$ (Table 5); the arrow ends at $s_{y(x)}$, the state in which the next letter is read. The letters are the cells $2x + y$, so the arrows into s_0 carry the inputs $0, 2$ and those into s_1 the inputs $1, 3$.

Since the edges of Λ^0_M are the two permutations s_0, s_1 of the finite level $M+1$, and the inverse of a permutation of a finite set is a positive power of it, Λ^0_M is strongly connected if and only if G acts transitively on level $M+1$. Likewise, by Definition 6.7, a vertex (ω, z) of W_M is the ordered pair of distinct *siblings* $\bar\omega = u\,\omega_0$ and $u\,z$ of level $M+1$, with common parent $u = \omega_M \cdots \omega_1$ of level M, and an edge applies the same generator to both. Hence:

Lemma 6.10. *W_M is strongly connected if and only if G acts transitively on ordered pairs of distinct siblings of level $M+1$. This holds if G is transitive on level M and, for one vertex u of level M, the level-1 image of $\{t|_u : t \in G,\ t(u) = u\}$ is 2-transitive on $\mathcal{A}$.*

Proof. The first statement is the remark above. For the second, given ordered sibling pairs (vx, vz) and $(v'x', v'z')$, move v and v' to u by elements of G, and then (x, z) to (x', z') by an element of the stabilizer of u, which acts on the children of u through its sections at u. □

Let $H = \langle s_0^2, s_1\rangle \le G$, and let $x_* = 1$ for $f = 187$ and $x_* = 3$ for $f = 119$. Words in $s_0^{\pm1}, s_1^{\pm1}$ are composed from right to left.

Lemma 6.11 (finite check: section identities)**.** *The following elements of H fix the letter x_* and have the stated sections at x_*:*

f	*element*	*section at x_**
187, $x_* = 1$	$t_1 = s_1 s_0^{-2} s_1^2$	s_1
	$t_2 = s_0^2 s_1 s_0^{-2} s_1$	s_0^2
119, $x_* = 3$	$t_1 = s_1$	s_1
	$t_2 = s_0^{-2} s_1^3 s_0^2$	$s_0^2 s_1$

Proof. All four elements are words in $s_0^{\pm2}$ and $s_1^{\pm1}$, so they lie in H. By the rules for sections, $s_\nu|_x = s_{y(x)}$, and $s_\nu^{-1}|_x = (s_{y(x')})^{-1}$ with $x' = \phi_\nu^{-1}(x)$. For $f = 119$ and $t_1 = s_1$: $\phi_1(3) = 3$ and $s_1|_3 = s_{y(3)} = s_1$. For the other three elements, Table 7 follows the letter x_* through the word, from the rightmost generator to the leftmost. Each row gives the generator applied, the letter before and after it (read off from Table 5), and the section of that generator at the letter before it. In each case the last letter is x_*. By the product rule, the section of the word at x_* is the product of the sections in the table, taken from the last row on the left to the first row on the right; the last line of each block gives this product and its reduction. □

187: $t_1 = s_1 s_0^{-2} s_1^2$		
step	letter	section
s_1	$1 \mapsto 3$	s_1
s_1	$3 \mapsto 2$	s_1
s_0^{-1}	$2 \mapsto 1$	s_1^{-1}
s_0^{-1}	$1 \mapsto 0$	s_0^{-1}
s_1	$0 \mapsto 1$	s_0
$s_0\, s_0^{-1}\, s_1^{-1}\, s_1\, s_1 = s_1$		

187: $t_2 = s_0^2 s_1 s_0^{-2} s_1$		
step	letter	section
s_1	$1 \mapsto 3$	s_1
s_0^{-1}	$3 \mapsto 3$	s_1^{-1}
s_0^{-1}	$3 \mapsto 3$	s_1^{-1}
s_1	$3 \mapsto 2$	s_1
s_0	$2 \mapsto 0$	s_0
s_0	$0 \mapsto 1$	s_0
$s_0\, s_0\, s_1\, s_1^{-1}\, s_1^{-1}\, s_1 = s_0^2$		

119: $t_2 = s_0^{-2} s_1^3 s_0^2$		
step	letter	section
s_0	$3 \mapsto 2$	s_1
s_0	$2 \mapsto 0$	s_0
s_1	$0 \mapsto 1$	s_0
s_1	$1 \mapsto 2$	s_1
s_1	$2 \mapsto 0$	s_0
s_0^{-1}	$0 \mapsto 2$	s_0^{-1}
s_0^{-1}	$2 \mapsto 3$	s_1^{-1}
$s_1^{-1} s_0^{-1}\, s_0\, s_1\, s_0\, s_0\, s_1 = s_0^2 s_1$		

TABLE 7. Proof of Lemma 6.11. The generators of each word are applied from right to left, starting at the letter x_* ($x_* = 1$ for 187, $x_* = 3$ for 119). A step s_ν maps the letter x to $\phi_\nu(x)$ with section $s_{y(x)}$; a step s_ν^{-1} maps x to $x' = \phi_\nu^{-1}(x)$ with section $s_{y(x')}^{-1}$.

Lemma 6.12 (finite check: level-1 groups)**.** *The level-1 image of H is $\langle \phi_0^2, \phi_1\rangle$, which is* $\mathrm{Sym}(\mathcal{A}) \cong S_4$ *for* 187 *and the alternating group A_4 for* 119*. Both are transitive and 2-transitive on $\mathcal{A}$.*

Proof. The level-1 images of s_0^2 and s_1 are ϕ_0^2 and ϕ_1; they are listed in Table 8. For 187, ϕ_0^2 is a 3-cycle and ϕ_1 a 4-cycle, so the order of the group they generate is divisible by 12, and the group contains an odd permutation; so it is S_4. For 119, ϕ_0^2 and ϕ_1 are even. Conjugating $\phi_0^2 = (0\,3)(1\,2)$ by ϕ_1 gives $(1\,3)(0\,2)$, and the product of the two is $(0\,1)(2\,3)$, so the group contains the Klein four-group and a 3-cycle; its order is divisible by 12, and it is A_4. Both S_4 and A_4 are 2-transitive on four points: they are transitive, and the stabilizer of a point contains a 3-cycle, which permutes the other three points cyclically. □

f	ϕ_0^2	ϕ_1	orders	signs	$\langle \phi_0^2, \phi_1 \rangle$
187	$(0\,2\,1)$	$(0\,1\,3\,2)$	3, 4	even, odd	S_4
119	$(0\,3)(1\,2)$	$(0\,1\,2)$	2, 3	even, even	A_4

TABLE 8. Proof of Lemma 6.12: the level-1 images of the generators s_0^2, s_1 of H, from Table 5.

Proposition 6.13. *For $f \in \{119, 187\}$:*

1. *H is self-replicating at x_*: for every $t \in H$ there is $t' \in H$ with $t'(x_*) = x_*$ and $t'|_{x_*} = t$; consequently, for every $m \geq 0$ and $t \in H$ there is $t' \in H$ with $t'(x_*^m) = x_*^m$ and $t'|_{x_*^m} = t$;*
2. *H, and hence G, acts transitively on every level, so Λ_M^0 is strongly connected for every M;*
3. *W_M is strongly connected for every $M \geq 0$.*

Proof. (1) On the stabilizer of x_* in H, the map $t \mapsto t|_{x_*}$ is a homomorphism, by the product rule for sections. By Lemma 6.11 its image contains s_1 and s_0^2 (for 119, $s_0^2 = (s_0^2 s_1)s_1^{-1}$), hence all of H. For $m \geq 1$ argue by induction: given t, choose $t' \in H$ fixing x_* with $t'|_{x_*} = t$, and $t'' \in H$ fixing x_*^m with $t''|_{x_*^m} = t'$; then t'' fixes x_*^{m+1} and $t''|_{x_*^{m+1}} = t'|_{x_*} = t$. (2) By induction on the level n. For $n = 1$ this is Lemma 6.12. Given words xv and $x'v'$ of level $n+1$, choose $t \in H$ with $t(x) = x_*$ and $t' \in H$ with $t'(x') = x_*$, so that the words become $x_* w$ and $x_* w'$; by (1) and the induction hypothesis some element of H fixing x_* has a section mapping w to w'. (3) By (2), G is transitive on level M. By (1), the sections at $u = x_*^M$ of the elements of H fixing u form all of H, whose level-1 image is 2-transitive by Lemma 6.12. Lemma 6.10 applies. □

In the terminology of [22], (1) and (2) say that H is self-replicating and level-transitive. For 119 the level-1 image of G is S_4, since ϕ_0 is a 4-cycle, while that of H is A_4; by Lemma 6.12 the smaller group suffices.

6.4. **The kernel of the offset.** By Proposition 6.13(3) and Corollary 6.9, $\mathrm{Aut}(X_F) = \langle \sigma \rangle \times \ker e$, and every $h \in \ker e$ reads only cells to its right: $h(c)_i = K(c_{[i,i+A]})$ with $A = A(h)$ and K bijective in its first argument. We show $\ker e = \langle F \rangle$. Throughout, h, h_1, h_2 denote elements of $\ker e$ with local rules K, K_1, K_2, and $F^{\pm 1} \in \ker e$ by Corollary 6.9(1).

Tops. Let $A = A(h) \geq 1$. For $\epsilon = (\epsilon_0, \ldots, \epsilon_{A-1}) \in \mathcal{A}^A$ let $\vartheta_\epsilon \colon \mathcal{A} \to \mathcal{A}$, $\vartheta_\epsilon(z) = K(\epsilon_0, \ldots, \epsilon_{A-1}, z)$, be the dependence of K on its last argument when the first A arguments are ϵ. Since A is minimal, some ϑ_ϵ is not constant.

Lemma 6.14 (diagonal top)**.** *Let $A(h) \geq 1$. For every $\epsilon \in \mathcal{A}^A$ there are maps $\alpha_\epsilon, \beta_\epsilon \colon \{0,1\} \to \{0,1\}$ with $\vartheta_\epsilon(z) = \big(\alpha_\epsilon(x(z)),\, \beta_\epsilon(y(z))\big)$.*

Proof. Let c have $c_{[1,A]} = \epsilon$, that is $c_{1+j} = \epsilon_j$, and $c_{A+1} = z$; the cell c_0 is chosen below. By (24),

$$(Fh)(c)_0 = \phi_{y(h(c)_1)}\big(h(c)_0\big) = \phi_{y(\vartheta_\epsilon(z))}\big(K(c_0, \epsilon)\big).$$

On the other side, $(hF)(c)_0 = K(F(c)_{[0,A]})$, and by (a) the cells $F(c)_j$ with $j < A$ do not involve z, while $F(c)_A = \phi_{y(z)}(c_A)$ involves z only through $y(z)$. Since K is bijective in its first argument, we may choose c_0 with $y(K(c_0,\epsilon)) = 1$. Then $\nu \mapsto \phi_\nu(K(c_0,\epsilon))$ is injective by (c), and $hF = Fh$ shows that $y(\vartheta_\epsilon(z))$ is a function of $y(z)$. For the x-bit use $hF^{-1} = F^{-1}h$: $(F^{-1}h)(c)_0 = \psi_{x(\vartheta_\epsilon(z))}(K(c_0,\epsilon))$, while $(hF^{-1})(c)_0$ involves z only through $x(z)$ by (a); choosing c_0 with $x(K(c_0,\epsilon)) = 1$ and using (c) shows that $x(\vartheta_\epsilon(z))$ is a function of $x(z)$. □

Put $\mathcal{X}_h = \{\epsilon : \alpha_\epsilon \text{ is not constant}\}$ and $\mathcal{Y}_h = \{\epsilon : \beta_\epsilon \text{ is not constant}\}$. By minimality of A, $\mathcal{X}_h \cup \mathcal{Y}_h \neq \emptyset$. Call h of *y-type* if $\mathcal{Y}_h \neq \emptyset = \mathcal{X}_h$, of *x-type* if $\mathcal{X}_h \neq \emptyset = \mathcal{Y}_h$, and of *both types* if both are nonempty. A nonconstant map $\{0,1\} \to \{0,1\}$ is a bijection. By (24), F has anticipation 1 and $\vartheta_\epsilon(z) = \phi_{y(z)}(\epsilon_0)$, whose x-bit $y(\epsilon_0)$ is constant; so F is of y-type, with $\mathcal{Y}_F = \{\epsilon_0 : y(\epsilon_0) = 1\}$ by (c). Likewise F^{-1} is of x-type with anticipation 1.

Lemma 6.15. (1) *If h is of x-type, then $A(hF) \le A(h)$, and if equality holds, hF is of y-type.*
(2) *If h is of y-type, then $A(hF^{-1}) \le A(h)$, and if equality holds, hF^{-1} is of x-type.*

Proof. (1) Let $A = A(h)$. $(hF)(c)_0 = K(F(c)_{[0,A]})$, and since h is of x-type, K reads its last argument $F(c)_A$ only through $x(F(c)_A) = y(c_A)$, by (b). The cells $F(c)_j$, $j < A$, read $c_{[0,A]}$, and c_A only through $y(c_A)$ (in $F(c)_{A-1} = \phi_{y(c_A)}(c_{A-1})$). So $(hF)(c)_0$ depends only on $c_{[0,A]}$, and on c_A only through $y(c_A)$. If $A(hF) = A$, its tops therefore do not depend on $x(z)$, so by Lemma 6.14 every α_ϵ is constant, and hF is of y-type. (2) is the mirror argument: K reads $F^{-1}(d)_A$ only through $y(F^{-1}(d)_A) = x(d_A)$, and $F^{-1}(d)_{A-1} = \psi_{x(d_A)}(d_{A-1})$. □

Lemma 6.16. *Let $A(h_1), A(h_2) \ge 1$.*
(1) *If $\mathcal{Y}_{h_1}$ and $\mathcal{Y}_{h_2}$ are nonempty, then $A(h_1h_2) = A(h_1) + A(h_2)$ and $\mathcal{Y}_{h_1h_2} \neq \emptyset$; if h_1 and h_2 are of y-type, so is h_1h_2.*
(2) *The same holds with $\mathcal{X}$ in place of $\mathcal{Y}$ and x-type in place of y-type.*

Proof. Write $A_j = A(h_j)$ and $A = A_1 + A_2$. $(h_1h_2)(c)_0 = K_1(h_2(c)_{[0,A_1]})$ reads only $c_{[0,A]}$, and of the cells $h_2(c)_j$, $0 \le j \le A_1$, only $h_2(c)_{A_1} = K_2(c_{[A_1,A]})$ involves c_A. Fix $c_{[0,A)}$ and let $z = c_A$ vary. Then $h_2(c)_{[0,A_1)} = \epsilon'$ is fixed, $h_2(c)_{A_1} = \vartheta^{(2)}_{\epsilon''}(z)$ with $\epsilon'' = c_{[A_1,A)}$, and the top of h_1h_2 at $c_{[0,A)}$ is $\vartheta^{(1)}_{\epsilon'} \circ \vartheta^{(2)}_{\epsilon''}$, whose x-part is $\alpha^{(1)}_{\epsilon'} \circ \alpha^{(2)}_{\epsilon''}$ and whose y-part is $\beta^{(1)}_{\epsilon'} \circ \beta^{(2)}_{\epsilon''}$, by Lemma 6.14. Choose $\epsilon'' \in \mathcal{Y}_{h_2}$. As $c_{[0,A_1)}$ varies with $c_{[A_1,A)} = \epsilon''$ fixed, $h_2(c)_{[0,A_1)}$ takes every value in $\mathcal{A}^{A_1}$: solve for $c_{A_1-1}, c_{A_1-2}, \dots, c_0$ in turn, using that K_2 is bijective in its first argument. So we may choose $\epsilon' \in \mathcal{Y}_{h_1}$. Then the y-part of the top is a composite of two bijections, so h_1h_2 depends on c_A, $A(h_1h_2) = A$, and $\mathcal{Y}_{h_1h_2} \neq \emptyset$. If both are of y-type, every $\alpha^{(j)}$ is constant, so every x-part of a top of h_1h_2 is constant and h_1h_2 is of y-type. (2) is identical. □

By Lemma 6.16 and induction, F^b is of y-type with $A(F^b) = b$ and F^{-b} is of x-type with $A(F^{-b}) = b$, for $b \ge 1$. In particular F has infinite order.

Lemma 6.17 (no both-type)**.** *No $h \in \ker e$ is of both types. Hence every $h \in \ker e$ with $A(h) \ge 1$ is of x-type or of y-type.*

Proof. Suppose h is of both types, so $A(h) \ge 1$. If $A(h^{-1}) = 0$, then h^{-1} is a cellwise permutation commuting with F, so $h^{-1} = \mathrm{id}$ by Lemma 6.2, contradicting $A(h) \ge 1$. Otherwise $\mathcal{X}_{h^{-1}}$ or $\mathcal{Y}_{h^{-1}}$ is nonempty, and Lemma 6.16, applied to $h_1 = h$ and $h_2 = h^{-1}$ with $\mathcal{X}$ or $\mathcal{Y}$ accordingly, gives $A(\mathrm{id}) = A(h) + A(h^{-1}) \ge 2$, whereas $A(\mathrm{id}) = 0$. □

Lemma 6.18 (no half step)**.** *There is no $h \in \ker e$ of y-type with $A(h) = A \ge 1$ such that $h' = hF^{-1}$ is of x-type with $A(h') = A$.*

Proof. Suppose there is. By Lemma 6.16, h^2 is of y-type with $A(h^2) = 2A$, and $h'^2 = h^2F^{-2}$ is of x-type with $A(h'^2) = 2A$. Let $\hat{h} = h^2F^{-1} = h'^2F$. Lemma 6.15(2) applied to h^2 gives $A(\hat{h}) \le 2A$, with $\hat{h}$ of x-type in case of equality; Lemma 6.15(1) applied to h'^2 gives $\hat{h}$ of y-type in case of equality. A map is not of both x-type and y-type, so $A(\hat{h}) \le 2A - 1$. We consider the three possibilities left by Lemma 6.17.

- $A(\hat{h}) = 0$. Then $\hat{h}$ is a cellwise permutation commuting with F, so $\hat{h} = \mathrm{id}$ by Lemma 6.2, and $h^2 = F$. But $A(F) = 1 < 2 \le 2A = A(h^2)$.
- $\hat{h}$ is of y-type. By Lemma 6.15(2), $A(h'^2) = A(\hat{h}F^{-1}) \le A(\hat{h}) < 2A$, a contradiction.
- $\hat{h}$ is of x-type. By Lemma 6.15(1), $A(h^2) = A(\hat{h}F) \le A(\hat{h}) < 2A$, a contradiction. □

Theorem 6.19. *For $f \in \{119, 187\}$, $\ker e = \langle F \rangle$, and $A(F^b) = |b|$.*

Proof. We show by strong induction on $A = A(h)$ that every $h \in \ker e$ is a power of F. If $A = 0$, then $h = \mathrm{id}$ by Lemma 6.2. Let $A \ge 1$; by Lemma 6.17, h is of y-type or of x-type.

h of y-type. Let $h' = hF^{-1} \in \ker e$. By Lemma 6.15(2), $A(h') \le A$, and if $A(h') = A$ then h' is of x-type, which Lemma 6.18 excludes. So $A(h') < A$, by induction $h' = F^{b'}$, and $h = F^{b'+1}$.

h of x-type. Let $h'' = hF \in \ker e$. By Lemma 6.15(1), $A(h'') \le A$, and if $A(h'') = A$ then h'' is of y-type with $h''F^{-1} = h$ of x-type and the same anticipation, which Lemma 6.18, applied to h'', excludes. So $A(h'') < A$, $h'' = F^{b''}$ by induction, and $h = F^{b''-1}$.

The formula $A(F^b) = |b|$ is the statement following Lemma 6.16. □

Proof of Theorem 6.1. By the conjugacies of Section 6.1 it suffices to take $f \in \{119, 187\}$. By Proposition 6.13(3), W_M is strongly connected for every M, so Corollary 6.9 gives $\mathrm{Aut}(X_F) = \langle \sigma \rangle \times \ker e$, and $\ker e = \langle F \rangle$ by Theorem 6.19. If $\sigma^a F^b = \mathrm{id}$, then $a = -e(\sigma^a F^b) = 0$ and $F^b = \mathrm{id}$, so $b = 0$ since $A(F^b) = |b|$. Lemma 2.2 translates this into a statement about the automorphism group of the spacetime subshift. □

6.5. **One-sided automorphisms.** The elements of $\ker e$ and their inverses read only cells to their right. Such maps are the automorphisms of the one-sided full shift, and Theorem 6.19 can be stated in these terms.

Definition 6.20. A block map h of $\mathcal{A}^{\mathbb{Z}}$ is *right-sided* if there are $A \ge 0$ and $K\colon \mathcal{A}^{A+1} \to \mathcal{A}$ with $h(c)_i = K(c_i, \dots, c_{i+A})$ for all c and i. A *one-sided automorphism* of $\mathcal{A}^{\mathbb{Z}}$ is an invertible block map h such that both h and h^{-1} are right-sided.

A composite of right-sided block maps is right-sided, so the one-sided automorphisms form a subgroup of $\mathrm{Aut}(\mathcal{A}^{\mathbb{Z}})$. The shift σ is right-sided, but σ^{-1} is not, so σ is not a one-sided automorphism. Let $\mathbb{N} = \{0, 1, 2, \dots\}$, let σ_+ be the shift on $\mathcal{A}^{\mathbb{N}}$, $\sigma_+(c)_i = c_{i+1}$, and let $\mathrm{Aut}(\mathcal{A}^{\mathbb{N}})$ be the group of homeomorphisms of $\mathcal{A}^{\mathbb{N}}$ commuting with σ_+; the automorphism groups of one-sided shifts of finite type are studied in [3]. A right-sided block map h with local rule K defines a continuous map $h_+\colon \mathcal{A}^{\mathbb{N}} \to \mathcal{A}^{\mathbb{N}}$, $h_+(c)_i = K(c_i, \dots, c_{i+A})$, which commutes with σ_+ and satisfies

$$h_+\big(c|_{\mathbb{N}}\big) = h(c)|_{\mathbb{N}} \qquad (c \in \mathcal{A}^{\mathbb{Z}}). \tag{25}$$

Lemma 6.21. (1) *If h and h' are right-sided block maps, then $(hh')_+ = h_+h'_+$, and $h_+ = h'_+$ implies $h = h'$.*

(2) *The map $h \mapsto h_+$ is an isomorphism from the group of one-sided automorphisms of $\mathcal{A}^{\mathbb{Z}}$ onto $\mathrm{Aut}(\mathcal{A}^{\mathbb{N}})$.*

(3) *If h is a one-sided automorphism and $h(c)_i = K(c_i, \dots, c_{i+A})$, then K is a bijection in its first argument when the others are fixed.*

Proof. (1) Every element of $\mathcal{A}^{\mathbb{N}}$ is $c|_{\mathbb{N}}$ for some $c \in \mathcal{A}^{\mathbb{Z}}$, and by (25), applied to h', to h and to hh',

$$h_+\big(h'_+(c|_{\mathbb{N}})\big) = h_+\big(h'(c)|_{\mathbb{N}}\big) = h(h'(c))|_{\mathbb{N}} = (hh')_+\big(c|_{\mathbb{N}}\big).$$

If $h_+ = h'_+$, then for every $c \in \mathcal{A}^{\mathbb{Z}}$ and $i \in \mathbb{Z}$, since h and h' commute with σ,

$$h(c)_i = h(\sigma^i c)_0 = h_+\big((\sigma^i c)|_{\mathbb{N}}\big)_0 = h'_+\big((\sigma^i c)|_{\mathbb{N}}\big)_0 = h'(c)_i.$$

(2) Let h be a one-sided automorphism. By (1), $h_+(h^{-1})_+ = (h^{-1})_+h_+ = \mathrm{id}_+$, and id_+ is the identity of $\mathcal{A}^{\mathbb{N}}$. So h_+ is a continuous bijection of the compact space $\mathcal{A}^{\mathbb{N}}$ commuting with σ_+, that is, $h_+ \in \mathrm{Aut}(\mathcal{A}^{\mathbb{N}})$, and $h \mapsto h_+$ is an injective homomorphism by (1). To see that it is surjective, let $\chi \in \mathrm{Aut}(\mathcal{A}^{\mathbb{N}})$. The map $c \mapsto \chi(c)_0$ is continuous on the compact space $\mathcal{A}^{\mathbb{N}}$, so it depends only on $c_0, \dots, c_A$ for some A, say $\chi(c)_0 = K(c_0, \dots, c_A)$; then $\chi(c)_i = \chi(\sigma_+^i c)_0 = K(c_i, \dots, c_{i+A})$. In the same way $\chi^{-1}(c)_i = K'(c_i, \dots, c_{i+A'})$. Let h and h' be the right-sided block maps of $\mathcal{A}^{\mathbb{Z}}$ with local rules K and K', so that $h_+ = \chi$ and $h'_+ = \chi^{-1}$. By (1), $(hh')_+ = (h'h)_+ = \mathrm{id}_+$, and hence $hh' = h'h = \mathrm{id}$. So h is a one-sided automorphism with $h_+ = \chi$.

(3) Let c and c' differ only at the site i, and suppose that $h(c)_i = h(c')_i$. For $j > i$ the cells $h(c)_j$ and $h(c')_j$ are read from sites $\geq j > i$, where $c = c'$. So hc and hc' agree at every site $\geq i$, and since h^{-1} is right-sided, $c_i = h^{-1}(hc)_i = h^{-1}(hc')_i = c'_i$, a contradiction. Hence K is injective, so bijective, in its first argument. □

By (24), F and F^{-1} are right-sided, for every rule of the form $f(a, b, c) = g(b, c)$. So F is a one-sided automorphism, and F_+, given by $F_+(c)_i = \phi_{y(c_{i+1})}(c_i)$ for $i \geq 0$, is an automorphism of the one-sided full shift on four letters.

Corollary 6.22. *Let $f \in \{119, 187\}$ and $F = F_f$.*

1. *The one-sided automorphisms of $\mathcal{A}^{\mathbb{Z}}$ that commute with F are exactly the elements of $\ker e$, that is, the powers of F.*
2. *F_+ has infinite order, and its centralizer in $\mathrm{Aut}(\mathcal{A}^{\mathbb{N}})$ is $\langle F_+ \rangle \cong \mathbb{Z}$.*

Proof. (1) The elements of $\ker e$ are one-sided automorphisms by Corollary 6.9(2), they commute with F, and $\ker e = \langle F \rangle$ by Theorem 6.19. Conversely, let h be a one-sided automorphism commuting with F. Then $h \in \mathrm{Aut}(X_F)$, and $e(h)$ is defined by Propositions 6.13(3) and 6.8. Let $Q = (c, c')$ be a pair with right edge p. For $i > p$ the cells $h(c)_i$ and $h(c')_i$ are read from sites $\geq i > p$, where $c = c'$, so $\lambda(hQ) \leq p$. Hence $e(h) \leq 0$. The same applies to h^{-1}, so $-e(h) = e(h^{-1}) \leq 0$ by Corollary 6.9(1). So $e(h) = 0$.

(2) By Lemma 6.21(2), $F_+^b = (F^b)_+$ for $b \in \mathbb{Z}$, and $F_+^b = \mathrm{id}$ only if $F^b = \mathrm{id}$, that is, only if $b = 0$, since $A(F^b) = |b|$ by Theorem 6.19. Let $\chi \in \mathrm{Aut}(\mathcal{A}^{\mathbb{N}})$ commute with F_+. By Lemma 6.21(2), $\chi = h_+$ for a one-sided automorphism h, and $(hF)_+ = \chi F_+ = F_+\chi = (Fh)_+$, so $hF = Fh$ by Lemma 6.21(1). By (1), $h = F^b$ for some b, and $\chi = F_+^b$. □

Since κ is a cellwise map, conjugation by κ_+ carries Corollary 6.22 over to the rules 238 and 221. For the four reflected rules 63, 252, 243 and 207 it holds with $\mathcal{A}^{\mathbb{N}}$ replaced by $\mathcal{A}^{-\mathbb{N}}$ and right-sided maps by left-sided ones.

Remark 6.23 (what the kernel argument uses). For the rules of Theorem 6.1, Section 6.4 uses the following about $\ker e$ and nothing else: it is a group containing F, all of whose elements commute with F; each of its elements is right-sided, with a local rule that is bijective in its first argument; the facts (a) and (b); from (c), only that $\nu \mapsto \phi_\nu(z)$ is injective for the cells z with $y(z) = 1$ and $\nu \mapsto \psi_\nu(z)$ for the cells with $x(z) = 1$; and Lemma 6.2. No window graph enters. The group of one-sided automorphisms commuting with F has these properties by Lemma 6.21, so Corollary 6.22(1) can also be proved without the window graphs, by applying Section 6.4 to that group. The condition that h^{-1} is right-sided cannot be dropped there: σ is right-sided and

commutes with F, and it is not a power of F. We make no claim for other rules that ignore their left argument. When a nontrivial cellwise permutation commutes with F, as κ does for rule $102 = b \oplus c$, the proofs of Lemmas 6.17 and 6.18 do not apply.

Remark 6.24 (automorphisms of small radius). Let $f \in \{119, 187\}$. By Theorem 6.19 and Corollary 6.9(2), $F^b(c)_i$ is a function of the cells $c_i, \dots, c_{i+|b|}$ that is bijective in c_i and, for $b \neq 0$, not independent of $c_{i+|b|}$. So $\sigma^a F^b(c)_i$ depends on both ends of the interval $[i + a,\, i + a + |b|]$ and on no cell outside it, and the least radius of $\sigma^a F^b$ is $\max(|a|, |a + |b||)$, which can be smaller than $|a| + |b|$. The pairs (a, b) with $[a, a + |b|] \subset [-R, R]$ are those with $-R \leq a \leq R$ and $|b| \leq R - a$, and there are $\sum_{a=-R}^{R}(2(R - a) + 1) = (2R + 1)^2$ of them. So X_F has exactly $(2R + 1)^2$ automorphisms of radius at most R; by the conjugacies of Section 6.1 the same count holds for all eight rules of Theorem 6.1 (for 63, which reads (a, b), the interval is $[a - |b|, a]$). For $R = 1, 2, 3$ this gives 9, 25 and 49 automorphisms, and these are exactly the invertible maps in the exhaustive enumeration of the block maps of radius at most 3 commuting with F in [11]. This enumeration and the other computations that confirm Theorem 6.1 are described in Appendix A.4; none of them is used in a proof.

Appendix A. Computations

This appendix specifies the finite computations of the paper. Every result marked as computer-assisted in the text is the output of a terminating exact search over a finite set. Section A.1 describes the enumeration of commuting block maps, Section A.2 the computations on which computer-assisted propositions rely, Section A.3 the computations whose results are stated in remarks, and Section A.4 computations that confirm results proved by hand. The programs accompany the paper as supplementary material, in its directory `code/`; the file `README.md` there states, for each program, what it computes and its expected output.

A.1. **Enumeration of commuting block maps.** *Commutation as a finite constraint.* Let h be a block map of radius R with local rule h_{loc}, and $F = F_f$ with f elementary. Both $h \circ F$ and $F \circ h$ are block maps of radius $R + 1$, so they are equal if and only if their local rules agree on every word $w \in \mathcal{A}^{2R+3}$. Write $n = 2R + 1$ and, for $w = (w_0, \dots, w_{2R+2})$, let v_1, v_2, v_3 be the three subwords of length n starting at positions $0, 1, 2$, and let v_F be the word of length n obtained by applying the local rule of F to w. Then

$$(h \circ F)(w) = h_{\mathrm{loc}}(v_F), \qquad (F \circ h)(w) = \Big(h_{\mathrm{loc}}(v_2)_y,\ h_{\mathrm{loc}}(v_2)_x \oplus f\big(h_{\mathrm{loc}}(v_1)_y, h_{\mathrm{loc}}(v_2)_y, h_{\mathrm{loc}}(v_3)_y\big)\Big),$$

where the subscripts x, y denote the two output bits. Writing X_v, Y_v for the two unknown bits of the table entry $h_{\mathrm{loc}}(v)$, each word w contributes the two constraints

$$X_{v_F} = Y_{v_2}, \qquad Y_{v_F} \oplus X_{v_2} = f(Y_{v_1}, Y_{v_2}, Y_{v_3}). \tag{26}$$

The first is two binary clauses; the second is a constraint on at most five variables, encoded by its truth table in at most sixteen clauses. Altogether there are $2 \cdot 4^{2R+1}$ variables and at most $18 \cdot 4^{2R+3}$ clauses. For an automaton of radius r of the form (16) the same encoding is used with words of length $2R + 2r + 1$ and the $2r + 1$ subwords of length n that f reads.

Enumeration and verification. All solutions are enumerated by repeated solving with blocking clauses, using PySAT [16] with CaDiCaL [2] or MiniSat [13]. Every table returned by the solver is re-checked, independently of the clauses, by evaluating $h \circ F$ and $F \circ h$ on all words. Each solution is then matched against the maps $\sigma^a F^b$ of radius at most R, which are built directly by composing local rules, and against the constant maps. The enumeration asserts that every $\sigma^a F^b$ of radius at most R occurs among the solutions, which checks the encoding against the maps known in advance to commute.

Invertibility. A block map is an automorphism of $\mathcal{A}^{\mathbb{Z}}$ if and only if it is injective. Injectivity of a block map of radius R is decided exactly by the pair graph: its vertices are the pairs (u, v) of words of length $2R$, with an edge $(u, v) \to (u', v')$ labelled $(a, b) \in \mathcal{A}^2$ whenever u' and v' are ua and vb with their first letters removed, and $h_{\mathrm{loc}}(ua) = h_{\mathrm{loc}}(vb)$. Then h fails to be injective if and only if a bi-infinite path passes through an off-diagonal vertex, which is decided by iteratively deleting vertices without a predecessor or without a successor. Where only the invertible commuting maps are wanted, balance is added to the clauses: a surjective block map of radius R is balanced [14], each output symbol occurring exactly 4^{2R} times in its table, so the balanced commuting maps are enumerated and each is submitted to the pair-graph test. The supplementary material contains `sat/centralizer.py` for the elementary rules, `sat/cent_r.py` for the automata (16) of radius r, and `sat/invertible_sat.py` for the enumeration with balance.

A.2. **Computations used in computer-assisted propositions.** *The certificate* (T_P). For given r, g and P, the sets Im_m of Section 4.10 are computed by iterating the successor relation on the 2^{2rP} column states until they stabilise at S_∞, and (T_P) is then tested. Increasing P gives a semi-decision procedure for a sufficient condition; we do not claim that (Λ) implies (T_P) for some P.

Proposition 4.40. For each of the 65 536 functions g of four variables, (T_P) is tested for $P = 1, 2, 3, 4$ in turn, up to the least P for which it holds. The supplementary material contains two independent programs for this, which represent the column states differently (`scan_r2.py` with `scan_r2b.py`, and `tp_direct.py`), and a third (`compare.py`) that compares their tables of least P for every g; the tables are identical. The program `scan_r2.py` also builds Λ_2 and Λ_3 from the cell rule for the functions with no (T_P), $P \leq 3$: for the 44 functions with least $P = 4$ the graphs Λ_2 and Λ_3 are strongly connected, and for $g = \xi_3$ and $g = 1 \oplus \xi_3$ the graph Λ_3 is not. For $g = \xi_3$ the graphs Λ_M, $2 \leq M \leq 5$, are built in the same way and are not strongly connected. The program `xcheck.py` compares the two (T_P) routines on 153 functions and counts the functions that satisfy (N) and (D).

Tables 1, 2 and 3, and Propositions 4.41 and 4.42. $J(g)$ and self-duality are computed from the truth table. (T_P) is tested for $P \leq 4$ at radius 2 and $P \leq 3$ at radius 3. Condition $(\mathrm{I})_q$ is tested by enumerating, for each n, the points of least spatial period n, computing the stabiliser of each $\mathbb{Z}^2$-orbit, and counting the orbits with each stabiliser; this gives $k(\mathcal{L})$ exactly (Lemma 4.3). The witness listed is the first in the order of increasing n, then increasing index nb, then increasing s; the enumeration goes up to $n = 5$. Each of these quantities is computed by two independent programs, with identical results; the supplementary material contains `examples.py` and `checks3.py`. In addition, Λ_M built from the cell rule is strongly connected for $M \leq 3$ for every rule of Tables 1 and 2, and for $M \leq 2$ for every rule of Table 3. The column "strip" of Table 1 is a SAT encoding of (20), as a system of constraints on the truth table of γ, whose enumeration stops after 20 solutions (`sat/strip.py`); the column "extra" is the enumeration of Section A.1 (`sat/cent_r.py`). The statements about rules 90 and 150 in Section 4.11 are computed in the same way.

The lift certificate (Proposition B.29). The program `lift_cert.py` computes the node sets $\mathcal{R}_k$ as specified in Section B.7, with the parameters $(f, n_{\max}, T_{\max})$ of the proposition, up to the depth at which the repetition stated there occurs for (1) and (2), and up to depth 105 for (3). Each new set is compared with all the sets before it; for (3) the comparison is through SHA-256 digests of the sorted lists of nodes, and the 106 digests are pairwise different. The consequences of Proposition B.29 for the holonomy are also proved by hand (Proposition B.28), and no other proof depends on this computation.

A.3. **Computer-assisted statements in remarks.** *Affine rules* (Remark 5.8). Λ_M and the eventual image of $\Gamma_{w,0}$ are built directly from the local rule and tested for strong connectivity.

For rule 90 the graph Λ_M is not strongly connected for $1 \le M \le 8$; this is the range quoted in Remark 5.8 and in Section 4.11.

A.4. **Confirmations of results proved by hand.** The following computations confirm results that the paper proves by hand. None of them is used in a proof.

Rule 30 *at radius at most* 4 (Theorem 3.1). The enumeration of Section A.1 for $f = 30$ and $R = 1, 2, 3, 4$ gives exactly $2R^2 + 2R + 3$ tables, that is 7, 15, 27 and 43: the $2R^2 + 2R + 1$ tables of the maps $\sigma^a F_{30}^b$ with $|a| + |b| \le R$, which are pairwise distinct and invertible, and the two constant maps onto $(0,0)^{\mathbb{Z}}$ and $(1,1)^{\mathbb{Z}}$. So $\mathcal{C}_4(F_{30})$ consists of the elements of $\langle \sigma, F_{30} \rangle$ of radius at most 4 and these two constants, in agreement with Theorem 3.1; this also determines the non-invertible commuting block maps of radius at most 4, which the theorem does not address. The enumeration is part of [11].

The elementary permutive rules (Theorem 5.1). For $f = 45, 75, 135$ and $R = 1$, and for $f = 45, 135$ and $R = 2$, the enumeration gives only the maps $\sigma^a F^b$ and constant maps. For the eight non-affine left-permutive rules, Λ_M built from the cell rule is strongly connected for $M \le 5$.

The two-copy window graph (Appendix B). For $f = 30$ and all (w, M) with $w + M \le 6$, and for $f = 45, 75, 135$ and $(w, M) \in \{(1,0), (2,0), (3,0), (4,0), (1,1), (2,1), (3,1), (1,2)\}$, the graph $\Gamma_{w,M}$ built from the cell rule has an eventual image of $2^{3w+2}4^M$ states, which is strongly connected and is reached by every walk of length $\lceil w/2 \rceil$.

The holonomy behind the front (Lemma B.25 and Proposition B.26). For 135 the graphs $\Gamma_{w,M}$ are built directly from the cell rule for $2 \le w \le 7$ with $M = 0$, and for $(w, M) = (2,1)$, $(3,1)$, $(4,1)$, $(5,1)$, $(6,1)$, $(2,2)$, $(3,2)$, $(4,2)$, $(2,3)$. In every case the eventual image has $2^{3w+2}4^M$ states, the graph G_w at the step $y[w]$, obtained from it by forgetting the layer $y[w]$, is strongly connected, and the values $\Psi_w(\varsigma)$ of its closed walks are all 32 elements of $(\mathbb{Z}/2)^5$. The same holds for the other seven non-affine left-permutive rules for $2 \le w \le 4$, $M = 0$. For the affine rules 90, 105 and 150 with $2 \le w \le 4$, $M = 0$, the values of all closed walks generate a group of order at most 8 (order 8 for 90, whose graph is not strongly connected; order 2 or 4 for 105 and 150). The supplementary material contains four programs for this, which use the cell rule only and neither the layer equation nor the conditions (C_j). The program `ref_check.py` enumerates all window states, iterates to the eventual image, and finds the exact set of values Ψ_w of the closed walks at a vertex by a search in $G_w \times (\mathbb{Z}/2)^5$; with the option `flip` it tests that the top flips and the swap of Lemma B.25 are automorphisms of G_w that change ψ_w as stated there. The program `ref_fast.py` builds the eventual image by a breadth-first search from one of its states, counts the strongly connected components, and computes the group generated by the values Ψ_w from the potentials of a spanning tree; it is used for the larger cases $w = 6, 7$ with $M = 0$ and $(w, M) = (5,1)$, $(6,1)$, $(4,2)$. The program `ref_affine.py` does the same over all components for the affine rules, and `ref_base.py` checks that the history below 2 of Section B.6 for 135 is a closed walk of length 3 in G_2 with $\Psi_2 = (1,0,1,1,1)$ for $M \le 3$, and that sampled closed walks of G_{d-1} with $\ell = (0,0)$ lift to closed walks of G_d of the same length.

Rules that read two neighbours (Section 6). The three finite checks, Lemmas 6.2, 6.11 and 6.12, are confirmed by direct computation (`halfradius/hr_checks.py`): the section identities by free reduction of the section words and by evaluating both sides on all words of length 7. The window graph W_M, built from the cell rule, is strongly connected for $M \le 6$, and the level-1 image of $\{t|_{x_*^M} : t \in G,\ t(x_*^M) = x_*^M\}$ is exactly A_4 for 119 when $1 \le M \le 10$ (and S_4 for $M = 0$), and S_4 for 187 when $0 \le M \le 10$. An exhaustive SAT enumeration (`halfradius/kernel_sat.py`) of the right-sided block maps of anticipation at most 6 that commute with F and are bijective in their first argument, as every right-sided automorphism with right-sided inverse is (Lemma 6.21(3)), gives exactly the powers F^b with $|b| \le 6$, for both 119 and 187. For rule 102 the same enumeration gives non-invertible right-sided block maps commuting with F. The enumeration of [11] gives,

for the classes of 63 and 187, exactly the $(2R+1)^2$ automorphisms $\sigma^a F^b$ of radius at most R for $R \le 3$ (Remark 6.24).

Appendix B. The two-copy window graph

This appendix is not used in the proof of any theorem of Sections 1–6. Those proofs follow one configuration along the front, through the graphs Λ_M. Here we follow both configurations of a front pair, through a *two-copy* window graph $\Gamma_{w,M}$ that also records w cells behind the front, and we determine its eventual image: its size, the number of steps after which every walk has entered it, and, for the rules 30, 45, 75 and 135, its strong connectivity (Theorem B.4 and Proposition B.30). Strong connectivity is reduced to a statement about the holonomy of closed walks at each layer behind the front. For 30, 45 and 75 this holonomy is read off from explicit periodic histories, one family for all depths. For 135 we know no such family, and the holonomy is obtained from a single closed walk of odd length by an induction on the depth (Proposition B.26).

Throughout this appendix $f(a,b,c) = a \oplus g(b,c)$ is a left-permutive elementary rule, with $g(b,c) = g_{11}bc \oplus g_{10}b \oplus g_{01}c \oplus g_{00}$ as in Section 5.1, and $F = F_f$. This is the automaton of Section 4 with $r = 1$, and we use the diagrams, pairs and front pairs of Section 4.2; for rule 30 they are those of Section 3.1. Every diagram satisfies

$$Y(t+1,i) \oplus Y(t-1,i) \oplus Y(t,i-1) = g\big(Y(t,i), Y(t,i+1)\big). \tag{27}$$

B.1. **Window states.** Fix integers $w \ge 1$ and $M \ge 0$. A *window state* is a pair $s = (a,b)$ of words $a = a_{p-w} \cdots a_{p+M}$ and $b = b_{p-w} \cdots b_{p+M}$ over $\mathcal{A}$, indexed by the sites $p-w, \dots, p+M$ for some integer p (the *front*), such that

$$a_i = b_i \quad (p < i \le p+M), \qquad y(a_p) \ne y(b_p), \qquad x(a_p) = x(b_p). \tag{28}$$

States that differ only by a translation of the indices are identified, and $\mathcal{W} = \mathcal{W}_{w,M}$ denotes the finite set of window states. The *window* of a front pair $P = (c, c')$ at p is

$$W(P) = \big(c_{p-w} \cdots c_{p+M},\ c'_{p-w} \cdots c'_{p+M}\big) \in \mathcal{W}.$$

Every $s \in \mathcal{W}$ is the window of some front pair: extend a and b by common cells.

For $s = (a,b) \in \mathcal{W}$ with front p and $\iota = (\iota_1, \iota_2) \in \mathcal{A}^2$, let $\tilde{a}, \tilde{b}$ be configurations that agree with a, b on $[p-w, p+M]$ and satisfy $\tilde{a}_{p+M+1} = \tilde{b}_{p+M+1} = \iota_1$ and $\tilde{a}_{p+M+2} = \tilde{b}_{p+M+2} = \iota_2$. Define $s \cdot \iota = (a', b')$ as the pair of words $a'_i = F(\tilde{a})_i$, $b'_i = F(\tilde{b})_i$ for $p+1-w \le i \le p+1+M$, with front $p+1$. Since F has radius 1, $s \cdot \iota$ does not depend on the other cells of $\tilde{a}, \tilde{b}$. The *window graph* $\Gamma = \Gamma_{w,M}$ has vertex set $\mathcal{W}$ and an edge $s \to s \cdot \iota$ for every $s \in \mathcal{W}$ and $\iota \in \mathcal{A}^2$.

Lemma B.1. *If $P = (c, c')$ is a front pair at p with $W(P) = s$, then FP is a front pair at $p+1$ and $W(FP) = s \cdot (c_{p+M+1}, c_{p+M+2})$. In particular $s \cdot \iota \in \mathcal{W}$ for all s, ι. For every $s \in \mathcal{W}$ and $\iota \in \mathcal{A}^2$ there is a front pair P with $W(P) = s$ and $W(FP) = s \cdot \iota$.*

Proof. FP is a front pair at $p+1$ by Lemma 4.12(1) with $r = 1$. The cells $F(c)_i$ with $p+1-w \le i \le p+1+M$ read only $c_{p-w}, \dots, c_{p+M+2}$, and $c_i = c'_i$ for $i > p$; this gives the formula for $W(FP)$. For the last statement, extend s by common cells with ι_1, ι_2 at $p+M+1, p+M+2$. □

Lemma B.2. *Let $s = (a,b) \in \mathcal{W}$ have front p, and let $\iota = (\iota_1, \iota_2) \in \mathcal{A}^2$. Then $s \cdot \iota$ depends on ι only through*

$$(u,v) = \big(y(\iota_1),\ x(\iota_1) \oplus f(y(a_{p+M}), y(\iota_1), y(\iota_2))\big) \in \{0,1\}^2,$$

and the map $\iota \mapsto (u,v)$ is 4-to-1 onto $\{0,1\}^2$. The new cell at site $p+M+1$ is (u,v) in a' and $(u, v \oplus [M=0])$ in b'. Hence every state has exactly four successors, which are distinct; we write them $s \cdot (u,v)$.

Proof. The new cells at sites $i \le p+M$ read only $\tilde{a}_{i-1}, \tilde{a}_i, \tilde{a}_{i+1}$; site $p+M+1$ enters only for $i = p+M$, and only through its y-bit u. The new cell at $p+M+1$ is $F(\tilde{a})_{p+M+1} = (u, v)$. In b the same formula applies with b_{p+M}. If $M \ge 1$, then $b_{p+M} = a_{p+M}$. If $M = 0$, then b_p differs from a_p only in the y-bit, which enters f as its left argument, hence through $\oplus$, so the new cell is $(u, v \oplus 1)$. For each u and each of the four values of ι_2, exactly one value of $x(\iota_1)$ gives a prescribed v, so each (u, v) arises from exactly four ι. Different (u, v) give different new cells at $p+M+1$, so the four successors are distinct. □

Layer coordinates. For a window state $s = (a, b)$ with front p and each copy $c \in \{a, b\}$ put

$$y_c[d] = y(c_{p-d}) \ \ (-M \le d \le w), \qquad x_c[e] = x(c_{p-e-1}) \ \ (-M-1 \le e \le w-1). \tag{29}$$

Thus the cell at site $p-d$ is $(x_c[d-1], y_c[d])$, and this is a bijection between words in $\mathcal{A}^{w+1+M}$ and assignments of bits to the layers $y[-M], \dots, y[w]$ and $x[-M-1], \dots, x[w-1]$. Write $\delta y[d] = y_a[d] \oplus y_b[d]$ and $\delta x[e] = x_a[e] \oplus x_b[e]$. In layer terms, (28) says

$$\delta y[d] = [d = 0] \ \ (-M \le d \le 0), \qquad \delta x[e] = 0 \ \ (-M-1 \le e \le -1). \tag{30}$$

The x-bit of the front cell is $x[-1]$.

Lemma B.3 (layer equation)**.** *Let $s' = s \cdot (u, v)$, with layers y'_c, x'_c. For each copy $c \in \{a, b\}$:*

1. *$x'_c[e] = y_c[e]$ for $-M \le e \le w-1$, and $x'_c[-M-1] = u$;*
2. *$y'_c[-M] = v_c$, where $v_a = v$ and $v_b = v \oplus [M = 0]$;*
3. *for $-M+1 \le d \le w$,*

$$y'_c[d] = y_c[d] \oplus x_c[d-2] \oplus g\big(y_c[d-1], y_c[d-2]\big), \tag{31}$$

with the convention $y_c[-M-1] = u$.

Proof. The new cell at site $i \le p+M$ is $(y(c_i),\ x(c_i) \oplus y(c_{i-1}) \oplus g(y(c_i), y(c_{i+1})))$, with $y(c_{p+M+1}) = u$. With front $p+1$, $x'_c[e]$ is the x-bit of the new cell at $p-e$, which is $y(c_{p-e}) = y_c[e]$ for $p-e \le p+M$; for $e = -M-1$ it is the x-bit u of the new cell at $p+M+1$ (Lemma B.2). Likewise $y'_c[-M]$ is the y-bit of the new cell at $p+M+1$. For $-M+1 \le d \le w$, $y'_c[d]$ is the y-bit of the new cell at $i = p+1-d \le p+M$, and $x(c_i) = x_c[d-2]$, $y(c_{i-1}) = y_c[d]$, $y(c_i) = y_c[d-1]$, $y(c_{i+1}) = y_c[d-2]$, where $y_c[-M-1]$ stands for $y(c_{p+M+1}) = u$. □

If P is a front pair at p and $z_t[d]$ denotes the y-bit of $F^t(c)$ at site $p+t-d$, then the window of $F^t P$ has y-layers z_t and x-layers z_{t-1}, and (31) is (27) written in these comoving coordinates. It is lower triangular: $y'[d]$ reads only layers with index at most d, and the leftmost x-layer $x[w-1]$ is never read.

B.2. **The eventual image.**

Theorem B.4. *Let f be rule 30, and let $w \ge 1$ and $M \ge 0$. There is a set $\mathcal{T} = \mathcal{T}_{w,M} \subseteq \mathcal{W}_{w,M}$ with $|\mathcal{T}| = 2^{3w+2} 4^M$ such that:*

1. *if $s \in \mathcal{T}$, then $s \cdot \iota \in \mathcal{T}$ for every $\iota \in \mathcal{A}^2$;*
2. *every walk in $\Gamma_{w,M}$ of length at least $\lceil w/2 \rceil$ ends in $\mathcal{T}$;*
3. *the subgraph of $\Gamma_{w,M}$ induced on $\mathcal{T}$ is strongly connected.*

Consequently $\mathcal{T}$ is the unique terminal strongly connected component of $\Gamma_{w,M}$, and for any $s, s' \in \mathcal{W}_{w,M}$ there are walks from s and from s' that end at a common state.

The consequences follow from (1)–(3). Every vertex has out-degree 4, so from s there is a walk of length $\lceil w/2 \rceil$, which ends at some $t \in \mathcal{T}$ by (2). Likewise s' reaches some $t' \in \mathcal{T}$, and by (3) there is a walk from t to t'. So both s and s' reach t'. By (1) and (3), $\mathcal{T}$ is a terminal strongly connected component. Any terminal component T contains the endpoints of walks of length $\lceil w/2 \rceil$ that start in it, so T meets $\mathcal{T}$, and hence $T = \mathcal{T}$.

B.3. **The tower of factor graphs.** In this subsection f is any left-permutive elementary rule, except in Lemmas B.16 and B.17, which are specific to rule 30. Thus parts (1) and (2) of Theorem B.4 and the size of $\mathcal{T}$ are proved for every such rule, and part (3) is reduced to the existence of closed walks with prescribed holonomy (Lemma B.15). Fix $w \geq 1$ and $M \geq 0$, and write $\Gamma = \Gamma_{w,M}$ and $\mathcal{W} = \mathcal{W}_{w,M}$. We use the layer coordinates (29). For a quantity Θ computed from one copy of a state we write Θ^a, Θ^b for its values on the two copies, and $\delta\Theta = \Theta^a \oplus \Theta^b$. A state of $\mathcal{W}$ is called *valid* to stress that it satisfies (30).

Backward rows. For one copy of a window state put $J_0 = [-M, w]$ and $J_m = [-M-1,\ w+1-2m]$ for $m \geq 1$, and define $\beta_m \colon J_m \to \{0,1\}$ by $\beta_0(d) = y[d]$, $\beta_1(e) = x[e]$ and, for $m \geq 1$ and $e \in J_{m+1}$,

$$\beta_{m+1}(e) = \beta_{m-1}(e+2) \oplus \beta_m(e+2) \oplus g\big(\beta_m(e+1), \beta_m(e)\big). \tag{32}$$

The arguments lie in the stated domains. For $e \in J_{m+1}$ we have $e+2 \leq w+1-2m$, so $e, e+1, e+2 \in J_m$. If $m \geq 2$, then $e+2 \in J_{m-1}$. If $m = 1$, then $e+2 \in [-M+1, w-1] \subseteq J_0$. Formula (32) is (31) solved for the earlier row, so β_m recomputes the history m steps back.

Lemma B.5. *Let $s_0 \to s_1 \to \cdots \to s_n$ be a walk in Γ with $s_{t+1} = s_t \cdot (u_t, v_t)$. For one copy, define $z_t[d]$ for $-1 \leq t \leq n$ as follows:*

- *$z_t[d] = y[d]$ of s_t for $0 \leq t \leq n$ and $-M \leq d \leq w$;*
- *$z_{t-1}[e] = x[e]$ of s_t for $0 \leq t \leq n$ and $-M-1 \leq e \leq w-1$.*

This is consistent, and $z_t[-M-1] = u_t$ for $0 \leq t \leq n-1$. The rows of s_n satisfy $\beta_m(e) = z_{n-m}[e]$ for $0 \leq m \leq n+1$ and $e \in J_m$.

Proof. For $1 \leq t \leq n$ and $-M \leq e \leq w-1$, Lemma B.3(1) gives $x[e]$ of $s_t = y[e]$ of s_{t-1}, so the two definitions agree; and $x[-M-1]$ of s_{t+1} is u_t. By Lemma B.3(3), for $0 \leq t \leq n-1$ and $-M+1 \leq d \leq w$,

$$z_{t+1}[d] = z_t[d] \oplus z_{t-1}[d-2] \oplus g\big(z_t[d-1], z_t[d-2]\big),$$

where $z_t[-M-1] = u_t$ is the convention of (31). The cases $m = 0, 1$ of the claim are the definitions. Let $1 \leq m \leq n$ and $e \in J_{m+1}$, and assume the claim for $m-1$ and m. The displayed relation at $t = n-m \in [0, n-1]$ and $d = e+2 \in [-M+1, w]$, solved for $z_{n-m-1}[e]$, reads $z_{n-m-1}[e] = z_{n-m+1}[e+2] \oplus z_{n-m}[e+2] \oplus g(z_{n-m}[e+1], z_{n-m}[e])$. By induction its right side is the right side of (32). □

The set $\mathcal{T}$. For $0 \leq j \leq w-1$ define (m_j, e_j) by

$$(m_0, e_0) = (1, 0), \qquad (m_j, e_j) = \left(\tfrac{j+3}{2}, -1\right) \ (j \text{ odd}), \qquad (m_j, e_j) = \left(\tfrac{j+2}{2}, 0\right) \ (j \geq 2 \text{ even}).$$

Then $j = e_j + 2m_j - 2$, and $e_j \in J_{m_j}$ because $e_j \leq w+1-2m_j$ is equivalent to $j \leq w-1$. The map $j \mapsto (m_j, e_j)$ is a bijection from $[0, w-1]$ onto the set of pairs (m, e) with $m \geq 1$, $e \in \{-1, 0\} \cap J_m$ and $(m, e) \neq (1, -1)$. Condition (C_j) on a window state is

$$(C_j) \qquad \delta\beta_{m_j}(e_j) = [e_j = 0],$$

and $\mathcal{T} = \{s \in \mathcal{W} :\ s \text{ satisfies } (C_j) \text{ for all } 0 \leq j \leq w-1\}$. By Lemma B.5, (C_j) says that the history recomputed from the state has its difference where the history of a front pair has it: a difference in the y-bit at the comoving front (layer 0) and none one site ahead of it (layer -1). The excluded pair $(1, -1)$ is the validity condition $\delta x[-1] = 0$.

Lemma B.6. *For every $e \in J_1$, $\beta_1(e) = x[e]$. For $m \geq 2$, $e \in J_m$ and $j = e + 2m - 2$,*

$$\beta_m(e) = y[j] \oplus x[j] \oplus \Phi_{m,e},$$

where $\Phi_{m,e}$ is a function of $y[i]$ for $e+2 \leq i \leq j-1$ and of $x[i]$ for $e \leq i \leq j-1$. In particular $\beta_m(e)$ reads only layers with index in $[e, j]$, and its y-layers have index in $[e+2, j]$.

Proof. Induction on m. For $m = 2$, $\beta_2(e) = y[e+2] \oplus x[e+2] \oplus g(x[e+1], x[e])$. Let $m \geq 2$, $e \in J_{m+1}$ and $j' = e + 2m$. In (32), by induction $\beta_m(e+2) = y[j'] \oplus x[j'] \oplus \Phi_{m,e+2}$, where $\Phi_{m,e+2}$ reads y in $[e+4, j'-1]$ and x in $[e+2, j'-1]$. The term $\beta_{m-1}(e+2)$ is $x[e+2]$ if $m = 2$, and otherwise it reads y in $[e+4, j'-2]$ and x in $[e+2, j'-2]$. The terms $\beta_m(e+1)$ and $\beta_m(e)$ read y in $[e+2, j'-1]$ and x in $[e, j'-1]$. So $\Phi_{m+1,e} = \beta_{m-1}(e+2) \oplus \Phi_{m,e+2} \oplus g(\beta_m(e+1), \beta_m(e))$ has the stated dependence. □

Lemma B.7. *Let $s \in \mathcal{T}$. Then $\delta\beta_m(e) = [e = 0]$ for all $m \geq 0$ and all $e \in J_m$ with $e \leq 0$.*

Proof. Strong induction on m. For $m = 0$ this is validity. For $m = 1$ and $e \leq -1$ it is validity, and for $e = 0$ it is (C_0); note $0 \in J_1$ since $w \geq 1$. For $m + 1 \geq 2$ and $e \in \{-1, 0\}$, it is (C_j) for the j with $(m_j, e_j) = (m+1, e)$. For $m + 1 \geq 2$ and $e \leq -2$, (32) gives $\delta\beta_{m+1}(e) = \delta\beta_{m-1}(e+2) \oplus \delta\beta_m(e+2) \oplus \delta g(\beta_m(e+1), \beta_m(e))$. Since $e + 2 \leq 0$, the first two terms are $[e+2 = 0] \oplus [e+2 = 0] = 0$ by induction. Both arguments of g have $\delta = 0$ by induction, as $e + 1 \leq -1$, so the last term vanishes. □

Lemma B.8. (1) *(C_0) is $\delta x[0] = 1$. For $1 \leq j \leq w-1$, with $\Phi_j = \Phi_{m_j,e_j}$, condition (C_j) is*

$$\delta x[j] = \delta y[j] \oplus \Phi_j^a \oplus \Phi_j^b \oplus [j \text{ even}],$$

and its right side depends only on $y_a[i], y_b[i]$ for $1 \leq i \leq j$ and on $x_a[i], x_b[i]$ for $-1 \leq i \leq j-1$. In particular the conditions (C_j) involve only the cells at sites $[p-w, p]$, so $\mathcal{T} = \mathcal{T}_w^\circ \times \mathcal{A}^M$ for a set $\mathcal{T}_w^\circ$ of pairs of words on $[p-w, p]$ that does not depend on M.

(2) *The map $s = (a, b) \mapsto \big(a, (\delta y[d])_{1 \leq d \leq w}\big)$ is a bijection from $\mathcal{T}$ onto $\mathcal{A}^{w+1+M} \times \{0,1\}^w$. Hence $|\mathcal{T}| = 2^{3w+2} 4^M$.*

Proof. (1) Apply δ to Lemma B.6; for $j \geq 1$ we have $m_j \geq 2$ and $[e_j = 0] = [j \text{ even}]$. Since $e_j \geq -1$, only layers with index ≥ -1 are read, and y-layers only with index $\geq e_j + 2 \geq 1$. The layer $x[e]$ with $e \geq -1$ and the layer $y[d]$ with $d \geq 1$ lie in cells at sites $p - e - 1$ and $p - d$, which are in $[p-w, p]$. The functions $\Phi_{m,e}$ do not involve M.

(2) Given a and $\delta y[1], \ldots, \delta y[w]$, validity fixes $\delta y[d]$ for $d \leq 0$ and $\delta x[e]$ for $e \leq -1$, and $(C_0), (C_1), \ldots, (C_{w-1})$ determine $\delta x[0], \ldots, \delta x[w-1]$ in turn, because the right side of (C_j) involves only $\delta y[j]$ and layers with index $< j$ of both copies, which are already fixed. So b exists and is unique. Thus $|\mathcal{T}| = 4^{w+1+M} 2^w = 2^{3w+2} 4^M$. □

Entrance, predecessors and closure. Let $\mathrm{Im}_n \subseteq \mathcal{W}$ be the set of endpoints of walks of length n in Γ, so that $\mathrm{Im}_0 = \mathcal{W}$.

Lemma B.9. *For $n \geq 1$, every $s \in \mathrm{Im}_n$ satisfies (C_j) for all $0 \leq j \leq \min(w-1, 2n-1)$. In particular $\mathrm{Im}_n \subseteq \mathcal{T}$ for $n \geq \lceil w/2 \rceil$.*

Proof. Take a walk $s_0 \to \cdots \to s_n = s$. By Lemma B.5, $\beta_{m_j}(e_j) = z_{n-m_j}[e_j]$ for both copies whenever $m_j \leq n+1$. If $e_j = 0$ and $m_j \leq n$, then $z_{n-m_j}[0]$ is the layer $y[0]$ of the valid state s_{n-m_j}, so its δ is 1. If $e_j = -1$ and $m_j \leq n+1$, then $z_{n-m_j}[-1]$ is the layer $x[-1]$ of the valid state s_{n-m_j+1}, so its δ is 0. For even j, $m_j \leq n$ is equivalent to $j \leq 2n-2$; for odd j, $m_j \leq n+1$ is equivalent to $j \leq 2n-1$. Finally $w - 1 \leq 2n - 1$ if and only if $n \geq w/2$. □

Lemma B.10. *Let $s \in \mathcal{T}$. There are exactly four pairs $(\hat{s}, (u, v))$ with $\hat{s} \in \mathcal{T}$ and $\hat{s} \cdot (u, v) = s$. In particular s has a predecessor in $\mathcal{T}$.*

Proof. Let $\hat{s} \in \mathcal{W}$ have layers $\hat{y}_c, \hat{x}_c$, and let y_c, x_c be the layers of s. By Lemma B.3, $\hat{s} \cdot (u, v) = s$ if and only if:

(i) $\hat{y}_c[e] = x_c[e]$ for $-M \leq e \leq w-1$, and $u = x_c[-M-1]$ (this value is common to both copies, since $\delta x[-M-1] = 0$);

(ii) $v = y_a[-M]$ (then $y_b[-M] = v \oplus [M = 0]$ holds automatically, since $\delta y[-M] = [M = 0]$);
(iii) for $-M+1 \le d \le w$, $\hat{x}_c[d-2] = y_c[d] \oplus \hat{y}_c[d] \oplus g(\hat{y}_c[d-1], \hat{y}_c[d-2])$, with $\hat{y}_c[-M-1] = u$.

So (u, v) is determined by s. For $d \le w-1$, (i) turns (iii) into $\hat{x}_c[d-2] = y_c[d] \oplus x_c[d] \oplus g(x_c[d-1], x_c[d-2]) = \beta_2^s(d-2)$, where β_m^s are the rows of s; the convention $\hat{y}_c[-M-1] = u = x_c[-M-1]$ is used when $d-2 = -M-1$. This determines $\hat{x}$ on $[-M-1, w-3]$. For $d = w$, (iii) gives $\hat{x}_c[w-2] = y_c[w] \oplus \hat{y}_c[w] \oplus g(x_c[w-1], x_c[w-2])$, where $\hat{y}_c[w]$ is free. Finally $\hat{x}_c[w-1]$ is unconstrained. Hence the candidates are parametrized by $\hat{y}_a[w], \hat{y}_b[w], \hat{x}_a[w-1], \hat{x}_b[w-1]$, and we must decide which give $\hat{s} \in \mathcal{T}$.

Validity of $\hat{s}$. For $-M \le d \le 0$, $\delta\hat{y}[d] = \delta x[d]$, which is 0 for $d \le -1$ by validity of s and 1 for $d = 0$ by (C_0) for s (note $0 \le w-1$). For $-M-1 \le e \le \min(-1, w-3)$, $\delta\hat{x}[e] = \delta\beta_2^s(e) = 0$ by Lemma B.7. If $w \ge 2$ this covers all $e \le -1$. If $w = 1$, the remaining condition is $\delta\hat{x}[-1] = 0$, where $-1 = w-2$.

Transfer of conditions. We claim $\beta_0^{\hat{s}}(e) = \beta_1^s(e)$ for $-M \le e \le w-1$, and $\beta_m^{\hat{s}}(e) = \beta_{m+1}^s(e)$ for $m \ge 1$ and $e \in J_{m+1}$. For $m = 0$ this is (i), and for $m = 1$ it is the formula for $\hat{x}$ on $J_2 = [-M-1, w-3]$ obtained above. For $m \ge 1$ and $e \in J_{m+2}$, apply (32) to both sides: the arguments are $\beta_{m-1}^{\hat{s}}(e+2)$, $\beta_m^{\hat{s}}(e+2)$, $\beta_m^{\hat{s}}(e+1)$, $\beta_m^{\hat{s}}(e)$ with $e, e+1, e+2 \in J_{m+1} \subseteq J_m$ (and $e+2 \in [-M, w-1]$ when $m-1 = 0$), so by induction they equal $\beta_m^s(e+2)$, $\beta_{m+1}^s(e+2)$, $\beta_{m+1}^s(e+1)$, $\beta_{m+1}^s(e)$, and (32) for s gives $\beta_{m+1}^{\hat{s}}(e) = \beta_{m+2}^s(e)$. Now $(m_j + 1, e_j) = (m_{j+2}, e_{j+2})$, and $e_j \in J_{m_j+1}$ if and only if $j \le w-3$. So for $j \le w-3$, condition (C_j) for $\hat{s}$ is (C_{j+2}) for s, which holds.

The two remaining conditions. By Lemma B.8(1) applied to $\hat{s}$, condition (C_j) for $\hat{s}$ reads $\delta\hat{x}[0] = 1$ if $j = 0$, and $\delta\hat{x}[j] = \delta\hat{y}[j] \oplus \hat{\Phi}_j^a \oplus \hat{\Phi}_j^b \oplus [j \text{ even}]$ if $j \ge 1$, where $\hat{\Phi}_j$ reads $\hat{y}[i]$ for $i \le j-1$ and $\hat{x}[i]$ for $i \le j-1$.

- $j = w-2$ (when $w \ge 2$). The left side is $\delta\hat{x}[w-2] = \delta y[w] \oplus \delta\hat{y}[w] \oplus \delta g(x[w-1], x[w-2])$, in which only $\delta\hat{y}[w]$ is not yet fixed, and it enters with coefficient 1. The right side involves $\hat{y}[w-2] = x[w-2]$ and layers of $\hat{s}$ with index at most $w-3$, namely $\hat{y}[i] = x[i]$ and $\hat{x}[i] = \beta_2^s(i)$, all determined by s. So this condition determines $\delta\hat{y}[w]$, and $\hat{y}_a[w]$ is free. For $w = 1$ the same argument applies to the validity condition $\delta\hat{x}[-1] = 0$.
- $j = w-1$. The left side is $\delta\hat{x}[w-1]$. The right side involves $\hat{y}[w-1] = x[w-1]$ and layers of $\hat{s}$ with index at most $w-2$. Among these, $\hat{x}_a[w-2]$ and $\hat{x}_b[w-2]$ depend on the values $\hat{y}_a[w], \hat{y}_b[w]$ chosen in the previous step, and everything else is determined by s; $\hat{x}[w-1]$ does not occur on the right. So, for each choice made in the previous step, this condition determines $\delta\hat{x}[w-1]$, and $\hat{x}_a[w-1]$ is free.

Hence $\hat{s} \in \mathcal{T}$ if and only if $\delta\hat{y}[w]$ and $\delta\hat{x}[w-1]$ take the values just described, which leaves exactly two choices of $\hat{y}_a[w]$ and two of $\hat{x}_a[w-1]$, and the input (u, v) is unique. □

Proposition B.11. (1) *$\mathcal{T} \subseteq \mathrm{Im}_n$ for every $n \ge 0$.*
(2) *$\mathrm{Im}_n = \mathcal{T}$ for every $n \ge \lceil w/2 \rceil$.*
(3) *If $s \in \mathcal{T}$, then $s \cdot (u, v) \in \mathcal{T}$ for all (u, v). Inside $\mathcal{T}$ every state has out-degree 4 and in-degree 4.*
(4) *Every walk in Γ of length at least $\lceil w/2 \rceil$ ends in $\mathcal{T}$.*

Proof. (1) Induction on n: if $\mathcal{T} \subseteq \mathrm{Im}_n$, then every $s \in \mathcal{T}$ has a predecessor in $\mathcal{T} \subseteq \mathrm{Im}_n$ by Lemma B.10, so $s \in \mathrm{Im}_{n+1}$. (2) Combine (1) with Lemma B.9. (3) With $n = \lceil w/2 \rceil$, $s \in \mathrm{Im}_n$ gives $s \cdot (u, v) \in \mathrm{Im}_{n+1} = \mathcal{T}$. The degrees come from Lemmas B.2 and B.10. (4) The endpoint of a walk of length n lies in Im_n. □

This proves parts (1) and (2) of Theorem B.4, and $|\mathcal{T}| = 2^{3w+2}4^M$ is Lemma B.8(2); so far f is any left-permutive elementary rule.

Strong connectivity. Order the layers as

$$L_1 = x[-M-1],\ L_2 = y[-M],\ L_3 = x[-M],\ L_4 = y[-M+1],\ \dots,\ x[w-1],\ L_{2(w+1+M)} = y[w],$$

that is, $x[-M-1]$ first and then, for $d = -M, \dots, w$, the layer $y[d]$ followed by $x[d]$ (when $d \le w-1$). There are $2(w+1+M)$ coordinates, and each takes values in pairs of bits (the a-bit and the b-bit). Let $P_\nu = \{L_1, \dots, L_\nu\}$, let π_ν be the restriction of a state to P_ν, and put $S_\nu = \pi_\nu(\mathcal{T})$.

Lemma B.12. (1) *S_ν is the set of assignments η of bit pairs to P_ν that satisfy those conditions of (30) that involve only coordinates in P_ν, and (C_j) for every j with $x[j] \in P_\nu$.*

(2) *If $L_{\nu+1}$ is an x-layer or a layer $y[d]$ with $d \le 0$, then each $\eta \in S_\nu$ has exactly two extensions in $S_{\nu+1}$, one for each value of the a-bit of $L_{\nu+1}$. If $L_{\nu+1} = y[d]$ with $d \ge 1$, then each $\eta \in S_\nu$ has exactly four extensions, one for each value of the pair $(y_a[d], y_b[d])$.*

(3) *For $s \in \mathcal{T}$, $\pi_\nu(s \cdot (u,v))$ depends only on $\pi_\nu(s)$ and (u,v).*

Proof. (1) By Lemma B.8(1), (C_j) has the form $\delta x[j] =$ (a function of $y[i]$, $i \le j$, and $x[i]$, $i \le j-1$), and all of these coordinates precede $x[j]$. So every element of S_ν satisfies the stated conditions. Conversely, let η satisfy them, and extend η one coordinate at a time: choose the a-bit arbitrarily; set $y_b[d] = y_a[d] \oplus [d = 0]$ for $d \le 0$, choose $y_b[d]$ arbitrarily for $d \ge 1$, set $x_b[e] = x_a[e]$ for $e \le -1$, and choose $x_b[j]$ for $j \ge 0$ so that (C_j) holds, which is possible because (C_j) prescribes $\delta x[j]$ in terms of earlier coordinates. The resulting state is valid and satisfies all (C_j), so it lies in $\mathcal{T}$ and restricts to η. (2) follows from (1) and the same description. (3) By Lemma B.3, the new value of each coordinate depends only on the input and on coordinates that precede it or equal it: $x'[e]$ reads $y[e]$ or u; $y'[-M]$ reads v; and $y'[d]$ reads $y[d]$, $x[d-2]$, $y[d-1]$, $y[d-2]$ (or u). □

Let Γ_ν be the graph with vertex set S_ν and, for each $\eta \in S_\nu$ and $(u,v) \in \{0,1\}^2$, an edge $\eta \to \eta \cdot (u,v) := \pi_\nu(s \cdot (u,v))$ labelled (u,v), where $s \in \mathcal{T}$ is any state with $\pi_\nu(s) = \eta$. This is well defined by Lemma B.12(3), and $\eta \cdot (u,v) \in S_\nu$ by Proposition B.11(3). For $\nu = 2(w+1+M)$, Γ_ν is the subgraph of Γ induced on $\mathcal{T}$. Every walk in Γ_ν lifts to a walk in $\Gamma_{\nu+1}$ with the same labels, starting from any lift of its initial vertex.

Lemma B.13. *Γ_1 and Γ_2 are complete digraphs with loops; in particular they are strongly connected.*

Proof. $S_1 = \{0,1\}$ is the common value of $x[-M-1]$, and the edge labelled (u,v) goes to u. $S_2 = \{0,1\}^2$ consists of the common value of $x[-M-1]$ and the value of $y_a[-M]$ (and $y_b[-M] = y_a[-M] \oplus [M = 0]$). By Lemma B.3(1),(2), the edge labelled (u,v) goes to (u,v). □

Lemma B.14. *Let $L_{\nu+1} = x[e]$ with $e \ge -M$. If Γ_ν is strongly connected, so is $\Gamma_{\nu+1}$.*

Proof. Write the vertices of $\Gamma_{\nu+1}$ as (η, ξ) with $\eta \in S_\nu$ and $\xi = x_a[e]$. The layer $x[e]$ is read by Lemma B.3 only in $y'[e+2]$, which is not in $P_{\nu+1}$, and its new value is $x'[e] = y[e]$ with $y[e] \in P_\nu$. So the edge labelled (u,v) goes from (η, ξ) to $(\eta \cdot (u,v),\ y_a[e](\eta))$. Let $(\eta_1, \xi_1), (\eta_2, \xi_2) \in S_{\nu+1}$. Lift (η_2, ξ_2) to some $s \in \mathcal{T}$. By Lemma B.10 there are $\hat{s} \in \mathcal{T}$ and (u,v) with $\hat{s} \cdot (u,v) = s$. Put $\hat{\eta} = \pi_\nu(\hat{s})$; then $\hat{\eta} \cdot (u,v) = \eta_2$ and $y_a[e](\hat{\eta}) = x_a[e](s) = \xi_2$ by Lemma B.3(1). Take a walk from η_1 to $\hat{\eta}$ in Γ_ν and lift it to $\Gamma_{\nu+1}$ starting at (η_1, ξ_1). It ends at $(\hat{\eta}, \xi')$ for some ξ', and the edge labelled (u,v) then leads to (η_2, ξ_2). □

Lemma B.15. *Let $L_{\nu+1} = y[d]$ with $d \ge -M+1$. Let $K = (\mathbb{Z}/2)^2$ and $z = (y_a[d], y_b[d])$ if $d \ge 1$, and let $K = \mathbb{Z}/2$ and $z = y_a[d]$ if $d \le 0$ (then $y_b[d] = y_a[d] \oplus [d = 0]$). For an edge ϵ of Γ_ν from η with label (u,v) put*

$$\varphi_c(\epsilon) = x_c[d-2] \oplus g(y_c[d-1], y_c[d-2]), \tag{33}$$

evaluated on η with $y_c[-M-1]=u$, and $\varphi(\epsilon)=(\varphi_a,\varphi_b)\in K$ if $d\geq 1$, $\varphi(\epsilon)=\varphi_a\in K$ if $d\leq 0$. Then:

1. *the edges of $\Gamma_{\nu+1}$ are $(\eta,z)\to(\eta\cdot(u,v),\, z+\varphi(\epsilon))$, and for $d\leq 0$, $\varphi_a=\varphi_b$;*
2. *for a walk ς in Γ_ν let $\ell(\varsigma)\in K$ be the sum of φ over its edges, and let* Hol *be the subgroup of K generated by the values $\ell(\varsigma)$ of closed walks. If Γ_ν is strongly connected and* $\mathrm{Hol}=K$*, then $\Gamma_{\nu+1}$ is strongly connected.*

Proof. (1) is (31). The layers $x[d-2]$, $y[d-1]$, $y[d-2]$ lie in P_ν. For $d\leq 0$ they have index at most -1, so they are common to both copies, as is u; hence $\varphi_a=\varphi_b$, and adding φ preserves the relation $y_b[d]=y_a[d]\oplus[d=0]$. (2) By Lemma B.12(2), $(\eta,z)\in S_{\nu+1}$ for every $\eta\in S_\nu$ and every $z\in K$. Fix $\eta_2\in S_\nu$, and let Hol_{η_2} be the set of values $\ell(\varsigma)$ of closed walks ς at η_2. It contains 0 and is closed under addition (concatenation), so it is a submonoid of the finite group K, hence a subgroup. For a closed walk ς at any η, choose walks α from η_2 to η and β from η to η_2. Then $\ell(\varsigma)=\ell(\alpha\varsigma\beta)-\ell(\alpha\beta)\in\mathrm{Hol}_{\eta_2}$, so $\mathrm{Hol}_{\eta_2}=\mathrm{Hol}=K$. Given (η_1,z_1) and (η_2,z_2), take a walk α from η_1 to η_2 and a closed walk ς at η_2 with $\ell(\varsigma)=z_2-z_1-\ell(\alpha)$. The lift of $\alpha\varsigma$ starting at (η_1,z_1) ends at (η_2,z_2). □

Lemma B.16 (stationary states)**.** *Let f be rule 30, so that $g(b,c)=b\vee c$. Let $L_{\nu+1}=y[d]$ with $d\geq -M+1$, so that P_ν consists of the layers up to $x[d-1]$. Let η be an assignment of bit pairs to P_ν that satisfies the conditions of (30) on P_ν, and suppose that for some bit χ and both copies c:*

- *$x_c[i]=y_c[i]$ for $-M\leq i\leq d-1$, and $x_c[-M-1]=\chi$;*
- *with $y_c[-M-1]:=\chi$, $y_c[i+1]\leq y_c[i]$ for $-M-1\leq i\leq d-3$.*

Then $\eta\in S_\nu$, the edge labelled $(\chi,y_a[-M])$ is a loop at η, and on this loop $\varphi_c=y_c[d-1]\wedge\neg y_c[d-2]$.

Proof. The loop. By Lemma B.3 with $u=\chi$ and $v=y_a[-M]$: $x'[i]=y[i]=x[i]$ for $-M\leq i\leq d-1$, and $x'[-M-1]=\chi$. Next, $y'_c[-M]=y_a[-M]\oplus[c=b][M=0]=y_c[-M]$ by validity. For $-M+1\leq i\leq d-1$, (31) and $x[i-2]=y[i-2]$ (for $i-2=-M-1$ both equal $\chi=u$) give

$$y'[i]=y[i]\oplus\big(y[i-2]\oplus(y[i-1]\vee y[i-2])\big)=y[i]\oplus\big(y[i-1]\wedge\neg y[i-2]\big)=y[i],$$

because $i-2\in[-M-1,d-3]$ and $y[i-1]\leq y[i-2]$. The same identity gives φ_c.

Membership. By Lemma B.12(1) we must check (C_j) for $0\leq j\leq d-1$. Let $Z(i)=y_c[i]$ for $-M-1\leq i\leq d-1$, so that $y_c=Z$ on $[-M,d-1]$ and $x_c=Z$ on $[-M-1,d-1]$. We claim that $\beta_m(e)=Z(e)$ whenever $e\geq -M-1$ (and $e\geq -M$ if $m=0$) and $e+\max(2m-2,0)\leq d-1$. For $m=0,1$ this holds by assumption. For $m+1\geq 2$, the arguments $\beta_{m-1}(e+2)$, $\beta_m(e+2)$, $\beta_m(e+1)$ and $\beta_m(e)$ of (32) all satisfy the claim's hypotheses, so $\beta_{m+1}(e)=Z(e+2)\oplus Z(e+2)\oplus(Z(e+1)\vee Z(e))=Z(e)$, because $e\leq d-1-2m\leq d-3$ gives $Z(e+1)\leq Z(e)$. Every (C_j) with $j\leq d-1$ has $e_j+2m_j-2=j\leq d-1$, so $\delta\beta_{m_j}(e_j)=\delta Z(e_j)$. For $e_j=0$ this is $\delta y[0]=1$. For $e_j=-1$ it is $\delta y[-1]=0$ if $M\geq 1$, and $\delta\chi=0$ if $M=0$. So (C_j) holds. □

Lemma B.17. *Let f be rule 30. For every $d\in[-M+1,w]$, at the step $L_{\nu+1}=y[d]$ we have* $\mathrm{Hol}=K$*.*

Proof. In each case we give states satisfying the hypotheses of Lemma B.16, with $x=y$ on $[-M,d-1]$ in both copies, and read off φ on their loops. Each loop is a closed walk, so its φ lies in Hol.

Case $d\leq 0$. All layers in P_ν have index at most -1; take them common, with $\chi=0$, $y[i]=0$ for $-M\leq i\leq d-2$, and $y[d-1]=1$. Validity holds since there are no differences, and monotonicity concerns only layers up to $d-2$, which are all 0. Then $\varphi=y[d-1]\wedge\neg y[d-2]=1$ (when $d-2=-M-1$, $y[d-2]=\chi=0$), which generates $K=\mathbb{Z}/2$.

Case $d=1$. Take $\chi=0$, all common layers ($-M\leq i\leq -1$) equal to 0, $y_a[0]=\sigma_0$ and $y_b[0]=1-\sigma_0$ for a bit σ_0. Then $\delta y[0]=1$, the x-bit of the front is common, and monotonicity

concerns only layers up to -1. Here $\varphi_c = y_c[0] \wedge \neg y_c[-1] = y_c[0]$. So $\sigma_0 = 1$ gives $(1,0)$ and $\sigma_0 = 0$ gives $(0,1)$, which generate $(\mathbb{Z}/2)^2$.

Case $d = 2$. Take $\chi = 1$, all common layers equal to 1, $y_a[0] = \sigma_0$, $y_b[0] = 1 - \sigma_0$, and $y_c[1] = \alpha_c$. Monotonicity is $1 \geq \cdots \geq 1 \geq y_c[0]$, which holds. Here $\varphi_c = \alpha_c \wedge \neg y_c[0]$. With $\sigma_0 = 0$, $\alpha_a = 1$, $\alpha_b = 0$ we get $(1,0)$; with $\sigma_0 = 1$, $\alpha_a = 0$, $\alpha_b = 1$ we get $(0,1)$.

Case $d \geq 3$. Take $\chi = 1$, all common layers equal to 1, $y_a[0] = 1$, $y_b[0] = 0$, $y_c[i] = 0$ for $1 \leq i \leq d-2$, and $y_c[d-1] = \alpha_c$. The sequence $\chi, 1, \ldots, 1, y_c[0], 0, \ldots, 0$ up to layer $d-2$ is nonincreasing. Here $\varphi_c = \alpha_c \wedge \neg y_c[d-2] = \alpha_c$, so $(\alpha_a, \alpha_b) = (1,0)$ and $(0,1)$ generate $(\mathbb{Z}/2)^2$. □

In the case $d = 2$ of this proof the common layers must equal 1. Monotonicity forces this, because one of $y_a[0], y_b[0]$ is 1. It is also forced by (C_1), which reads $\delta x[1] = \delta y[1] \oplus \neg x[-1]$ (from $\beta_2(-1) = y[1] \oplus x[1] \oplus (x[0] \vee x[-1])$ and $\delta x[0] = 1$), so a state with $x[1] = y[1]$ in both copies must have $x[-1] = 1$.

Proof of Theorem B.4. Parts (1) and (2) and the size of $\mathcal{T}$ were proved above. For (3), we show by induction on ν that Γ_ν is strongly connected. Lemma B.13 covers $\nu = 1, 2$. Each later coordinate is either $x[e]$ with $e \geq -M$, covered by Lemma B.14, or $y[d]$ with $d \geq -M+1$, covered by Lemmas B.15 and B.17. For the last value of ν, Γ_ν is the subgraph of Γ induced on $\mathcal{T}$. □

B.4. **Closed walks from periodic histories.** For the rules other than 30, the stationary states of Lemma B.16 are replaced by time-periodic solutions of the layer equation. In this subsection $f = a \oplus g(b,c)$ is a left-permutive elementary rule with $g_{11} = 1$, and $\partial g(b,c) = g(b,c) \oplus g(1 \oplus b, c) = c \oplus g_{10}$. Fix $d \in \mathbb{Z}$ and let $T \geq 1$.

Definition B.18. A *periodic history below* d of period T is a pair $Z = (Z^a, Z^b)$ of functions $Z^c \colon \mathbb{Z} \times (-\infty, d-1] \to \{0,1\}$, $(t,i) \mapsto Z^c_t(i)$, with $Z^c_{t+T} = Z^c_t$, such that for $c \in \{a,b\}$:

(Z1) $Z^c_{t+1}(i) = Z^c_t(i) \oplus Z^c_{t-1}(i-2) \oplus g\big(Z^c_t(i-1), Z^c_t(i-2)\big)$ for all $t \in \mathbb{Z}$ and $i \leq d-1$;

(Z2) $Z^a_t(i) = Z^b_t(i)$ for all t and $i \leq -1$, and, if $d \geq 1$, $Z^a_t(0) \neq Z^b_t(0)$ for all t.

If $d \leq 0$ the two copies coincide by (Z2), and we write Z as one function. More generally, for any d (in particular $d = 1$), a *single periodic history below* d of period T is one function $Y \colon \mathbb{Z} \times (-\infty, d-1] \to \{0,1\}$ with $Y_{t+T} = Y_t$ that satisfies (Z1); no condition (Z2) is imposed. For $d \leq 0$ the single histories below d are exactly the periodic histories below d, with $Z^a = Z^b = Y$. For $d = 1$ a single history is not a periodic history below 1, since (Z2) requires the copies to differ at layer 0; Lemma B.20(4) turns it into one. For a single history we write φ_t, $\ell(Y)$ and $N(Y,i)$ for the quantities below, computed from Y. For $c \in \{a,b\}$, $t \in \mathbb{Z}$ and $i \leq d-1$ put

$$\varphi^c_t = Z^c_{t-1}(d-2) \oplus g\big(Z^c_t(d-1), Z^c_t(d-2)\big), \qquad \ell_c(Z) = \sum_{t=0}^{T-1} \varphi^c_t, \qquad N_c(Z,i) = \sum_{t=0}^{T-1} \big(Z^c_t(i) \oplus g_{10}\big),$$

with sums in $\mathbb{Z}/2$.

Thus (Z1) is (31) for the y-layers of a comoving history, with x-layers equal to the y-layers at the preceding time, as in Lemma B.5. The bit φ^c_t is the increment that a layer d would have to follow, and $\ell_c(Z)$ is its sum over a period. If Z extends to a periodic history below $d+1$, then $\ell_c(Z) = 0$.

Lemma B.19. *Let $w \geq 1$, $M \geq 0$ and $-M+1 \leq d \leq w$, and let Γ_ν be the graph of the tower of Section B.3 for f at the step $L_{\nu+1} = y[d]$, so that P_ν consists of the layers $x[-M-1], y[-M], \ldots, y[d-1], x[d-1]$. Let Z be a periodic history below d of period T. For $t \in \mathbb{Z}$ let η_t be the assignment to P_ν given by*

$$y_c[i] = Z^c_t(i) \quad (-M \leq i \leq d-1), \qquad x_c[e] = Z^c_{t-1}(e) \quad (-M-1 \leq e \leq d-1).$$

Then $\eta_t \in S_\nu$, Γ_ν has an edge $\eta_t \to \eta_{t+1}$ with label $(u_t, v_t) = (Z^a_t(-M-1),\, Z^a_{t+1}(-M))$, and the closed walk $\eta_0 \to \eta_1 \to \cdots \to \eta_T = \eta_0$ has $\ell = (\ell_a(Z), \ell_b(Z))$ if $d \geq 1$ and $\ell = \ell_a(Z)$ if $d \leq 0$, in the notation of Lemma B.15. In particular these values lie in Hol. *Moreover S_ν, the edges of Γ_ν and their increments (33) do not depend on $w \geq d$.*

Proof. Validity. By (Z2), $\delta y[i] = [i = 0]$ for $-M \leq i \leq \min(0, d-1)$ and $\delta x[e] = 0$ for $e \leq -1$. These are the conditions of (30) that involve only P_ν.

Membership. By Lemma B.12(1) it remains to check (C_j) for $0 \leq j \leq d-1$. By Lemma B.6, $\beta_m(e)$ reads only layers with index in $[e, e+2m-2]$, so it is defined on η_n whenever $e + \max(2m-2, 0) \leq d-1$; such e lie in J_m because $d \leq w$. We claim that for such m, e (with $e \geq -M-1$, and $e \geq -M$ if $m = 0$), the backward rows of η_n satisfy $\beta_m(e) = Z_{n-m}(e)$ in each copy. For $m = 0, 1$ this is the definition of η_n. For $m \geq 1$ the four arguments $(m-1, e+2)$, $(m, e+2)$, $(m, e+1)$ and (m, e) in the recursion for $\beta_{m+1}(e)$ satisfy the same constraint, since $e + 2m \leq d-1$. So by induction

$$\beta_{m+1}(e) = Z_{n-m+1}(e+2) \oplus Z_{n-m}(e+2) \oplus g\big(Z_{n-m}(e+1), Z_{n-m}(e)\big) = Z_{n-m-1}(e),$$

the last step being (Z1) at time $n-m$ and layer $e+2 \in [-M+1, d-1]$. Now (C_j) with $j = e_j + 2m_j - 2 \leq d-1$ reads $\delta\beta_{m_j}(e_j) = \delta Z_{n-m_j}(e_j) = [e_j = 0]$, by (Z2), since $e_j \in \{-1, 0\}$.

Edges. By Lemma B.12(3), the edge of Γ_ν from η_t with label (u, v) leads to the assignment computed from η_t and (u, v) by the formulas of Lemma B.3 restricted to P_ν. With $(u, v) = (u_t, v_t)$, these formulas give $x'[e] = y[e] = Z_t(e)$ and $x'[-M-1] = u_t = Z_t(-M-1)$. They give $y'_c[-M] = v_c = Z^c_{t+1}(-M)$; for $c = b$ and $M = 0$ this uses (Z2), and for $M \geq 1$ the layer is common. For $-M+1 \leq i \leq d-1$, (31) with the convention $y_c[-M-1] = u_t = Z_t(-M-1)$ gives $y'_c[i] = Z^c_t(i) \oplus Z^c_{t-1}(i-2) \oplus g(Z^c_t(i-1), Z^c_t(i-2)) = Z^c_{t+1}(i)$ by (Z1). So the edge leads to η_{t+1}. Its increment (33) is $Z^c_{t-1}(d-2) \oplus g(Z^c_t(d-1), Z^c_t(d-2)) = \varphi^c_t$, again with the convention when $d - 2 = -M - 1$. Summing over a period gives ℓ.

Independence of w. By Lemma B.12(1) and Lemma B.8(1), S_ν is cut out by conditions involving only P_ν, and these do not depend on w. The same holds for the formulas of Lemma B.3 on P_ν. □

Lemma B.20. *Let Z be a periodic history below d of period T.*

1. (Swap.) *Exchanging the two copies gives a periodic history below d with (ℓ_a, ℓ_b) exchanged.*
2. (Top flip.) *Let $\Sigma \subseteq \{a, b\}$ be nonempty, with $\Sigma = \{a, b\}$ if $d - 1 \leq 0$. Complementing $Z^c_t(d-1)$ for all t and all $c \in \Sigma$ gives a periodic history Z' below d with $\ell_c(Z') = \ell_c(Z) \oplus N_c(Z, d-2)$ for $c \in \Sigma$ and $\ell_c(Z') = \ell_c(Z)$ for $c \notin \Sigma$.*
3. (Restriction.) *For $d' \leq d$, the restriction of Z to layers $i \leq d' - 1$ is a periodic history below d'.*
4. (Splitting the front.) *If Y is a single periodic history below 1, then (Y, Y'), where Y' equals Y except that $Y'_t(0) = 1 \oplus Y_t(0)$, is a periodic history below 1 with $\ell_a \oplus \ell_b = N(Y, -1)$.*

Proof. (1) and (3) are clear. (2) Among the equations (Z1) for $i \leq d-1$, layer $d-1$ occurs only in the one for $i = d-1$, and only through $Z_{t+1}(d-1) \oplus Z_t(d-1)$, which complementation does not change. Condition (Z2) is preserved: if $d - 1 \leq 0$, both copies are complemented. In φ^c_t, $Z_{t-1}(d-2)$ is unchanged and $g(1 \oplus z, Z_t(d-2)) = g(z, Z_t(d-2)) \oplus \partial g(z, Z_t(d-2)) = g(z, Z_t(d-2)) \oplus Z_t(d-2) \oplus g_{10}$, because $g_{11} = 1$. Summing over t gives the formula. (4) The pair (Y, Y) is not a history below 1, since it violates (Z2), so (2) does not apply to it. But the computation in the proof of (2) is made in one copy at a time and uses only (Z1). Applied with $d = 1$ to the single history Y, it shows that Y' satisfies (Z1) and that $\ell(Y') = \ell(Y) \oplus N(Y, -1)$. So $\ell_b = \ell_a \oplus N(Y, -1)$ for the pair (Y, Y'). Condition (Z2) holds because Y and Y' agree below layer 0 and differ at layer 0. □

Lemma B.21. *Let $c \in \mathcal{A}^{\mathbb{Z}}$ satisfy $\sigma^n c = c$ and $F^L c = c$, and let $p \in \mathbb{Z}$. Then $Y_t(i) = y\big((F^t c)_{p+t-i}\big)$ is periodic in t with a period dividing $\operatorname{lcm}(n, L)$, satisfies $Y_t(i+n) = Y_t(i)$, and satisfies (Z1) for every $i \in \mathbb{Z}$. So it is a single periodic history below every $d \le 0$, and below 1.*

Proof. Periodicity is clear. With $s = p + t + 1 - i$, (Z1) at (t, i) reads $Y(t+1, s) = Y(t, s-1) \oplus Y(t-1, s) \oplus g(Y(t, s), Y(t, s+1))$ for the diagram $Y = Y_c$, which is (27). □

We call such a Y the comoving diagram of the *tail* c with *phase* p. At time 0 the front sits at site p, and layer i at time t is site $p + t - i$.

Corollary B.22. *Let $w \ge d$, and let $M \ge 0$ with $-M + 1 \le d$. Then* Hol $= K$ *at the step $L_{\nu+1} = y[d]$ in each of the following cases:*

(a) *$d \le 1$, and there is a single periodic history Y below d with $N(Y, d-2) = 1$;*
(b) *$d \ge 1$, and there is a periodic history Z below d with $\ell_a(Z) \ne \ell_b(Z)$;*
(c) *$d \ge 2$, and there is a periodic history Z below d with $N_a(Z, d-2) \ne N_b(Z, d-2)$.*

If the history is the restriction of a tail or of a history defined on all layers $\le D - 1$, then the conclusion holds for every such M and w at once, for all $d \le D$ covered by the case.

Proof. (b) By Lemma B.19, $(\ell_a, \ell_b) \in \{(1,0), (0,1)\}$ lies in Hol, and by Lemma B.20(1) so does the other element of this set. These two elements generate $K = (\mathbb{Z}/2)^2$. (a) For $d \le 0$, Y and its top flip (Lemma B.20(2) with $\Sigma = \{a, b\}$) have ℓ differing by 1, so one of them gives $\ell = 1$, which generates $K = \mathbb{Z}/2$. For $d = 1$, Lemma B.20(4) gives a history with $\ell_a \ne \ell_b$, and (b) applies. (c) If $\ell_a(Z) \ne \ell_b(Z)$, use (b). Otherwise flip the top layer $d - 1 \ge 1$ of the one copy c with $N_c(Z, d-2) = 1$. By Lemma B.20(2) this changes ℓ_c and not the other, and (b) applies. The last sentence holds because the history is the same for all M and w. □

B.5. **Histories ahead of the front.** For $f = 30$ the loops of Lemma B.17 are time-constant histories in the sense of Definition B.18. For 45, 75 and 135 we give histories explicitly. The conventions below are also used in Section B.6. Tails are given by their x-row and y-row, with the phase stated by the site at which the front sits at time 0. Every series below is listed over $t = 0, 1, \dots, T-1$. The column φ lists the bits $Z_{t-1}(i-2) \oplus g(Z_t(i-1), Z_t(i-2))$ for $t = 0, \dots, T-1$. Condition (Z1) at layer i says exactly that $Z_{t+1}(i) = Z_t(i) \oplus \varphi_t$, which can be checked row by row. Layers that come from a tail satisfy (Z1) by Lemma B.21, and need no check.

Ahead of the front and layer 1. For $d \le 1$ we use case (a) of Corollary B.22 with a tail whose $N(Y, i)$ is odd for every i. Any phase may be used.

- $f = 45$, $g_{10} = 0$. The tail is $x = y = \cdots 111 \cdots$, the configuration $(1,1)^{\mathbb{Z}}$. It is fixed by F, since $f(1,1,1) = 0$. So $Y \equiv 1$, $T = 1$, and $N(Y, i) = 1$.
- $f = 75$, $g_{10} = 1$. The tail is $x = y = \cdots 0101 \cdots$, with $x_s = y_s = [s \text{ odd}]$. It is fixed by F: for even s, $y'_s = x_s \oplus y_{s-1} \oplus g(y_s, y_{s+1}) = 0 \oplus 1 \oplus g(0,1) = 0$, and for odd s, $y'_s = 1 \oplus 0 \oplus g(1,0) = 1$. So $Y_t(i) = [p + t - i \text{ odd}]$, $T = 2$, and $N(Y, i) = (Y_0(i) \oplus 1) + (Y_1(i) \oplus 1) = 1$.
- $f = 135$, $g_{10} = 0$. The tail is $x = y = \cdots 0001\,0001 \cdots$, with $x_s = y_s = [s \equiv 3 \bmod 4]$. Its F-orbit has length 8:

  $$(0001;0001) \to (0001;0110) \to (0110;1001) \to (1001;0100)$$
  $$\to (0100;0100) \to (0100;1001) \to (1001;0110) \to (0110;0001) \to (0001;0001),$$

  written over the sites $0, 1, 2, 3$. With phase $p = 0$ the comoving period is $T = 8$, and the layers $i \equiv -1, -2, -3, -4 \pmod 4$ have series 01101000, 00110100, 10000110, 01000011. Each contains three 1's, so $N(Y, i) = 1$ for all i. The same holds for the other three phases.

(For $f = 30$ the tail $(0,0)^{\mathbb{Z}}$, with $Y \equiv 0$ and $g_{10} = 1$, serves the same purpose.)

Proposition B.23. *Let $f \in \{30, 45, 75, 135\}$, $w \geq 1$ and $M \geq 0$. Then* $\mathrm{Hol} = K$ *at every step $L_{\nu+1} = y[d]$ with $-M + 1 \leq d \leq \min(1, w)$.*

Proof. For 30 this is the cases $d \leq 0$ and $d = 1$ of Lemma B.17. For 45, 75 and 135, the tails above give, by Lemma B.21, single periodic histories below every $d \leq 1$ with $N(Y, d-2) = 1$, so Corollary B.22(a) applies. Each tail is defined on all layers, so it serves every M and w at once. □

B.6. **Histories behind the front.** The steps $y[d]$ with $d \geq 2$ are behind the front. We prove that they have full holonomy for 45, 75 and 135 (Proposition B.28). For 45 and 75 the proof uses explicit histories, which are the time-periodic replacements for the stationary states of Lemma B.16; for 135 it uses one explicit history and Proposition B.26. The conventions are those of Section B.5.

Rule 75 behind the front: a stationary wall. Let $d \geq 2$, and take $T = 1$, so that (Z1) reads $Z(i-2) = g(Z(i-1), Z(i-2))$ for $i \leq d-1$. Here $g = \neg b \vee c$, so $g(b,1) = 1$ and $g(b,0) = \neg b$. Put

$$Z(i) = 1 \quad (i \leq -1), \qquad Z^a(i) = [i \text{ even}], \quad Z^b(i) = [i \text{ odd}] \quad (0 \leq i \leq d-2),$$

and $Z^a(d-1) = Z^b(d-1) = 0$. So $Z^a = (\dots 1,1 \mid 1 \mid 0,1,0,\dots)$ and $Z^b = (\dots 1,1 \mid 0 \mid 1,0,1,\dots)$, read from layer -1 through the front at layer 0, with the top layer $d-1$ set to 0. Condition (Z1) asks $Z(k) = g(Z(k+1), Z(k))$ for $k \leq d-3$. This holds when $Z(k) = 1$. When $Z(k) = 0$, we have $0 \leq k \leq d-3$, and the alternating pattern gives $Z(k+1) = 1$ and $g(1,0) = 0$. Condition (Z2) is clear. Now $\varphi^c = Z^c(d-2) \oplus g(0, Z^c(d-2)) = Z^c(d-2) \oplus 1$, and $Z^a(d-2) \neq Z^b(d-2)$, so $\ell_a \neq \ell_b$. Corollary B.22(b) applies for every $d \geq 2$.

Rule 45 at $d = 2$: a loop of period 4. Here $g = b \vee \neg c$. The tail is $x = y = \cdots 0101 \cdots$, with $x_s = y_s = [s \text{ odd}]$, and phase $p = 0$, so the front sits at a site where the cell is $(0,0)$. Its F-orbit is the 4-cycle O of Lemma 5.5, and the comoving period is $T = 4$. For $i \leq -1$ the layer series is 0011 for even i and 1100 for odd i. The history below 2 is

layer i	$Z^a(i)$	φ^a	$Z^b(i)$	φ^b
-2	0011	0101	0011	0101
-1	1100	0101	1100	0101
0	0011	0101	1100	0101
1	0011	0101	0111	1001
step $d = 2$		0110		0001

Copy a is the tail on layers ≤ 0. Its layer 1 is the complement of the tail's (1100), so it is a top flip. Copy b is the tail with layer 0 complemented (Lemma B.20(4)). Only layer 1 of copy b needs checking. There, $\varphi^b_t = Z_{t-1}(-1) \oplus g(Z^b_t(0), Z_t(-1))$ equals $0 \oplus g(1,1)$, $1 \oplus g(1,1)$, $1 \oplus g(0,0)$, $0 \oplus g(0,0)$, that is, $1,0,0,1$. And 0111 does follow these increments. The last row gives $\ell = (0,1)$, so Corollary B.22(b) applies at $d = 2$.

Rule 45 at $d \geq 3$: a time-periodic wall. Take $T = 2$ and the all-ones tail on layers ≤ -1. Since $Z_{t-1} = Z_{t+1}$, condition (Z1) at layer i says that both entries of $\varphi(i) = \big(Z_1(i-2) \oplus g(Z_0(i-1), Z_0(i-2)),\ Z_0(i-2) \oplus g(Z_1(i-1), Z_1(i-2))\big)$ equal $Z_0(i) \oplus Z_1(i)$. The series $(Z_0(i), Z_1(i))$ are:

layer i	≤ -1	0	1	2	3	4	5	6	7	$\cdots$
$Z^a(i)$	11	11	11	00	01	10	00	00	01	period 4 from $i=2$
$\varphi^a(i)$	00	00	00	00	11	11	00	00	11	
$Z^b(i)$	11	00	01	10	00	00	01	10	00	period 4 from $i=0$
$\varphi^b(i)$	00	00	11	11	00	00	11	11	00	

For example, $\varphi^a(3) = (Z_1(1) \oplus g(Z_0(2), Z_0(1)),\ Z_0(1) \oplus g(Z_1(2), Z_1(1))) = (1 \oplus g(0,1),\ 1 \oplus g(0,1)) = (1,1)$, which matches $Z_0^a(3) \oplus Z_1^a(3) = 1$. Condition (Z1) at layer i reads only layers $i-2, i-1, i$. By the stated periodicities, these three layers coincide with the three layers four lower when $i-6 \ge 2$ in copy a and when $i-6 \ge 0$ in copy b. So the checks at $i \ge 8$ (copy a) and $i \ge 6$ (copy b) repeat checks four layers lower, and every other check, at $i \le 7$ and $i \le 5$ respectively, is visible in the table. (The triples differ at $i = 6, 7$ in copy a and at $i = 5$ in copy b, so these indices cannot be lowered.) Condition (Z2) is clear. With $g_{10} = 0$, $N_c(Z, i) = Z_0^c(i) \oplus Z_1^c(i)$. For $i \ge 1$ this is 1 for copy a exactly when $i \equiv 3, 0 \pmod 4$, and 1 for copy b exactly when $i \equiv 1, 2 \pmod 4$. So $N_a(Z, i) \ne N_b(Z, i)$ for every $i \ge 1$. For every $d \ge 3$, the restriction of Z to layers $\le d-1$ (Lemma B.20(3)) satisfies Corollary B.22(c). At $d = 2$ it does not, since $N_a(Z, 0) = N_b(Z, 0) = 0$; that step is covered by the period-4 loop above.

Rule 135 at $d = 2, 3$. Here $g = \neg(b \wedge c)$. The tail is $x = y = \cdots 011\,011 \cdots$, with $x_s = y_s = [s \not\equiv 0 \bmod 3]$, and phase $p = 0$, so the front sits at a site where the cell is $(0, 0)$. The tail is fixed by F. For $s \equiv 0, 1, 2 \pmod 3$, $y'_s = x_s \oplus y_{s-1} \oplus g(y_s, y_{s+1})$ equals $0 \oplus 1 \oplus g(0,1) = 0$, $1 \oplus 0 \oplus g(1,1) = 1$ and $1 \oplus 1 \oplus g(1,0) = 1$. Its comoving period is $T = 3$, and the layers $i \equiv 0, -1, -2 \pmod 3$ have series 011, 110, 101. Two histories, below 2 and below 3:

	below 2				below 3			
layer i	Z^a	φ^a	Z^b	φ^b	Z^a	φ^a	Z^b	φ^b
-2	101	110	101	110	101	110	101	110
-1	110	011	110	011	110	011	110	011
0	011	101	100	101	011	101	100	101
1	010	110	111	000	010	110	000	000
2					000	000	011	101
step d		000		001		110		111

In both histories, copy a is the tail on layers ≤ 0, and its layer 1 is a top flip of the tail's (101). Copy b is the tail with layer 0 complemented. The remaining rows are checked against the φ columns. For example, for copy b at layer 1, $\varphi_t^b = Z_{t-1}(-1) \oplus g(Z_t^b(0), Z_t(-1))$ equals $0 \oplus g(1,1)$, $1 \oplus g(0,1)$, $1 \oplus g(0,0)$, that is, $0, 0, 0$, so layer 1 is constant (111 or 000). The step rows give $\ell = (0, 1)$ in both cases, so Corollary B.22(b) applies at $d = 2$ and $d = 3$. Both histories have the odd period 3; the history below 2 supplies hypothesis (ii) of Proposition B.26 for 135. The same works for the other two phases of this tail.

Remark B.24 (stationary fronts). Call a periodic history *stationary* if $T = 1$. A stationary history below d with $d \ge 2$ can exist only if $g_{10}(1 + g_{01}) + g_{00} = 0$. Indeed, (Z1) at $i = 1$ reads $Z(-1) = g(Z^c(0), Z(-1))$ for both copies, and $Z^a(0) \ne Z^b(0)$. So $\partial g(\cdot, Z(-1)) = Z(-1) \oplus g_{10}$ vanishes, that is, $Z(-1) = g_{10}$. Then $g_{10} = g(0, g_{10}) = g_{01}g_{10} \oplus g_{00}$. The quantity $g_{10}(1+g_{01})+g_{00}$ is invariant under κ. It is 0 for the classes of 30 and 75 and 1 for those of 45 and 135, so for the latter two classes the histories behind the front must be time-periodic. For 30 and 75, stationary histories for every $d \ge 2$ are given explicitly (Lemma B.17 and the wall above). We do not claim the converse in general.

All depths from one closed walk of odd length. The histories above reach a fixed depth, or every depth when there is a wall. The next proposition needs neither: one closed walk of odd length at the step $y[2]$ gives full holonomy at every step behind it. Fix a left-permutive rule $f = a \oplus g(b, c)$ with $g_{11} = 1$, and integers $w \geq 2$ and $M \geq 0$. For $1 \leq d \leq w$ let G_d be the graph Γ_ν of the tower of Section B.3 with $L_{\nu+1} = y[d]$. Its vertices are the elements of S_ν, which assign bit pairs to the layers $x[-M-1], y[-M], \ldots, y[d-1], x[d-1]$. For $d \geq 2$ these are the layers of G_{d-1} together with $y[d-1]$ and $x[d-1]$. Restricting the vertices of a walk ς in G_d to the layers of G_{d-1} gives a walk $\bar{\varsigma}$ in G_{d-1} with the same labels (Lemma B.12(3)), the *projection* of ς. For an edge ϵ of G_d from η put

$$\psi_d(\epsilon) = \big(1,\ \varphi_a(\epsilon),\ \varphi_b(\epsilon),\ y_a[d-1](\eta) \oplus g_{10},\ y_b[d-1](\eta) \oplus g_{10}\big) \in (\mathbb{Z}/2)^5,$$

where $\varphi = (\varphi_a, \varphi_b)$ is the increment (33) at the step $y[d]$, as in Lemma B.15. For a walk ς in G_d let $\Psi_d(\varsigma)$ be the sum of ψ_d over its edges. We write

$$\Psi_d(\varsigma) = \big(T(\varsigma),\ \ell_a(\varsigma),\ \ell_b(\varsigma),\ N_a(\varsigma),\ N_b(\varsigma)\big),$$

so that $T(\varsigma)$ is the length of ς modulo 2, $\ell(\varsigma) = (\ell_a, \ell_b)$ is the quantity of Lemma B.15, and $N_c(\varsigma)$ counts, modulo 2, the vertices of ς, the last one excepted, at which $y_c[d-1] \neq g_{10}$. Let $I_d \subseteq (\mathbb{Z}/2)^5$ be the set of values $\Psi_d(\varsigma)$ of closed walks ς in G_d. If ς is the closed walk of a periodic history Z below d of period T (Lemma B.19), then $\Psi_d(\varsigma) = \big(T, \ell_a(Z), \ell_b(Z), N_a(Z, d-1), N_b(Z, d-1)\big)$.

Lemma B.25. *Let $2 \leq d \leq w$.*

(1) *If G_d is strongly connected, then I_d is a subgroup of $(\mathbb{Z}/2)^5$.*
(2) (Top flip.) *For $c \in \{a, b\}$ let τ_c complement the two bits $y_c[d-1]$ and $x_c[d-1]$ of an assignment. Then τ_c maps the vertex set of G_d onto itself and maps every edge to an edge with the same label. For a closed walk ς in G_d with projection $\bar{\varsigma}$,*

$$\Psi_d(\tau_a\varsigma) = \Psi_d(\varsigma) + \big(0,\ N_a(\bar{\varsigma}),\ 0,\ T(\varsigma),\ 0\big), \qquad \Psi_d(\tau_b\varsigma) = \Psi_d(\varsigma) + \big(0,\ 0,\ N_b(\bar{\varsigma}),\ 0,\ T(\varsigma)\big),$$

where $N_c(\bar{\varsigma})$ is computed in G_{d-1}.
(3) (Swap.) *The exchange μ of the two copies maps the vertex set of G_d onto itself and edges to edges. It maps a closed walk ς to a closed walk $\mu\varsigma$ with $\Psi_d(\mu\varsigma) = (T, \ell_b, \ell_a, N_b, N_a)(\varsigma)$ and $N_a(\overline{\mu\varsigma}) = N_b(\bar{\varsigma})$, $N_b(\overline{\mu\varsigma}) = N_a(\bar{\varsigma})$.*
(4) (Lifting.) *Let ω be a closed walk in G_{d-1} with $\ell(\omega) = (0, 0)$, where ℓ is taken at the step $y[d-1]$. Then there is a closed walk $\hat{\omega}$ in G_d with projection ω; in particular $\hat{\omega}$ has the same length as ω.*

Proof. (1) Ψ_d is additive under concatenation of walks. So the argument in the proof of Lemma B.15(2) applies with Ψ_d in place of ℓ. The values of closed walks at a fixed vertex η_2 form a submonoid of a finite group, hence a subgroup. For a closed walk ς at any vertex η, choose walks α from η_2 to η and β from η to η_2; then $\Psi_d(\varsigma) = \Psi_d(\alpha\varsigma\beta) - \Psi_d(\alpha\beta)$ lies in that subgroup. Conversely that subgroup is contained in I_d, so I_d is equal to it.

(2) Let $\Gamma_\nu = G_d$. By Lemma B.12(1), S_ν is defined by the conditions of (30), which involve only layers $y[i]$ with $i \leq 0$ and $x[e]$ with $e \leq -1$, and by (C_j) for $0 \leq j \leq d-1$. By Lemmas B.6 and B.8(1), (C_0) is $\delta x[0] = 1$, and for $j \geq 1$ condition (C_j) reads $\delta x[j] \oplus \delta y[j] = \Phi_j^a \oplus \Phi_j^b \oplus [j \text{ even}]$, where Φ_j reads only layers of index less than j. Since $d - 1 \geq 1$, the layer $d-1$ occurs only in (C_{d-1}), and there only through $\delta x[d-1] \oplus \delta y[d-1]$, which τ_c does not change. So $\tau_c(S_\nu) = S_\nu$. By Lemma B.3, the layer $x[d-1]$ is read only by $y'[d+1]$, and the layer $y[d-1]$ is read by $x'[d-1] = y[d-1]$, by $y'[d-1]$, in which it is a summand, and by $y'[d]$ and $y'[d+1]$. The layers $y[d]$ and $y[d+1]$ do not belong to G_d. The other layers of G_d are neither changed by τ_c nor computed from the layer $d-1$. Since $\eta \cdot (u, v)$ is computed from η and (u, v) by these formulas (Lemma B.12(3)), $\tau_c(\eta \cdot (u, v)) = \tau_c(\eta) \cdot (u, v)$ for every vertex η and every label. Now compare

ψ_d on an edge from η and on its image. The increment $\varphi_c = x_c[d-2] \oplus g(y_c[d-1], y_c[d-2])$ changes by $\partial g(y_c[d-1], y_c[d-2]) = y_c[d-2] \oplus g_{10}$, because $g_{11} = 1$ (here $d-2 \geq 0 > -M-1$, so the convention $y_c[-M-1] = u$ does not enter); the increment of the other copy does not change; the entry $y_c[d-1] \oplus g_{10}$ changes by 1. Summing over the edges of ς gives the two formulas, since $\sum_t \big(y_c[d-2](\eta_t) \oplus g_{10}\big) = N_c(\bar{\varsigma})$, the sum being over the vertices η_t of ς, the last one excepted.

(3) The conditions (30) and (C_j) prescribe values of δ, so they are symmetric in the two copies. The formulas of Lemma B.3 are the same for both copies, except that $v_b = v \oplus [M = 0]$. So the exchange maps the edge with label (u, v) to the edge with label $(u, v \oplus [M = 0])$, and it exchanges the entries of ψ_d and of ψ_{d-1} belonging to the two copies.

(4) Let $G_{d-1} = \Gamma_{\nu'}$, so that $L_{\nu'+1} = y[d-1]$, $L_{\nu'+2} = x[d-1]$ and $G_d = \Gamma_{\nu'+2}$. Recall that a walk lifts to the next graph of the tower, uniquely, from any lift of its initial vertex. Let $\omega = (\eta_0 \to \cdots \to \eta_T = \eta_0)$; we may assume $T \geq 1$. Since $d-1 \geq 1$, every pair (η_0, z) with $z \in (\mathbb{Z}/2)^2$ is a vertex of $\Gamma_{\nu'+1}$ (Lemma B.12(2)). By Lemma B.15(1), the lift of ω to $\Gamma_{\nu'+1}$ starting at (η_0, z) ends at $(\eta_0, z + \ell(\omega)) = (\eta_0, z)$, so it is a closed walk ω_1. Lift ω_1 to $\Gamma_{\nu'+2}$, starting at any lift of its initial vertex. The lift ends at a vertex ζ over the same vertex, and by Lemma B.3(1) the layer $x[d-1]$ of ζ is the layer $y[d-1]$ of the last vertex but one of ω_1. Lift ω_1 once more, starting at ζ. The new lift ends at a vertex over the same vertex with the same layer $x[d-1]$, that is, at ζ. This closed walk is $\hat{\omega}$. □

Proposition B.26. *Let $f = a \oplus g(b, c)$ be left-permutive with $g_{11} = 1$, and let $w \geq 2$ and $M \geq 0$. Suppose that*

(i) $\mathrm{Hol} = K$ *at every step $L_{\nu+1} = y[d]$ with $-M+1 \leq d \leq 1$, and*
(ii) G_2 *has a closed walk of odd length.*

Then for every $2 \leq d \leq w$ the graph G_d is strongly connected and $I_d = (\mathbb{Z}/2)^5$. In particular $\mathrm{Hol} = K$ *at every step $L_{\nu+1} = y[d]$ with $2 \leq d \leq w$.*

Proof. The last statement follows from the first, since $I_d = (\mathbb{Z}/2)^5$ contains elements with every value of (ℓ_a, ℓ_b). We prove the first by induction on d.

The case $d = 2$. By Lemmas B.13, B.14 and B.15(2) and hypothesis (i), every graph of the tower up to G_2 is strongly connected, as in the proof of Theorem B.4. So I_2 is a group by Lemma B.25(1). Let ς be a closed walk of odd length in G_2, and put $n_c = N_c(\bar{\varsigma})$, computed in G_1. The top y-layer of G_1 is $y[0]$, and $y_b[0] = y_a[0] \oplus 1$ at every vertex, so $n_a + n_b = T(\varsigma) = 1$. By Lemma B.25(2), I_2 contains $\Psi_2(\tau_a\varsigma) - \Psi_2(\varsigma) = (0, n_a, 0, 1, 0)$ and $\Psi_2(\tau_b\varsigma) - \Psi_2(\varsigma) = (0, 0, n_b, 0, 1)$. The same applied to $\mu\varsigma$ (Lemma B.25(3)) gives $(0, n_b, 0, 1, 0)$ and $(0, 0, n_a, 0, 1)$. Since $\{n_a, n_b\} = \{0, 1\}$, these four elements are

$$(0,1,0,1,0), \qquad (0,0,0,1,0), \qquad (0,0,1,0,1), \qquad (0,0,0,0,1). \tag{34}$$

They span $\{0\} \times (\mathbb{Z}/2)^4$. Together with $\Psi_2(\varsigma)$, whose first entry is 1, they generate $(\mathbb{Z}/2)^5$.

The step from $d-1$ to d, for $3 \leq d \leq w$. Assume that G_{d-1} is strongly connected and $I_{d-1} = (\mathbb{Z}/2)^5$. Then $\mathrm{Hol} = K$ at the step $y[d-1]$, so G_d is strongly connected by Lemmas B.15(2) and B.14, and I_d is a group. Choose closed walks ω and ω' in G_{d-1} with

$$\Psi_{d-1}(\omega) = (1,0,0,1,1), \qquad \Psi_{d-1}(\omega') = (1,0,0,0,0).$$

Both have $\ell = (0, 0)$, so they have lifts $\hat{\omega}, \hat{\omega}'$ to closed walks in G_d (Lemma B.25(4)), of odd length. By Lemma B.25(2), I_d contains

$$\Psi_d(\tau_a\hat{\omega}) - \Psi_d(\hat{\omega}) = (0,1,0,1,0), \qquad \Psi_d(\tau_b\hat{\omega}) - \Psi_d(\hat{\omega}) = (0,0,1,0,1),$$

because $N_a(\omega) = N_b(\omega) = 1$, and likewise $(0,0,0,1,0)$ and $(0,0,0,0,1)$ from $\hat{\omega}'$, because $N_a(\omega') = N_b(\omega') = 0$. These are the elements (34). Together with $\Psi_d(\hat{\omega})$, whose first entry is 1, they generate $(\mathbb{Z}/2)^5$. □

Remark B.27. (1) Hypothesis (ii) enters through the entry T: on a closed walk of even length the top flip does not change N_c. It holds as soon as there is a periodic history below 2 of odd period, by Lemma B.19; such a history serves every M and every $w \geq 2$. For 30 and 75 the stationary histories below 2 (the case $d = 2$ of Lemma B.17, and the wall above) have period 1, so Proposition B.26 gives a second proof of the case $d \geq 2$ for these rules.

(2) The proof uses $g_{11} = 1$ only in the top flip. For an affine rule, $\partial g = g_{10}$ is constant, the top flip changes Ψ_d by $(0, g_{10}T, 0, T, 0)$, and the argument gives nothing. The conclusion can fail: for rule 150 with $w = 3$ and $M = 0$ the step $y[3]$ does not have $\mathrm{Hol} = K$, since the eventual image of $\Gamma_{3,0}$ is not strongly connected while that of $\Gamma_{2,0}$ is (Remark 5.8): the tower of $\Gamma_{3,0}$ through the step $x[2]$ is that of $\Gamma_{2,0}$ followed by one memory step (the last part of Lemma B.19, whose proof does not use $g_{11} = 1$), so the failure occurs at $y[3]$.

(3) The proof is not constructive. The walks ω and ω' exist because I_{d-1} is the whole group; they depend on M, and their bottom layers are arbitrary periodic input sequences rather than the layers of a tail.

Proposition B.28. *Let $f \in \{30, 45, 75, 135\}$, $w \geq 1$, $M \geq 0$, and $-M + 1 \leq d \leq w$. Then $\mathrm{Hol} = K$ at the step $L_{\nu+1} = y[d]$.*

Proof. For 30 this is Lemma B.17. For the others it follows from Proposition B.23 for $d \leq 1$. Let $d \geq 2$. For 75 and 45 we use Corollary B.22 with the histories of this subsection: for 75, the wall; for 45, the period-4 loop for $d = 2$ and the restricted wall for $d \geq 3$. All of them are defined on all layers below their top, so they serve every M. For 135 we use Proposition B.26. Its hypothesis (i) is Proposition B.23. For (ii), the history below 2 given above for 135 has period 3, so by Lemma B.19 it gives a closed walk of length 3 in G_2, for every $M \geq 0$ and $w \geq 2$. □

B.7. **The lift certificate.** This subsection is not used in any other proof. Proposition B.26 gives full holonomy for 135 at every depth without exhibiting histories. The following finite computation produces explicit histories depth by depth. It confirms the behind-layer case of Proposition B.28 by an independent route: for 30 and 45 at every depth, and for 135 at the depths $d \leq 105$. Its parameters are f, $n_{\max}$, $T_{\max}$ and the largest depth D (Appendix A.2).

Nodes. A node of depth $k \geq 0$ is a pair (X, V). Here X is a time series, taken up to rotation, of the four bits $(Z^a_t(k-1), Z^a_t(k), Z^b_t(k-1), Z^b_t(k))$ over its minimal period m, and $V \geq 0$ is an integer. The node is *realized* by a periodic history Z on layers $\leq k$ (that is, below $k+1$) of minimal period T_Z if the top two layers of Z have joint series X, up to a time shift of Z, and V is the 2-adic valuation $v_2(T_Z)$. Then $m \mid T_Z$.

Start. $\mathcal{R}_0$ consists of one node of depth 0 for each F-orbit of a configuration c with $\sigma^n c = c$, $n \leq n_{\max}$. It is realized by the comoving diagram of c with phase 0 in copy a and the same diagram with layer 0 complemented in copy b (Lemmas B.21 and B.20(4)). Here V is v_2 of the minimal comoving period of the diagram.

Test. The bits φ^c_t for the step $y[k+1]$ read only the top two layers, so they are m-periodic. Over the period T_Z, $\ell_c(Z) = (T_Z/m) \sum_{t<m} \varphi^c_t$, and T_Z/m is odd exactly when $v_2(m) = V$. The test $\mathrm{test}(\mathcal{R}_k)$ is true if some node of $\mathcal{R}_k$ has $v_2(m) = V$ and $\sum_{t<m} \varphi^a_t \neq \sum_{t<m} \varphi^b_t$. Then any realization Z of that node has $\ell_a(Z) \neq \ell_b(Z)$, and Corollary B.22(b) gives $\mathrm{Hol} = K$ at the step $y[k+1]$, for every M.

Lift. By (Z1), a layer $k+1$ extending Z in copy c is a series with $Z_{t+1}(k+1) = Z_t(k+1) \oplus \varphi^c_t$. If $\sum_{t<m} \varphi^c_t = 0$ there are two such series, of period m, one for each initial value. If the sum is 1, there are again two such series, each of period $2m$ and not of period m; so the minimal period m' of either one divides $2m$ and does not divide m, which gives $v_2(m') = v_2(m) + 1$. (The minimal period need not be $2m$: for $m = 3$ the series 010101 has minimal period 2.) Only this valuation is used. In this case only the series with initial value 0 is kept. The new history has

minimal period $\operatorname{lcm}(T_Z, m_a, m_b)$, where m_c is the minimal period of the new layer in copy c, and its 2-adic valuation is $\max(V, v_2(m_a), v_2(m_b))$. For each node of $\mathcal{R}_k$ and each choice of new layers, form the joint series of layers $(k, k+1)$; if its length is at most $T_{\max}$, it gives the node $(X', \max(V, v_2(m_a), v_2(m_b)))$. The set of these nodes is $\mathcal{R}_{k+1}$. Each node of $\mathcal{R}_{k+1}$ is realized, by extending a realization of its parent; condition (Z2) is unaffected since $k+1 \geq 1$.

By construction $\mathcal{R}_k$ is a *subset* of the nodes realized by periodic histories: lifts of period above $T_{\max}$ are omitted, and when both copies have odd increment sum, only one of the two relative phases of the new layers is included. This suffices, since only the existence of one good node per depth is used.

Why a repetition covers all depths. The new node set $\mathcal{R}_{k+1} = \mathcal{N}(\mathcal{R}_k)$ is computed by a map $\mathcal{N}$ that does not depend on k, and the test $\operatorname{test}(\mathcal{R}_k)$ depends only on $\mathcal{R}_k$. Hence if $\mathcal{R}_{k_0+\varrho} = \mathcal{R}_{k_0}$ for some $k_0 \geq 0$ and $\varrho \geq 1$, then $\mathcal{R}_{k+\varrho} = \mathcal{R}_k$ for all $k \geq k_0$, by induction, and $\{\mathcal{R}_k : k \geq 0\} = \{\mathcal{R}_0, \ldots, \mathcal{R}_{k_0+\varrho-1}\}$. If $\operatorname{test}(\mathcal{R}_k)$ holds for $0 \leq k < k_0 + \varrho$, it holds for all $k \geq 0$. Then $\mathrm{Hol} = K$ at every step $y[d]$ with $d \geq 1$, for every M.

Proposition B.29 (computer-assisted)**.** (1) *For $f = 45$, $n_{\max} = 2$, $T_{\max} = 4$: $\operatorname{test}(\mathcal{R}_k)$ holds for $0 \leq k \leq 22$, and $\mathcal{R}_{23} = \mathcal{R}_{19}$ (here $k_0 = 19$, $\varrho = 4$, so the tests needed are those with $k \leq 22$). Hence $\mathrm{Hol} = K$ at $y[d]$ for every $d \geq 1$.*

(2) *For $f = 30$, $n_{\max} = 2$, $T_{\max} = 1$: $\operatorname{test}(\mathcal{R}_k)$ holds for $0 \leq k \leq 2$, and $\mathcal{R}_3 = \mathcal{R}_2$.*

(3) *For $f = 135$, $n_{\max} = 4$, $T_{\max} = 12$: $\operatorname{test}(\mathcal{R}_k)$ holds for $0 \leq k \leq 104$, so $\mathrm{Hol} = K$ at $y[d]$ for $1 \leq d \leq 105$, and the sets $\mathcal{R}_k$, $0 \leq k \leq 105$, are pairwise distinct.*

This repeats the case $d \geq 1$ of Proposition B.28 by an independent route, for 45 and 30 at every depth and for 135 at the depths $d \leq 105$.

Proposition B.30. *For $f \in \{30, 45, 75, 135\}$ the conclusions of Theorem B.4 hold for all $w \geq 1$ and $M \geq 0$.*

Proof. Parts (1), (2) and the size of $\mathcal{T}$ hold for every left-permutive f (Proposition B.11 and Lemma B.8), and part (3) holds once $\mathrm{Hol} = K$ at every step $y[d]$ with $-M + 1 \leq d \leq w$, by the proof of Theorem B.4. This is Proposition B.28. The consequences listed in Theorem B.4 follow from (1)–(3) as shown after that theorem. □

B.8. **The class of 135 behind the front.** For 135 the holonomy behind the front is full at every depth (Proposition B.28), and the conclusions of Theorem B.4 hold for every w (Proposition B.30). The proof differs from that for the other three rules in one respect: it gives no uniform family of histories (Remark B.27(3)). By Corollary B.22(b), full holonomy at depth d is witnessed by one periodic history below d with $\ell_a \neq \ell_b$. For 30 and 75 stationary histories do this at every depth $d \geq 2$. For 45 a history of period 4 does so at $d = 2$, and for $d \geq 3$ a wall of period 2 does so after a top flip (Corollary B.22(c)). For 135 the certificate of Section B.7 gives such histories at every depth $d \leq 105$, and the node sets $\mathcal{R}_0, \ldots, \mathcal{R}_{105}$ of Proposition B.29(3) are pairwise distinct. We do not know whether rule 135 has a wall like that of 45, that is, a periodic history Z defined on all layers $i \in \mathbb{Z}$ with $N_a(Z, i) \neq N_b(Z, i)$ for all large i. Nothing in this paper depends on the answer.

Use of AI tools. Large language models (Anthropic's Claude models) were used extensively in this work: to propose and develop proofs, to design, write and run the computer searches and their independent re-implementations, to check proofs and computations, and to draft the manuscript. The models are not authors. The author has reviewed the content and takes full responsibility for it. All code and computational output accompany the paper (Appendix A).

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